

Fourier Analysis, Distributions, and Constant-Coefficient Linear PDE

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Contents

- 0. Introduction
- 1. Fourier series
- 2. Harmonic functions and holomorphic functions in the plane
- 3. The Fourier transform
- 4. Distributions and tempered distributions
- 5. The classical evolution equations
- 6. Radial distributions, polar coordinates, and Bessel functions
- 7. The method of images and Poisson's summation formula
- 8. Homogeneous distributions and principal value distributions
- 9. Elliptic operators
- 10. Local solvability of constant-coefficient PDE
- 11. The discrete Fourier transform
- 12. The fast Fourier transform
- A. The mighty Gaussian and the sublime gamma function

Introduction

Fourier analysis is perhaps the most important single tool in the study of linear partial differential equations. It serves in several ways, the most basic—and historically the first—being to give specific formulas for solutions to various linear PDE with constant coefficients, particularly the three classics, the Laplace, wave, and heat equations:

$$(0.1) \quad \Delta u = f, \quad \frac{\partial^2 u}{\partial t^2} - \Delta u = f, \quad \frac{\partial u}{\partial t} - \Delta u = f,$$

with $\Delta = \partial^2/\partial x_1^2 + \cdots + \partial^2/\partial x_n^2$. The Fourier transform accomplishes this by transforming the operation of $\partial/\partial x_j$ to the algebraic operation of multiplication by $i\xi_j$. Thus the equations (0.1) are transformed to algebraic equations and to ODE with parameters.

Before introducing the Fourier transform of functions on Euclidean space $\mathbb{R}^n$, we discuss the Fourier series associated to functions on the torus $\mathbb{T}^n$ in §1. Methods developed to establish the Fourier inversion formula for

Fourier series, in the special case of the circle $S^1 = \mathbb{T}^1$, provide for free a development of the basic results on harmonic functions in the plane, and we give such results in §2, noting that these results specialize further to yield standard basic results in the theory of holomorphic functions of one complex variable, such as power-series expansions and Cauchy's integral formula.

In §3 we define the Fourier transform of functions on $\mathbb{R}^n$ and prove the Fourier inversion formula. The proof shares with the argument for Fourier series in §1 the property of simultaneously yielding explicit solutions to a PDE, this time the heat equation.

It turns out that representations of solutions to such PDE as listed in (0.1) are most naturally done in terms of objects more general than functions, called distributions. We develop the theory of distributions in §4; Fourier analysis works very naturally with the class of distributions known as tempered distributions.

Section 5, in some sense the heart of this chapter, derives explicit solutions to the classical linear PDE (0.1) via Fourier analysis. The use of Fourier analysis and distribution theory to represent solutions to these PDE gives rise to numerous interesting identities, involving both elementary functions and “special functions,” such as the gamma function and Bessel functions, and we present some of these identities here, only a smattering from a rich area of classical analysis. Further development of harmonic analysis in Chapter 8 will bring in additional studies of special functions.

Fourier analysis and distribution theory are also useful tools for general investigations of linear PDE, in cases where explicit formulas might not be obtainable. We illustrate a couple of cases of this in the present chapter, discussing the existence and behavior of “parametrices” for elliptic PDE with constant coefficients, and applications to smoothness of solutions to such PDE, in §9, and proving local solvability of general linear PDE with constant coefficients in §10. Fourier analysis and distribution theory will acquire further power in the next chapter as tools for investigations of existence and qualitative properties of solutions to various classes of PDE, with the development of Sobolev spaces.

Sections 11 and 12 deal with the discrete Fourier transform, particularly with Fourier analysis on finite cyclic groups. We study this both as an approximation to Fourier analysis on the torus and Euclidean space, sometimes useful for numerical work, and as a subject with its intrinsic interest, and with implications for number theory. In §12 we give a brief description of “fast” algorithms for computing discrete Fourier transforms.

1. Fourier series

Let f be an integrable function on the torus $\mathbb{T}^n$, naturally isomorphic to $\mathbb{R}^n/\mathbb{Z}^n$ and to the Cartesian product of n copies $S^1 \times \cdots \times S^1$ of the circle.

Its Fourier series is by definition a function on $\mathbb{Z}^n$ given by

$$(1.1) \quad \hat{f}(k) = \frac{1}{(2\pi)^n} \int_{\mathbb{T}^n} f(\theta) e^{-ik \cdot \theta} d\theta,$$

where $k = (k_1, \dots, k_n)$, $k \cdot \theta = k_1\theta_1 + \dots + k_n\theta_n$. We use the notation

$$(1.2) \quad \mathcal{F}f(k) = \hat{f}(k).$$

Clearly, we have a continuous linear map

$$(1.3) \quad \mathcal{F} : L^1(\mathbb{T}^n) \longrightarrow \ell^\infty(\mathbb{Z}^n),$$

where $\ell^\infty(\mathbb{Z}^n)$ denotes the space of bounded functions on $\mathbb{Z}^n$, with the sup norm. If $f \in C^\infty(\mathbb{T}^n)$, then we can integrate by parts to get

$$(1.4) \quad k^\alpha \hat{f}(k) = \frac{1}{(2\pi)^n} \int_{\mathbb{T}^n} (D^\alpha f)(\theta) e^{-ik \cdot \theta} d\theta,$$

where $k^\alpha = k_1^{\alpha_1} \dots k_n^{\alpha_n}$, and

$$(1.5) \quad D^\alpha = D_1^{\alpha_1} \dots D_n^{\alpha_n}, \quad D_j = \frac{1}{i} \frac{\partial}{\partial \theta_j}.$$

It follows easily that

$$(1.6) \quad \mathcal{F} : C^\infty(\mathbb{T}^n) \longrightarrow s(\mathbb{Z}^n),$$

where $s(\mathbb{Z}^n)$ consists of functions u on $\mathbb{Z}^n$ which are rapidly decreasing, in the sense that, for each N ,

$$(1.7) \quad p_N(u) = \sup_{k \in \mathbb{Z}^n} \langle k \rangle^N |u(k)| < \infty.$$

Here, we use the notation

$$\langle k \rangle = (1 + |k|^2)^{1/2},$$

where $|k|^2 = k_1^2 + \dots + k_n^2$. If we use the inner product

$$(1.8) \quad (f, g) = (f, g)_{L^2} = \frac{1}{(2\pi)^n} \int_{\mathbb{T}^n} f(\theta) \overline{g(\theta)} d\theta,$$

for $f, g \in C^\infty(\mathbb{T}^n)$, or more generally for $f, g \in L^2(\mathbb{T}^n)$, and if on $s(\mathbb{Z}^n)$, or more generally on $\ell^2(\mathbb{Z}^n)$, the space of square summable functions on $\mathbb{Z}^n$, we use the inner product

$$(1.9) \quad (u, v) = (u, v)_{\ell^2} = \sum_{k \in \mathbb{Z}^n} u(k) \overline{v(k)},$$

we have the formula

$$(1.10) \quad (\mathcal{F}f, u)_{\ell^2} = (f, \mathcal{F}^*u)_{L^2},$$

valid for $f \in C^\infty(\mathbb{T}^n)$, $u \in s(\mathbb{Z}^n)$, where

$$(1.11) \quad \mathcal{F}^* : s(\mathbb{Z}^n) \longrightarrow C^\infty(\mathbb{T}^n)$$

is given by

$$(1.12) \quad (\mathcal{F}^* u)(\theta) = \sum_{k \in \mathbb{Z}^n} u(k) e^{ik \cdot \theta}.$$

Another identity that we will find useful is

$$(1.13) \quad \frac{1}{(2\pi)^n} \int_{\mathbb{T}^n} e^{ik \cdot \theta} e^{-i\ell \cdot \theta} d\theta = \delta_{k\ell},$$

where $\delta_{k\ell} = 1$ if $k = \ell$ and $\delta_{k\ell} = 0$ otherwise.

Our main goal here is to establish the Fourier inversion formula

$$(1.14) \quad f(\theta) = \sum_{k \in \mathbb{Z}^n} \hat{f}(k) e^{ik \cdot \theta},$$

the sum on the right in (1.14) converging in the appropriate function space, depending on the nature of f . Let us single out another space of functions on $\mathbb{T}^n$, the *trigonometric polynomials*:

$$(1.15) \quad \mathcal{TP} = \left\{ \sum_{k \in \mathbb{Z}^n} a(k) e^{ik \cdot \theta} : a(k) = 0 \text{ except for finitely many } k \right\}.$$

Clearly,

$$(1.16) \quad \mathcal{F} : \mathcal{TP} \longrightarrow c_{00}(\mathbb{Z}^n),$$

where $c_{00}(\mathbb{Z}^n)$ consists of functions on $\mathbb{Z}^n$ which vanish except at a finite number of points; this follows from (1.13). The formula (1.12) gives

$$(1.17) \quad \mathcal{F}^* : c_{00}(\mathbb{Z}^n) \longrightarrow \mathcal{TP},$$

and the formula (1.13) easily yields

$$(1.18) \quad \mathcal{F}\mathcal{F}^* = I \text{ on } c_{00}(\mathbb{Z}^n),$$

and even

$$(1.19) \quad \mathcal{F}\mathcal{F}^* = I \text{ on } s(\mathbb{Z}^n).$$

By comparison, the inversion formula (1.14) states

$$(1.20) \quad \mathcal{F}^*\mathcal{F} = I,$$

on $C^\infty(\mathbb{T}^n)$, or some other space of functions on $\mathbb{T}^n$, as specified below. Before getting to this, let us note one other implication of (1.13), namely, if

$$(1.21) \quad f_j(\theta) = \sum_k \varphi_j(k) e^{ik \cdot \theta}$$

are elements of $\mathcal{TP}$, or more generally, if $\varphi_j \in s(\mathbb{Z}^n)$, then we have the Parseval identity

$$(1.22) \quad (f_1, f_2)_{L^2} = \sum_{k \in \mathbb{Z}^n} \varphi_1(k) \overline{\varphi_2(k)};$$

in particular, the Plancherel identity

$$(1.23) \quad \|f_j\|_{L^2}^2 = \sum_{k \in \mathbb{Z}^n} |\varphi_j(k)|^2,$$

for $f_j \in \mathcal{TP}$, or more generally for any f_j of the form (1.21) with $\varphi_j \in s(\mathbb{Z}^n)$. In particular, the map $\mathcal{F}^*$ given by (1.12), and satisfying (1.11) and (1.17), has a unique continuous extension to $\ell^2(\mathbb{Z}^n)$, and

$$(1.24) \quad \mathcal{F}^* : \ell^2(\mathbb{Z}^n) \longrightarrow L^2(\mathbb{T}^n)$$

is an isometry of $\ell^2(\mathbb{Z}^n)$ onto its range. Part of the inversion formula will be that the map (1.24) is also surjective.

Let us note that if $f_j \in \mathcal{TP}$, satisfying (1.21), then (1.13) implies $\hat{f}_j(k) = \varphi_j(k)$, so we have directly in this case:

$$(1.25) \quad \mathcal{F}^* \mathcal{F} = I \text{ on } \mathcal{TP}.$$

One approach to more general inversion formulas would be to establish that $\mathcal{TP}$ is dense in various function spaces, on which $\mathcal{F}^* \mathcal{F}$ can be shown to act continuously. For more details on this approach, see the exercises at the end of §1 and §2 in the Functional Analysis appendix. Here, we will take a superficially different approach. We will make use of such basic results from real analysis as the denseness of $C(\mathbb{T}^n)$ in $L^p(\mathbb{T}^n)$, for $1 \leq p < \infty$.

Our approach to (1.14) will be to establish the following *Abel summability* result. Consider

$$(1.26) \quad J_r f(\theta) = \sum_{k \in \mathbb{Z}^n} \hat{f}(k) r^{|k|} e^{ik \cdot \theta},$$

where $|k| = |k_1| + \cdots + |k_n|$, $r \in [0, 1)$. We will show that

$$(1.27) \quad J_r f \rightarrow f, \text{ as } r \nearrow 1,$$

in the appropriate spaces. The operator J_r in (1.26) is defined for any $f \in L^1(\mathbb{T}^n)$, if $r < 1$, and we have the formula

$$(1.28) \quad J_r f(\theta) = (2\pi)^{-n} \int_{\mathbb{T}^n} f(\theta') \sum_{k \in \mathbb{Z}^n} r^{|k|} e^{ik \cdot (\theta - \theta')} d\theta'.$$

The sum over $\mathbb{Z}^n$ inside the integral can be written as

$$(1.29) \quad \begin{aligned} \sum_{k \in \mathbb{Z}^n} r^{|k|} e^{ik \cdot (\theta - \theta')} &= P_n(r, \theta - \theta') \\ &= p(r, \theta_1 - \theta'_1) \cdots p(r, \theta_n - \theta'_n), \end{aligned}$$

where

$$\begin{aligned}
 p(r, \theta) &= \sum_{k=-\infty}^{\infty} r^{|k|} e^{ik\theta} \\
 (1.30) \qquad &= 1 + \sum_{k=1}^{\infty} (r^k e^{ik\theta} + r^k e^{-ik\theta}) \\
 &= \frac{1 - r^2}{1 - 2r \cos \theta + r^2}.
 \end{aligned}$$

Then we have the explicit integral formula

$$\begin{aligned}
 J_r f(\theta) &= (2\pi)^{-n} \int_{\mathbb{T}^n} f(\theta') P_n(r, \theta - \theta') d\theta' \\
 (1.31) \qquad &= (2\pi)^{-n} \int_{\mathbb{T}^n} f(\theta - \theta') P_n(r, \theta') d\theta'.
 \end{aligned}$$

Let us examine $p(r, \theta)$. It is clear that the numerator and denominator on the right side of (1.30) are positive, so $p(r, \theta) > 0$ for each $r \in [0, 1)$, $\theta \in S^1$. Of course, as $r \nearrow 1$, the numerator tends to 0; as $r \nearrow 1$, the denominator tends to a nonzero limit, except at $\theta = 0$. Since it is clear that

$$(1.32) \qquad (2\pi)^{-1} \int_{S^1} p(r, \theta) d\theta = (2\pi)^{-1} \int_{-\pi}^{\pi} \sum r^{|k|} e^{ik\theta} d\theta = 1,$$

we see that, for r close to 1, $p(r, \theta)$ as a function of θ is highly peaked near $\theta = 0$ and small elsewhere, as in Fig. 1.1.

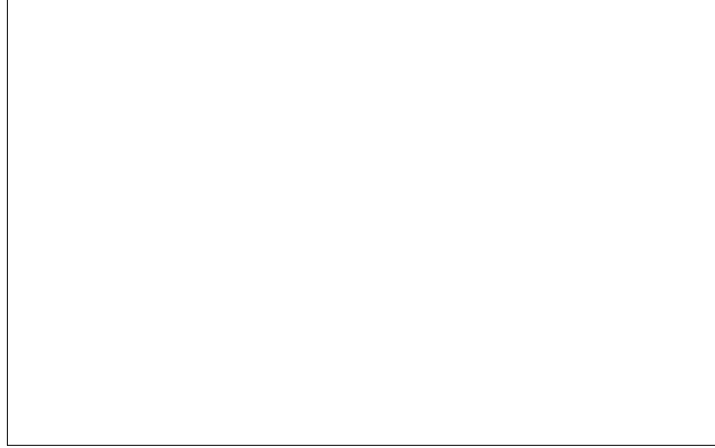


FIGURE 1.1

We are now prepared to prove the following result giving Abel summability (1.27).

Proposition 1.1. *If $f \in C(\mathbb{T}^n)$, then*

$$(1.33) \quad J_r f \rightarrow f \text{ uniformly on } \mathbb{T}^n \text{ as } r \nearrow 1.$$

Furthermore, for any $p \in [1, \infty)$, if $f \in L^p(\mathbb{T}^n)$, then

$$(1.34) \quad J_r f \rightarrow f \text{ in } L^p(\mathbb{T}^n) \text{ as } r \nearrow 1.$$

The proof of (1.33) is an immediate consequence of (1.31) and the peaked nature of $p(r, \theta)$ near $\theta = 0$ discussed above, together with the observation that, if f is continuous at θ , then it does not vary very much near θ . The convergence in (1.34) is in the L^p -norm, defined by

$$(1.35) \quad \|g\|_{L^p} = \left[(2\pi)^{-n} \int_{\mathbb{T}^n} |g(\theta)|^p d\theta \right]^{1/p}.$$

We have the well-known triangle inequality in such a norm:

$$(1.36) \quad \|g_1 + g_2\|_{L^p} \leq \|g_1\|_{L^p} + \|g_2\|_{L^p},$$

and this implies, via (1.31) and (1.32),

$$(1.37) \quad \begin{aligned} \|J_r f\|_{L^p} &= (2\pi)^{-n} \left\| \int_{\mathbb{T}^n} P_n(r, \theta') \tau_{\theta'} f d\theta' \right\|_{L^p} \\ &\leq (2\pi)^{-n} \int_{\mathbb{T}^n} P_n(r, \theta') \|\tau_{\theta'} f\|_{L^p} d\theta' \\ &= \|f\|_{L^p}, \end{aligned}$$

where

$$(1.38) \quad \tau_{\theta'} f(\theta) = f(\theta - \theta'),$$

which implies $\|\tau_{\theta'} f\|_{L^p} = \|f\|_{L^p}$. In other words,

$$(1.39) \quad \|J_r\|_{\mathcal{L}(L^p)} \leq 1, \quad 1 \leq p < \infty,$$

where we are using the operator norm on L^p :

$$(1.40) \quad \|T\|_{\mathcal{L}(L^p)} = \sup \{ \|Tf\|_{L^p} : \|f\|_{L^p} \leq 1 \}.$$

Using this, we can deduce (1.34) from (1.33), and the denseness of $C(\mathbb{T}^n)$ in each space $L^p(\mathbb{T}^n)$, for $1 \leq p < \infty$. Indeed, given $f \in L^p(\mathbb{T}^n)$, and given $\varepsilon > 0$, find $g \in C(\mathbb{T}^n)$ such that $\|f - g\|_{L^p} < \varepsilon$. Note that, generally, $\|g\|_{L^p} \leq \|g\|_{\text{sup}}$. Now we have

$$(1.41) \quad \begin{aligned} \|J_r f - f\|_{L^p} &\leq \|J_r(f - g)\|_{L^p} + \|J_r g - g\|_{L^p} + \|g - f\|_{L^p} \\ &< \varepsilon + \|J_r g - g\|_{L^\infty} + \varepsilon, \end{aligned}$$

making use of (1.39). By (1.33), the middle term is $< \varepsilon$ if r is close enough to 1, so this proves (1.34).

Corollary 1.2. *If $f \in C^\infty(\mathbb{T}^n)$, then the Fourier inversion formula (1.14) holds.*

Proof. In such a case, as noted, we have $\hat{f} \in s(\mathbb{Z}^n)$, so certainly the right side of (1.14) is absolutely convergent to some $f^\# \in C(\mathbb{T}^n)$. In such a case, one a fortiori has

$$(1.42) \quad \lim_{r \nearrow 1} \sum_{k \in \mathbb{Z}^n} \hat{f}(k) r^{|k|} e^{ik \cdot \theta} = f^\#(\theta).$$

But now Proposition 1.1 implies (1.42) is equal to $f(\theta)$ (i.e., $f^\# = f$), so the inversion formula is proved for $f \in C^\infty(\mathbb{T}^n)$.

As a result, we see that

$$(1.43) \quad \mathcal{F}^* : s(\mathbb{Z}^n) \longrightarrow C^\infty(\mathbb{T}^n)$$

is surjective, as well as injective, with two-sided inverse $\mathcal{F} : C^\infty(\mathbb{T}^n) \rightarrow s(\mathbb{Z}^n)$. This of course implies that the map (1.24) has dense range in $L^2(\mathbb{T}^n)$; hence

$$(1.44) \quad \mathcal{F}^* : \ell^2(\mathbb{Z}^n) \longrightarrow L^2(\mathbb{T}^n) \text{ is unitary.}$$

Another way of stating this is

$$(1.45) \quad \{e^{ik \cdot \theta} : k \in \mathbb{Z}^n\} \text{ is an orthonormal basis of } L^2(\mathbb{T}^n),$$

with inner product given by (1.8). Also, the inversion formula

$$(1.46) \quad \mathcal{F}^* \mathcal{F} = I \text{ on } C^\infty(\mathbb{T}^n)$$

implies

$$(1.47) \quad \|\mathcal{F}f\|_{\ell^2} = \|f\|_{L^2},$$

so therefore $\mathcal{F}$ extends by continuity from $C^\infty(\mathbb{T}^n)$ to a map

$$(1.48) \quad \mathcal{F} : L^2(\mathbb{T}^n) \longrightarrow \ell^2(\mathbb{Z}^n), \text{ unitary,}$$

inverting (1.44). The denseness $C^\infty(\mathbb{T}^n) \subset L^2(\mathbb{T}^n) \subset L^1(\mathbb{T}^n)$ implies that this $\mathcal{F}$ coincides with the restriction to $L^2(\mathbb{T}^n)$ of the map (1.3). Note that the fact that (1.44) and (1.48) are inverses of each other extends the inversion result of Corollary 1.2.

We devote a little space to conditions implying that the Fourier series (1.14) is absolutely convergent, weaker than the hypothesis that $f \in C^\infty(\mathbb{T}^n)$. Note that since $|e^{ik \cdot \theta}| = 1$, the absolute convergence of (1.14) implies uniform convergence. By (1.4), we see that

$$(1.49) \quad f \in C^\ell(\mathbb{T}^n) \implies |\hat{f}(k)| \leq C \langle k \rangle^{-\ell},$$

which in turn clearly gives absolute convergence provided

$$(1.50) \quad \ell \geq n + 1.$$

Using Plancherel's identity and Cauchy's inequality, we can do somewhat better:

Proposition 1.3. *If $f \in C^\ell(\mathbb{T}^n)$, then the Fourier series for f is absolutely convergent provided*

$$(1.51) \quad \ell > \frac{n}{2}.$$

Proof. We have

$$(1.52) \quad \begin{aligned} \sum_k |\hat{f}(k)| &= \sum_k \langle k \rangle^{-\ell} \langle k \rangle^\ell |\hat{f}(k)| \\ &\leq \left[\sum_k \langle k \rangle^{-2\ell} \right]^{1/2} \cdot \left[\sum_k \langle k \rangle^{2\ell} |\hat{f}(k)|^2 \right]^{1/2} \\ &\leq C \left[\sum_k \langle k \rangle^{2\ell} |\hat{f}(k)|^2 \right]^{1/2}, \end{aligned}$$

as long as (1.51) holds. The square of the right side is dominated by

$$(1.53) \quad \begin{aligned} C' \sum_k \sum_{|\gamma| \leq \ell} |k^\gamma \hat{f}(k)|^2 &= C' \sum_{|\gamma| \leq \ell} \|D^\gamma f\|_{L^2}^2 \\ &\leq C'' \|f\|_{C^\ell}^2, \end{aligned}$$

so the proposition is proved.

Sharper results on absolute convergence of Fourier series will be given in Chapter 4. See also some of the exercises below for more on convergence when $n = 1$.

Exercises

1. Given $f, g \in L^1(\mathbb{T}^n)$, show that

$$\hat{f}(k) \hat{g}(k) = \hat{u}(k),$$

with

$$u(\theta) = (2\pi)^{-n} \int_{\mathbb{T}^n} f(\varphi) g(\theta - \varphi) d\varphi.$$

2. Given $f, g \in C(\mathbb{T}^n)$, show that

$$(\widehat{fg})(k) = \sum_m \hat{f}(k-m) \hat{g}(m).$$

3. Using the proof of Proposition 1.3, show that every $f \in \text{Lip}(S^1)$ has an absolutely convergent Fourier series.

4. Show that for any $f \in L^1(\mathbb{T}^n)$, $\hat{f}(k) \rightarrow 0$ as $|k| \rightarrow \infty$.
(Hint: Given $\varepsilon > 0$, pick $f_\varepsilon \in C^\infty(\mathbb{T}^n)$, $\|f - f_\varepsilon\|_{L^1} < \varepsilon$. Compare $\hat{f}_\varepsilon(k)$ and $\hat{f}(k)$.)
 This result is known as the Riemann-Lebesgue lemma.
5. For $f \in L^1(S^1)$, set

$$(1.54) \quad S_N f(\theta) = \sum_{k=-N}^N \hat{f}(k) e^{ik\theta}.$$

Show that $S_N f(\theta) = (1/2\pi) \int_{-\pi}^{\pi} f(\theta - \varphi) D_N(\varphi) d\varphi$, where

$$(1.55) \quad D_N(\theta) = \sum_{k=-N}^N e^{ik\theta} = \frac{\sin(N + \frac{1}{2})\theta}{\sin \frac{1}{2}\theta}.$$

(Hint: To evaluate the sum, recall how to sum a finite geometrical series.)
 $D_N(\theta)$ is called the Dirichlet kernel. See Fig. 1.2.



FIGURE 1.2

6. Let $f \in L^1(S^1)$ have the following property of “vanishing” at $\theta = 0$:

$$\frac{f(\theta)}{\sin \frac{1}{2}\theta} = g(\theta) \in L^1(-\pi, \pi).$$

Show that $S_N f(0) \rightarrow 0$ as $N \rightarrow \infty$.

(Hint. Adapt the Riemann-Lebesgue lemma to show that

$$g \in L^1(-\pi, \pi) \Rightarrow \int_{-\pi}^{\pi} g(\theta) \sin(N + \frac{1}{2})\theta d\theta \rightarrow 0 \text{ as } N \rightarrow \infty.)$$

7. Deduce that if $f \in L^1(S^1)$ is Lipschitz continuous at θ_0 , then $S_N f(\theta_0) \rightarrow f(\theta_0)$ as $N \rightarrow \infty$. Furthermore, if f is Lipschitz on an open interval $J \subset S^1$, then $S_N f \rightarrow f$ uniformly on compact subsets of J .

8. Let $f \in L^\infty(S^1)$ be piecewise Lipschitz, with a finite number of simple jumps. Show that $S_N f(\theta) \rightarrow f(\theta)$ at points of continuity. If f has a jump at θ_j , with limiting values $f_\pm(\theta_j)$, show that

$$(1.56) \quad S_N f(\theta_j) \rightarrow \frac{1}{2} [f_+(\theta_j) + f_-(\theta_j)],$$

as $N \rightarrow \infty$.

(Hint: By Exercise 7, it remains only to establish (1.56). Show that this can be reduced to the case $\theta_j = \pi$, $f(\theta) = \theta$, for $-\pi \leq \theta < \pi$. Verify that this function has Fourier series

$$2 \sum_{k=1}^{\infty} \frac{(-1)^k}{k} \sin k\theta.$$

Alternative: Reduce to the case $\theta_j = 0$ and note that $S_N f(0)$ depends only on the even part of f , $(1/2)[f(\theta) + f(-\theta)]$.

9. Work out the Fourier series of the function $f \in \text{Lip}(S^1)$ given by

$$f(\theta) = |\theta|, \quad -\pi \leq \theta \leq \pi.$$

Examining this at $\theta = 0$, establish that

$$(1.57) \quad \sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}.$$

10. One can obtain Fourier coefficients of functions θ^k and $|\theta|^k$ on $[-\pi, \pi]$ in terms of the Fourier coefficients of

$$q_k(\theta) = \begin{cases} \theta^k & \text{on } [0, \pi], \\ 0 & \text{on } [-\pi, 0]. \end{cases}$$

Show that, for $n \neq 0$,

$$\sum_{k=0}^{\infty} \frac{1}{k!} \hat{q}_k(n) (is)^k = -\frac{1}{2\pi i n} \left[(-1)^n \cdot e^{\pi i s} - 1 \right] \left(1 - \frac{s}{n} \right)^{-1},$$

and use this to work out the Fourier series for these functions. Apply this to Exercise 9, and to the calculation at the end of Exercise 8.

11. Assume that $g \in L^1(S^1)$ has uniformly convergent Fourier series ($S_N g \rightarrow g$) on compact subsets of an open interval $J \subset S^1$. Show that whenever $f \in L^1(S^1)$ and $f = g$ on J , then f also has uniformly convergent Fourier series on compact subsets of J .

(Hint: Apply Exercise 7 to $f - g$.)

This result is called the localization principle for Fourier series.

12. Suppose f is Hölder continuous on S^1 , that is, $f \in C^r(S^1)$, for some $r \in (0, 1)$, which means

$$|f(\varphi + \theta) - f(\varphi)| \leq C|\theta|^r.$$

Show that f has uniformly convergent Fourier series on S^1 .

(Hint: We have

$$\left| \frac{f(\theta + \varphi) - f(\varphi)}{\sin \frac{1}{2}\theta} \right| \leq C \left| \sin \frac{1}{2}\theta \right|^{-(1-r)}.$$

Apply Exercise 6.)

13. If $\omega : [0, \infty) \rightarrow [0, \infty)$ is continuous and increasing and $\omega(0) = 0$, we say a function f on S^1 is continuous with modulus of continuity ω provided

$$|f(\varphi + \theta) - f(\varphi)| \leq C\omega(|\theta|).$$

Formulate the most general condition you can to establish uniform convergence of the Fourier series of a function with such a modulus of continuity.

Note that Exercise 12 deals with the case $\omega(s) = s^r$, $r \in (0, 1)$.

14. Consider the *Cesaro sum* of the Fourier series of f :

$$C_N f(\theta) = \sum_{k=-N}^N \left(1 - \frac{|k|}{N}\right) \hat{f}(k) e^{ik\theta} = \frac{1}{N} \sum_{\ell=0}^{N-1} S_\ell f(\theta).$$

Show that $C_N f(\theta) = (1/2\pi) \int_{-\pi}^{\pi} f(\theta - \varphi) F_N(\varphi) d\varphi$, where

$$(1.58) \quad F_N(\theta) = \frac{1}{N} \sum_{\ell=0}^{N-1} D_\ell(\theta) = \frac{1}{N \sin \frac{1}{2}\theta} \sum_{\ell=0}^{N-1} \sin\left(\ell + \frac{1}{2}\right)\theta = \frac{1}{N} \left(\frac{\sin \frac{N}{2}\theta}{\sin \frac{1}{2}\theta} \right)^2.$$

The function $F_N(\theta)$ is called the Fejer kernel (see Fig. 1.3). Modify the proof of Proposition 1.1 to show that

$$C_N f \rightarrow F \text{ in } B, \text{ for } f \in B,$$

where B is one of the Banach spaces $C(S^1)$ or $L^p(S^1)$, $1 \leq p < \infty$.

(Hint: To evaluate the second sum in (1.58), use $\sin(\ell + \frac{1}{2})\theta = \operatorname{Im} e^{i\theta/2} e^{i\ell\theta}$ and sum a finite geometrical series. Also use the identity $2 \sin^2 z = 1 - \cos 2z$.)

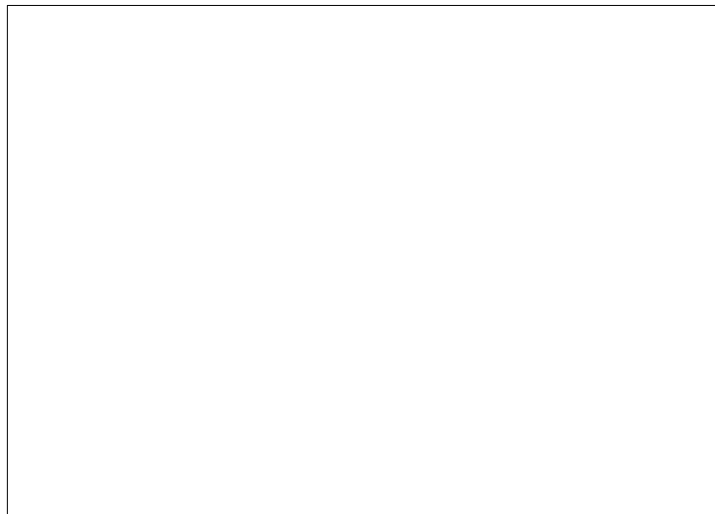


FIGURE 1.3

2. Harmonic functions and holomorphic functions in the plane

The method of proof of the Abel summability (1.26)–(1.27) of Fourier series, specialized to $\mathbb{T}^1 = S^1$, has important implications for the theory of harmonic functions on a domain $\Omega \subset \mathbb{R}^2$, which we will discuss here. In the case of S^1 , let us rewrite (1.26),

$$(2.1) \quad J_r f(\theta) = \sum_{k=-\infty}^{\infty} \hat{f}(k) r^{|k|} e^{ik\theta},$$

as

$$(2.2) \quad (\text{PI } f)(r, \theta) = \sum_{k=-\infty}^{\infty} \hat{f}(k) r^{|k|} e^{ik\theta}.$$

The function $u(r, \theta) = \text{PI } f(r, \theta)$ is called the *Poisson integral* of f . If we use polar coordinates in the complex plane $\mathbb{C} = \mathbb{R}^2$,

$$(2.3) \quad z = r e^{i\theta},$$

then (2.2) becomes

$$(2.4) \quad \begin{aligned} (\text{PI } f)(z) &= \sum_{k=0}^{\infty} \hat{f}(k) z^k + \sum_{k=1}^{\infty} \hat{f}(-k) \bar{z}^k \\ &= (\text{PI}_+ f)(z) + (\text{PI}_- f)(z), \end{aligned}$$

defined on the unit disk $|z| < 1$. Note also, from (1.30), that

$$(2.5) \quad \text{PI } f(z) = \frac{1 - |z|^2}{2\pi} \int_{S^1} \frac{f(w)}{|w - z|^2} ds(w),$$

the integral being with respect to arclength on S^1 . Recall that if $f \in L^1(S^1)$, the function $\hat{f}(k)$ is bounded, so both power series in (2.4) have radius of convergence at least 1. Clearly, on the unit disk, $v(z) = (\text{PI}_+ f)(z)$ is holomorphic and $w(z) = (\text{PI}_- f)(z)$ is antiholomorphic. In other words, v and w belong to $C^\infty(\mathcal{D})$, where $\mathcal{D} = \{z \in \mathbb{C} : |z| < 1\}$, and

$$(2.6) \quad \frac{\partial v}{\partial \bar{z}} = 0, \quad \frac{\partial w}{\partial z} = 0 \quad \text{on } \mathcal{D},$$

where

$$(2.7) \quad \frac{\partial}{\partial \bar{z}} = \frac{1}{2} \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right), \quad \frac{\partial}{\partial z} = \frac{1}{2} \left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right).$$

Note that

$$\frac{\partial}{\partial z} \frac{\partial}{\partial \bar{z}} = \frac{\partial}{\partial \bar{z}} \frac{\partial}{\partial z} = \frac{1}{4} \Delta,$$

where Δ is the Laplace operator on $\mathbb{R}^2$, a special case of the Laplace operator introduced in Chapter 2. Since $v, w \in C^\infty(\mathcal{D})$, we have $\Delta v = 0$ and $\Delta w = 0$, and hence

$$(2.8) \quad \Delta(\text{PI } f) = 0.$$

In light of the results of §1, we have the following.

Proposition 2.1. *If $f \in C(S^1)$, then*

$$(2.9) \quad u = \text{PI } f(z) \in C^\infty(\mathcal{D}) \cap C(\overline{\mathcal{D}})$$

is harmonic, with boundary value f , that is, u solves the Dirichlet problem

$$(2.10) \quad \Delta u = 0 \text{ in } \mathcal{D}, \quad u|_{\partial\mathcal{D}} = f.$$

One should expect that if f has extra smoothness on S^1 , so does $\text{PI } f$ on $\overline{\mathcal{D}}$. The following result is crude compared to results established in Chapters 4 and 5, but it will be of some interest.

Proposition 2.2. *For $\ell = 1, 2, 3, \dots$, we have*

$$(2.11) \quad \text{PI}: C^{\ell+1}(S^1) \longrightarrow C^\ell(\overline{\mathcal{D}}).$$

Proof. We begin with the case $\ell = 1$. Since we know from (2.4) that $\text{PI } f \in C^\infty(\mathcal{D})$, we need merely check smoothness near $\partial\mathcal{D} = S^1$. Clearly,

$$(2.12) \quad \frac{\partial}{\partial\theta} \text{PI } f = \text{PI } \frac{\partial f}{\partial\theta},$$

so if $f \in C^1(S^1)$, then $\partial f / \partial\theta \in C(S^1)$ and we have $(\partial / \partial\theta) \text{PI } f$ continuous on $\overline{\mathcal{D}}$. Also, by (2.2),

$$(2.13) \quad r \frac{\partial}{\partial r} \text{PI } f = \text{PI } (Nf),$$

where Nf is characterized by the Fourier series representation

$$(2.14) \quad Nf(\theta) = \sum_{k=-\infty}^{\infty} \hat{f}(k) |k| e^{ik\theta}.$$

Thus

$$(2.15) \quad Nf = -i \frac{\partial}{\partial\theta} Hf = -iH \frac{\partial f}{\partial\theta},$$

where H has the Fourier series representation

$$(2.16) \quad Hg(\theta) = \sum_{k=-\infty}^{\infty} (\text{sgn } k) \hat{g}(k) e^{ik\theta}.$$

We claim that, for $\ell \geq 0$,

$$(2.17) \quad H : C^{\ell+1}(S^1) \longrightarrow C^\ell(S^1),$$

and hence $N : C^{\ell+2}(S^1) \rightarrow C^\ell(S^1)$. Given this, for $f \in C^2(S^1)$, the quantity (2.13) is seen to belong to $C(\overline{\mathcal{D}})$, and this finishes the $\ell = 1$ case of (2.11).

In turn, since H commutes with $\partial/\partial\theta$, it suffices to establish the $\ell = 0$ case of (2.17). Now, by Proposition 1.3, the Fourier series for $g \in C^1(S^1)$ is absolutely convergent, giving (2.17).

To prove the general case of (2.11), a short calculation yields

$$(2.18) \quad \left(r \frac{\partial}{\partial r}\right)^j \left(\frac{\partial}{\partial \theta}\right)^k \text{PI } f = \text{PI} \left(\left(\frac{\partial}{\partial \theta}\right)^k N^j f \right).$$

Note that $N^j = (-i)^j (\partial/\partial\theta)^j H^{o(j)}$, where $o(j)$ is zero if j is even and one if j is odd, so, for $\ell \geq 0$,

$$N^j : C^{\ell+j+1}(S^1) \longrightarrow C^\ell(S^1),$$

the left side being improved to $C^{\ell+j}$ if j is even. Therefore, if $f \in C^{\ell+1}(S^1)$ and $j + k = \ell$, the right side of (2.18) is $\text{PI } f_{jk}$ with $f_{jk} \in C(S^1)$, which proves (2.11) in general.

The implication $f \in C^\ell(S^1) \Rightarrow \text{PI } f \in C^\ell(\overline{\mathcal{D}})$ does not quite work, as we will see later, essentially because H does not map $C^\ell(S^1)$ to itself. It is true that $f \in C^{\ell,\alpha}(S^1) \Rightarrow \text{PI } f \in C^{\ell,\alpha}(\overline{\mathcal{D}})$, for $\alpha \in (0, 1)$. This is a special case of Hölder estimates that will be established in §7 of Chapter 13. Similarly there are “sharp” results on regularity of $\text{PI } f$ in Sobolev spaces, discussed in Chapter 4, and in much greater generality, in Chapter 5.

It is important to know that $\text{PI } f$ provides the *unique* solution to the Dirichlet problem (2.10). We will establish several general uniqueness results, starting with the following.

Proposition 2.3. *Let $\Omega \subset \mathbb{R}^n$ be a bounded region with smooth boundary, say, $\Omega = \mathcal{D}$. Suppose $u, v \in C^2(\overline{\Omega})$, with $u = v = f$ on $\partial\Omega$, and $\Delta u = \Delta v = 0$ in Ω . Then $u = v$ on all of Ω .*

Proof. Set $w = u - v \in C^2(\overline{\Omega})$; $w = 0$ on $\partial\Omega$. We can apply the Green identity (3.15) of Chapter 2, to write

$$(2.19) \quad (dw, dw) = -(\Delta w, w) + \int_{\partial\Omega} w \frac{\partial \overline{w}}{\partial \nu} dS.$$

By hypothesis the right side of (2.19) is 0. Thus w is constant on each component of Ω , and the boundary condition forces $w = 0$.

In view of Proposition 2.2, we could apply this to $u = \text{PI } f$ if $f \in C^3(S^1)$, but this is not a satisfactory result, and we will do much better below.

A result related to our uniqueness question is the *mean-value property*, a special case of which is the following.

Proposition 2.4. *If $f \in C(S^1)$, $u = PI f$, then*

$$(2.20) \quad u(0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\theta) d\theta.$$

Proof. It follows from the series (2.2) that $u(0) = \hat{f}(0)$, which gives (2.20).

A more general result is the following.

Proposition 2.5. *If $B_R \subset \mathbb{R}^n$ is the open ball of radius R , centered at the origin, with $\partial B_R = S_R$, of area $A(R)$, then for $u \in C^2(B_R) \cap C(\overline{B}_R)$, $\Delta u = 0$, we have*

$$(2.21) \quad u(0) = \frac{1}{A(R)} \int_{S_R} u(x) dS.$$

Proof. We apply Green's identity

$$(2.22) \quad \int_{\Omega} [u\Delta v - v\Delta u] dx = \int_{\partial\Omega} \left[u \frac{\partial v}{\partial \nu} - v \frac{\partial u}{\partial \nu} \right] dS,$$

to $\Omega = B_r$, $0 < r < R$, $v(x) = |x|^2$, with $\Delta v = 2n$, to get, when $\Delta u = 0$,

$$(2.23) \quad n \int_{B_r} u(x) dx = r \int_{S_r} u(x) dS,$$

noting that substituting $v = 1$ in (2.22) gives $\int_{\partial\Omega} (\partial u / \partial \nu) dS = 0$. If we let $\varphi(r) = \int_{B_r} u(x) dx$, this implies $\varphi'(r) = (n/r)\varphi(r)$, and hence $\varphi(r) = Kr^n$, i.e., $V(r)^{-1} \int_{B_r} u(x) dx$ is constant. Passing to the limit $r \rightarrow 0$ gives (2.21).

Second Proof. Define $v \in C^2(B_R) \cap C(\overline{B}_R)$ by

$$v(x) = \int_{SO(n)} u(gx) dg,$$

where dg is Haar measure on the rotation group $SO(n)$, defined in §5 of Appendix B. The Laplace operator is invariant under rotations, so $\Delta v = 0$ on B_R . The function v is radial; $v(x) = \tilde{v}(|x|)$. The formula

$$\Delta = \frac{\partial^2}{\partial r^2} + \frac{n-1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \Delta_S,$$

(cf. (4.17) of Chapter 2), for Δ in polar coordinates, where Δ_S is the Laplace operator on the unit sphere S^{n-1} , gives

$$\left(\frac{d^2}{dr^2} + \frac{n-1}{r} \frac{d}{dr}\right) \tilde{v}(r) = 0.$$

This is an Euler equation, whose solutions are

$$\begin{aligned} A + Br^{2-n}, \quad n \geq 3, \\ A + B \log r, \quad n = 2. \end{aligned}$$

Since v does not blow up at 0, we have $B = 0$, so v is constant. Clearly $v(x)$ equals the right side of (2.21) for $|x| = R$, and $v(0) = u(0)$, so we again have (2.21).

Corollary 2.6. *For any $\Omega \subset \mathbb{R}^n$ open, any $u \in C^2(\Omega)$ harmonic, any ball B_p , centered at p and contained in Ω , we have*

$$(2.24) \quad u(p) = \text{Avg}_{\partial B_p} u(z).$$

We can now prove the following important *maximum principle* for harmonic functions. Much more general versions of this will be given in Chapter 5.

Proposition 2.7. *Let $\Omega \subset \mathbb{R}^n$ be connected and open, and let $u \in C^2(\Omega)$ be harmonic and real-valued. Then u has no interior maximum, or minimum, unless u is constant. In particular, if Ω is bounded and $u \in C(\bar{\Omega})$, then*

$$(2.25) \quad \sup_{p \in \Omega} u(p) = \sup_{q \in \partial \Omega} u(q).$$

Also, even for u complex-valued,

$$(2.26) \quad \sup_{p \in \Omega} |u(p)| = \sup_{q \in \partial \Omega} |u(q)|.$$

Proof. That a non-constant, real harmonic function has no interior extremum is an obvious consequence of (2.24), and the other consequences, (2.25) and (2.26), follow immediately.

Corollary 2.8. *The uniqueness result of Proposition 2.3 holds for any bounded open $\Omega \subset \mathbb{R}^n$, with no smoothness on $\partial \Omega$, and for any harmonic*

$$(2.27) \quad u, v \in C^2(\Omega) \cap C(\bar{\Omega}).$$

Proof. Apply the maximum principle to $u - v$.

Here, we have used the mean-value property to prove the maximum principle. The more general maximum principle established in Chapter 5 will not use the mean-value property; indeed, together with a symmetry argument, it can be made a basis for a proof of the mean-value property. We will leave these considerations until Chapter 5.

With our uniqueness result in hand, we can easily establish the following interior regularity result for harmonic functions.

Proposition 2.9. *Let $\Omega \subset \mathbb{R}^2$ be open, and let $u \in C^2(\Omega)$ be harmonic. Then in fact, $u \in C^\infty(\Omega)$; u is even real analytic on Ω .*

Proof. By translations and dilations, we can reduce to the case $\Omega = \mathcal{D}$, $u \in C^2(\overline{\mathcal{D}})$. The uniqueness result of Corollary 2.8 implies

$$(2.28) \quad u = \text{PI } f, \quad \text{where } f = u|_{S^1}.$$

Then the conclusion that u is real analytic on $\mathcal{D}$ follows directly from the power-series expansion (2.4).

Parenthetically, we remark that, by Corollary 2.8, the identity (2.28) holds for any $u \in C^2(\mathcal{D}) \cap C(\overline{\mathcal{D}})$ harmonic in $\mathcal{D}$.

Using the results we have developed, via Fourier series, about harmonic functions, we can quickly draw some basic conclusions about holomorphic functions. If $\Omega \subset \mathbb{C}$ is open, $f : \Omega \rightarrow \mathbb{C}$ is by definition holomorphic if and only if $f \in C^1(\Omega)$ and $\partial f / \partial \bar{z} = 0$, where $\partial / \partial \bar{z}$ is given by (2.7). Clearly, if $f \in C^2(\Omega)$ is holomorphic, then it is also harmonic, and so are its real and imaginary parts. Suppose $u \in C^2(\mathcal{D}) \cap C(\overline{\mathcal{D}})$ is holomorphic in $\mathcal{D}$. Then the series representation (2.4) is valid, since (2.28) holds. This series is a sum of two terms:

$$(2.29) \quad \begin{aligned} u(z) &= \sum_{k=0}^{\infty} \hat{f}(k) z^k + \sum_{k=1}^{\infty} \hat{f}(-k) \bar{z}^k \\ &= u_1(z) + u_2(z), \end{aligned}$$

where $\partial u_1 / \partial \bar{z} = 0$ and $\partial u_2 / \partial z = 0$. But if we are given that $\partial u / \partial \bar{z} = 0$, then also $\partial u_2 / \partial \bar{z} = 0$, so $\partial u_2 / \partial x = \partial u_2 / \partial y = 0$. Thus u_2 is constant, and since $u_2(0) = 0$, this forces $u_2 = 0$. In other words, the holomorphic function $u(z)$ has the power series

$$(2.30) \quad u(z) = \sum_{k=0}^{\infty} a_k z^k, \quad z \in \mathcal{D},$$

where $a_k = \hat{f}(k)$, $k \geq 0$. Note that differentiation of (2.30) gives

$$(2.31) \quad a_k = \frac{1}{k!} \left(\frac{\partial}{\partial z} \right)^k u(0).$$

By the usual method of translating and dilating coordinates, we deduce the following.

Proposition 2.10. *If $\Omega \subset \mathbb{C}$ is open, $u \in C^2(\Omega)$ holomorphic, $p \in \Omega$, and $\mathcal{D}_p$ a disk centered at p , $\overline{\mathcal{D}_p} \subset \Omega$, then on $\mathcal{D}_p$, $u(z)$ is given by a convergent power-series expansion*

$$(2.32) \quad u(z) = \sum_{k=0}^{\infty} b_k (z - p)^k.$$

We can relax the C^2 -hypothesis to $u \in C^1(\Omega)$. As much stronger and more general results are given in Chapter 5, we omit the details here.

For further use, we record the following result, whose proof is trivial.

Lemma 2.11. *If $u \in C^\infty(\Omega)$ is holomorphic, and*

$$P = \sum a_{jk} D_x^j D_y^k$$

is any constant-coefficient differential operator, then Pu is holomorphic in Ω .

Proof. $(\partial/\partial\bar{z})Pu = P(\partial u/\partial\bar{z})$.

We can use the power-series representation (2.30)–(2.32) to prove the fundamental result on uniqueness of analytic continuation, which we give below. Here is the first result, of a very general nature.

Proposition 2.12. *Let $\Omega \subset \mathbb{R}^n$ be open and connected, and let u be a real-analytic function on Ω . If $p \in \Omega$ and all derivatives $D^\alpha u(p) = 0$, then $u = 0$ on all of Ω .*

Proof. Let $\mathcal{K} = \{x \in \Omega : D^\alpha u(x) = 0 \text{ for all } \alpha \geq 0\}$. Since $u \in C^\infty(\Omega)$, $\mathcal{K}$ is closed in Ω . However, for each $p \in \mathcal{K}$, since u is given in a neighborhood of p by a power series

$$u(q) = \sum_{\alpha \geq 0} \frac{u^{(\alpha)}(p)}{\alpha!} (q - p)^\alpha,$$

we also see that $\mathcal{K}$ is open in Ω . This proves the proposition.

Our basic corollary for holomorphic functions is the following.

Corollary 2.13. *Let $\Omega \subset \mathbb{C}$ be open and connected, u holomorphic on Ω . Let γ be a line segment contained in Ω . If $u|_\gamma = 0$, then $u = 0$ on Ω .*

Proof. Translating and rotating, we can assume γ is a segment in the real axis, with $0 \in \gamma$. Near 0, $u(z)$ has a power-series expansion of the form (2.30), with a_k given by (2.31). Using Lemma 2.11, we see that

$$(2.33) \quad \left(\frac{\partial}{\partial z}\right)^k u(0) = \left(\frac{\partial}{\partial x}\right)^k u(0),$$

which vanishes for all k . Thus $u = 0$ on a nonempty open set in Ω , and the rest follows by Proposition 2.12.

Actually, a much stronger result is true. If $\Omega \subset \mathbb{C}$ is connected, $p_j \in \Omega$ are distinct, $p_j \rightarrow p \in \Omega$, u is holomorphic in Ω , and $u(p_j) = 0$ for each j , then u must vanish identically. In other words, u can have only isolated zeros if it does not vanish identically. Indeed, say $u(p) = 0$. If u is not identically zero, some coefficient in the series (2.32) is nonzero; let b_m be the first such coefficient:

$$(2.34) \quad \begin{aligned} u(z) &= (z-p)^m \sum_{k=0}^{\infty} b_{m+k} (z-p)^k \\ &= (z-p)^m v(z), \end{aligned}$$

where $v(z)$ is holomorphic on $\mathcal{D}_p$ and $v(0) = b_m \neq 0$. Thus, by continuity, $v(\zeta) \neq 0$ for $|\zeta - p| < \varepsilon$ if ε is sufficiently small, which implies $u(\zeta) \neq 0$ if $\zeta \neq p$ but $|\zeta - p| < \varepsilon$.

A typical use of Corollary 2.13 is in computations of integrals. We will see an example of this in the next section.

We end this section by recalling the classical Cauchy integral theorem and integral formula. Throughout, Ω will be a bounded open domain in $\mathbb{C}$ with smooth boundary. Stokes' formula, proved in Chapter 1, §13, states

$$(2.35) \quad \iint_{\Omega} d\alpha = \int_{\partial\Omega} \alpha,$$

for a 1-form α with coefficients in $C^1(\overline{\Omega})$. If $\alpha = p \, dx + q \, dy$, this gives the classical Green's formula

$$(2.36) \quad \int_{\partial\Omega} p \, dx + q \, dy = \iint_{\Omega} \left(\frac{\partial q}{\partial x} - \frac{\partial p}{\partial y} \right) dx \, dy.$$

If $u(x, y) \in C^1(\overline{\Omega})$ is a complex-valued function, we consequently have

$$\begin{aligned}
 \int_{\partial\Omega} u \, dz &= \int_{\partial\Omega} (u \, dx + iu \, dy) \\
 (2.37) \qquad &= \iint_{\Omega} \left(i \frac{\partial u}{\partial x} - \frac{\partial u}{\partial y} \right) dx \, dy \\
 &= 2i \iint_{\Omega} \frac{\partial u}{\partial \bar{z}} dx \, dy.
 \end{aligned}$$

In the special case when u is holomorphic in Ω , we have the Cauchy integral theorem:

Theorem 2.14. *If $\Omega \subset \mathbb{C}$ is bounded with smooth boundary and $u \in C^1(\overline{\Omega})$ is holomorphic, then*

$$(2.38) \qquad \int_{\partial\Omega} u(z) \, dz = 0.$$

Using various limiting arguments, one can relax the hypotheses on smoothness of $\partial\Omega$ and of u near $\partial\Omega$; we won't go into this here. Next we prove Cauchy's integral formula.

Proposition 2.15. *With Ω as above, $u \in C^2(\overline{\Omega})$ holomorphic in Ω , we have*

$$(2.39) \qquad u(\zeta) = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{u(z)}{z - \zeta} \, dz, \quad \text{for } \zeta \in \Omega.$$

Proof. Write

$$\begin{aligned}
 (2.40) \qquad u(z)(\zeta - z)^{-1} &= (\zeta - z)^{-1} [u(z) - u(\zeta)] + u(\zeta)(\zeta - z)^{-1} \\
 &= v(z) + u(\zeta)(\zeta - z)^{-1}.
 \end{aligned}$$

By the series expansion for $u(z)$ about ζ , we see that $v(z)$ is holomorphic near ζ ; clearly, it is holomorphic on the rest of Ω , and it belongs to $C^1(\overline{\Omega})$, so $\int_{\partial\Omega} v(z) \, dz = 0$. Thus, to prove (2.39), it suffices to show that

$$(2.41) \qquad \int_{\partial\Omega} (z - \zeta)^{-1} \, dz = 2\pi i, \quad \text{for } \zeta \in \Omega.$$

Indeed, if ε is small enough that $B(\zeta, \varepsilon) = \{z \in \mathbb{C} : |z - \zeta| \leq \varepsilon\}$ is contained in Ω , then Cauchy's theorem implies that the left side of (2.41) is equal to

$$(2.42) \qquad \int_{\partial B(\zeta, \varepsilon)} (z - \zeta)^{-1} \, dz$$

since $(\zeta - z)^{-1}$ is holomorphic in z for $z \neq \zeta$. Making a change of variable, we see that (2.42) is equal to

$$\int_0^{2\pi} (\varepsilon e^{i\theta})^{-1} i\varepsilon e^{i\theta} d\theta = 2\pi i,$$

so the proof is complete.

As stated before, the C^2 -hypothesis can be relaxed to C^1 .

A function $u(z)$ of the form $u(z) = v(z)/(z - a)^k$, $k \in \mathbb{Z}^+$, where v is holomorphic on a neighborhood $\mathcal{O}$ of a , is said to have a pole of order k at $z = a$ if $v(a) \neq 0$. In such a case, a variant of the preceding calculations yields

$$\frac{1}{2\pi i} \int_{\gamma} u(z) dz = \frac{1}{(k-1)!} v^{(k-1)}(a),$$

the coefficient of $(z - a)^{k-1}$ in the power series of $v(z)$ about $z = a$, if γ is a smooth, simple, closed curve about a such that v is holomorphic on a neighborhood of the closed region bounded by γ . This quantity is called the *residue* of $u(z)$ at $z = a$.

One can often evaluate integrals by evaluating residues. We give a simple illustration here; others are given in (2.48), (3.32), (A.14), and (A.15). Here we evaluate

$$(2.43) \quad \int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \lim_{R \rightarrow \infty} \int_{\gamma_R} \frac{dz}{1+z^2},$$

where γ_R is the closed curve, going from $-R$ to R along the real axis, then from R to $-R$ counterclockwise on the circle of radius R centered at 0, that is, $\gamma_R = \partial\mathcal{O}_R$, where $\mathcal{O}_R = \{z : \operatorname{Re} z > 0, |z| < R\}$. There is just one pole of $(1+z^2)^{-1}$ in $\mathcal{O}_R$, located at $z = i$. Since $(1+z^2)^{-1} = (z+i)^{-1}(z-i)^{-1}$, we see that the residue of $(1+z^2)^{-1}$ at $z = i$ is $1/2i$, so

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \pi.$$

Exercises

1. Suppose u satisfies the following Neumann boundary problem in the disk $\mathcal{D}$:

$$(2.44) \quad \Delta u = 0 \text{ in } \mathcal{D}, \quad \frac{\partial u}{\partial r} = g \text{ on } S^1.$$

If $u = \operatorname{PI}(f)$, show that f and g must be related by $\hat{g}(k) = |k|\hat{f}(k)$, for all $k \in \mathbb{Z}$, that is,

$$(2.45) \quad g = Nf,$$

with N defined by (2.13).

2. Define the function k_N by

$$k_N(\theta) = \sum_{k \neq 0} |k|^{-1} e^{ik\theta}.$$

Show that $k_N \in L^2(S^1) \subset L^1(S^1)$. Also show that, provided $g \in L^2(S^1)$ and

$$(2.46) \quad \int_{S^1} g(\theta) d\theta = 0,$$

a solution to (2.45) is given by

$$(2.47) \quad f(\theta) = (2\pi)^{-1} \int_{S^1} k_N(\theta - \varphi) g(\varphi) d\varphi = Tg(\theta).$$

3. If T is defined by (2.47), show that $T : L^p(S^1) \rightarrow L^p(S^1)$ for $p \in [1, \infty)$ and $T : C^\ell(S^1) \rightarrow C^\ell(S^1)$ for $\ell = 0, 1, 2, \dots$.
4. Given $g \in C^1(S^1)$, show that (2.44) has a solution $u \in C^1(\overline{\mathcal{D}})$ if and only if (2.46) holds. If $g \in C^\ell(S^1)$, show that (2.44) has a solution $u \in C^\ell(\overline{\mathcal{D}})$.
Note: Regularity results of a more precise nature are given in Chapter 4, §4, in Exercise 1. See also Chapter 5, §7, for more general results.
5. Let $\Omega \subset \mathbb{R}^2$ be a smooth, bounded, connected region. Show that if $w \in C^2(\overline{\Omega})$, $\Delta w = 0$ on Ω , and $\partial w / \partial \nu = 0$ on $\partial\Omega$, then w is constant. (*Hint:* Use (2.19).)
Note: One can weaken the C^2 -hypothesis to $w \in C^1(\overline{\Omega})$; compare Proposition 2.2 of Chapter 5. For another type of relaxation, see Chapter 4, §4, Exercise 3.
6. Show that a C^1 -function $f : \Omega \rightarrow \mathbb{C}$ is holomorphic if and only if, at each $z \in \Omega$, $Df(z)$, a priori a real linear map on $\mathbb{R}^2$, is in fact complex linear on $\mathbb{C}$.
Note: This exercise has already been given in Chapter 1, §1.
7. Let f be a holomorphic function on $\Omega \subset \mathbb{C}$, with $f : \Omega \rightarrow \mathcal{O}$, and let u be harmonic on $\mathcal{O}$. Show that $v = u \circ f$ is harmonic on Ω . (*Hint:* For a short proof, write u locally as a sum of a holomorphic and an anti-holomorphic function.)
8. Let $g(z) = \sum_{k=1}^{\infty} a_k z^k$, and form the harmonic function $u = 2 \operatorname{Re} g = g + \bar{g}$. Show that, under appropriate hypotheses on (a_k) , $g|_{S^1} = P_+(u|_{S^1})$, where P_+ is given by

$$P_+ f(\theta) = \sum_{k=0}^{\infty} \hat{f}(k) e^{ik\theta}.$$

9. Find a holomorphic function on $\mathcal{D}$ that is *unbounded* but whose real part has a continuous extension to $\overline{\mathcal{D}}$.
Reconsider this problem after reading §6 of Chapter 5.
10. Hence show that P_+ does not map $C(S^1)$ to itself, nor does it map any $C^\ell(S^1)$ to itself, for any integer $\ell \geq 0$.
11. Hence find $f \in C^1(S^1)$ such that $\operatorname{PI} f \notin C^1(\overline{\mathcal{D}})$.
12. Use the method of residues to calculate

$$(2.48) \quad \int_{-\infty}^{\infty} \frac{e^{i\xi x}}{1+x^2} dx, \quad \xi \geq 0.$$

(*Hint*: Write this as $\lim_{R \rightarrow \infty} \int_{\gamma_R} e^{i\xi z} / (1 + z^2) dz$, with γ_R as in (2.43), given $\xi \geq 0$. Then find the residue of $e^{i\xi z} / (1 + z^2)$ at $z = i$.)

13. Use the Poisson integral formula (2.5) to prove the following. Let $u \in C^2(\mathcal{D}) \cap C(\overline{\mathcal{D}})$ be harmonic in the disk $\mathcal{D} = \{z \in \mathbb{C} : |z| < 1\}$. Assume $u \geq 0$ on $\mathcal{D}$ and $u(0) = 1$. Then

$$(2.49) \quad |z| = a \in [0, 1) \implies u(z) \geq \frac{1-a}{1+a}.$$

This result is known as a Harnack inequality. *Hint*. Use the inequality

$$|w| = 1, |z| = a \in [0, 1) \implies \frac{1 - |z|^2}{|w - z|^2} \geq \frac{1-a}{1+a}.$$

Note. By translating and scaling, if u is harmonic and ≥ 0 on $\mathcal{D}_R(p) = \{z \in \mathbb{C} : |z - p| < R\}$, then

$$|z - p| = a \in [0, R) \implies u(z) \geq \frac{R-a}{R+a} u(p).$$

14. Using Exercise 13, show that if u is harmonic in the entire plane $\mathbb{C}$ and $u \geq 0$ on $\mathbb{C}$, then u is constant. More generally, if there exists a constant K such that $u \geq K$ on $\mathbb{C}$, then u is constant. This is a version of Liouville's theorem. See Proposition 4.6 for another version.
15. Using Exercise 13, show that there exists $A \in (0, \infty)$ with the following property. Let u be harmonic on $\mathcal{D}_R(0)$. Assume

$$u(0) = 0, \quad u(z) \leq M \quad \text{on } \mathcal{D}_R(0).$$

Then

$$u(z) \geq -AM \quad \text{on } \mathcal{D}_{R/2}(0).$$

(*Hint*. Set $v(z) = M - u(z)$, so $v(z) \geq 0$ on $\mathcal{D}_R(0)$, $v(0) = M$. Say $p \in \mathcal{D}_{R/2}(0)$, $u(p) = \inf_{\mathcal{D}_{R/2}(0)} u$. Deduce that

$$v(z) \geq \frac{1}{3}(M - u(p)) \quad \text{on } \mathcal{D}_{R/4}(p),$$

and from there that

$$\text{Avg}_{\mathcal{D}_R(0)} v \geq \frac{1}{16} \cdot \frac{1}{3}(M - u(p)),$$

while this average is equal to $v(0) = M$.)

16. Assume u is harmonic on $\mathbb{C}$ and

$$u(z) \leq C_0 + C_1|z|^k, \quad \forall z \in \mathbb{C},$$

with $C_0, C_1 \in (0, \infty)$. Take A from Exercise 15. Show that there exists C_2 such that

$$u(z) \geq -C_2 - AC_1|z|^k, \quad \forall z \in \mathbb{C}.$$

Note. In conjunction with Proposition 4.6, one gets that $u(z)$ must be a polynomial of degree $\leq k$ in x and y .

17. Let Ω be the strip $\Omega = \{z \in \mathbb{C} : 0 < \text{Re } z < 1\}$, with closure $\overline{\Omega}$. Let f be continuous on $\overline{\Omega}$ and holomorphic on Ω . Show that if f is bounded,

$$|f(z)| \leq A \quad \text{on } \partial\Omega \implies |f(z)| \leq A \quad \text{on } \Omega.$$

(*Hint.* First prove the implication under the additional hypothesis that $|f(z)| \rightarrow 0$ as $|z| \rightarrow \infty$. Then consider $f_\varepsilon(z) = e^{\varepsilon z^2} f(z)$.)

18. In the setting of Exercise 17, show that if $\theta \in (0, 1)$,

$$|f(iy)| \leq A, \quad |f(1+iy)| \leq B, \quad \forall y \in \mathbb{R} \implies |f(\theta+iy)| \leq A^{1-\theta} B^\theta, \quad \forall y \in \mathbb{R}.$$

This result is known as the Hadamard three-lines lemma. (*Hint.* Consider $g(z) = A^{z-1} B^{-z} f(z)$.)

3. The Fourier transform

The Fourier transform is defined by

$$(3.1) \quad \mathcal{F}f(\xi) = \hat{f}(\xi) = (2\pi)^{-n/2} \int f(x) e^{-ix \cdot \xi} dx$$

when $f \in L^1(\mathbb{R}^n)$. It is clear that

$$(3.2) \quad \mathcal{F} : L^1(\mathbb{R}^n) \longrightarrow L^\infty(\mathbb{R}^n).$$

This is analogous to (1.3). The analogue for $C^\infty(\mathbb{T}^n)$, and simultaneously for $s(\mathbb{Z}^n)$, of §1, in this case is the Schwartz space of rapidly decreasing functions:

$$(3.3) \quad \mathcal{S}(\mathbb{R}^n) = \{u \in C^\infty(\mathbb{R}^n) : x^\beta D^\alpha u \in L^\infty(\mathbb{R}^n) \text{ for all } \alpha, \beta \geq 0\},$$

where $x^\beta = x_1^{\beta_1} \cdots x_n^{\beta_n}$, $D^\alpha = D_1^{\alpha_1} \cdots D_n^{\alpha_n}$, with $D_j = -i\partial/\partial x_j$. It is easy to verify that

$$(3.4) \quad \mathcal{F} : \mathcal{S}(\mathbb{R}^n) \longrightarrow \mathcal{S}(\mathbb{R}^n)$$

and

$$(3.5) \quad \xi^\alpha D_\xi^\beta \mathcal{F}f(\xi) = (-1)^{|\beta|} \mathcal{F}(D^\alpha x^\beta f)(\xi).$$

We define $\mathcal{F}^*$ by

$$(3.6) \quad \mathcal{F}^* f(\xi) = \tilde{f}(\xi) = (2\pi)^{-n/2} \int f(x) e^{ix \cdot \xi} dx,$$

which differs from (3.1) only in the sign of the exponent. It is clear that $\mathcal{F}^*$ satisfies the mapping properties (3.2), (3.4), and

$$(3.7) \quad (\mathcal{F}u, v) = (u, \mathcal{F}^*v),$$

for $u, v \in \mathcal{S}(\mathbb{R}^n)$, where (u, v) denotes the usual L^2 -inner product, $(u, v) = \int u(x) \overline{v(x)} dx$.

As in the theory of Fourier series, the first major result is the Fourier inversion formula. The following is our first version.

Proposition 3.1. *We have the inversion formula*

$$(3.8) \quad \mathcal{F}^* \mathcal{F} = \mathcal{F} \mathcal{F}^* = I \quad \text{on } \mathcal{S}(\mathbb{R}^n).$$

As in the proof of the inversion formula for Fourier series, via Proposition 1.1, in the present proof we will sneak up on the inversion formula by throwing in a convergence factor that will allow interchange of orders of integration (in the proof of Proposition 1.1, the orders of an integral and an infinite series were interchanged). Also, as we will see in §5, this method will have serendipitous applications to PDE. So, let us write, for $f \in \mathcal{S}(\mathbb{R}^n)$,

$$(3.9) \quad \begin{aligned} \mathcal{F}^* \mathcal{F} f(x) &= (2\pi)^{-n} \int \left[\int f(y) e^{-iy \cdot \xi} dy \right] e^{ix \cdot \xi} d\xi \\ &= (2\pi)^{-n} \lim_{\varepsilon \searrow 0} \iint f(y) e^{-\varepsilon |\xi|^2} e^{i(x-y) \cdot \xi} dy d\xi. \end{aligned}$$

We can interchange the order of integration on the right for any $\varepsilon > 0$, to obtain

$$(3.10) \quad \mathcal{F}^* \mathcal{F} f(x) = \lim_{\varepsilon \searrow 0} \int f(y) p(\varepsilon, x - y) dy,$$

where

$$(3.11) \quad p(\varepsilon, x) = (2\pi)^{-n} \int e^{-\varepsilon |\xi|^2 + ix \cdot \xi} d\xi.$$

Note that

$$(3.12) \quad p(\varepsilon, x) = \varepsilon^{-n/2} q(\varepsilon^{-1/2} x),$$

where $q(x) = p(1, x)$. In a moment we will show that

$$(3.13) \quad p(\varepsilon, x) = (4\pi\varepsilon)^{-n/2} e^{-|x|^2/4\varepsilon}.$$

The derivation of this identity will also show that

$$(3.14) \quad \int_{\mathbb{R}^n} q(x) dx = 1.$$

From this, it follows as in the proof of Proposition 1.1 that

$$(3.15) \quad \lim_{\varepsilon \searrow 0} \int f(y) p(\varepsilon, x - y) dy = f(x),$$

for any $f \in \mathcal{S}(\mathbb{R}^n)$, even for f bounded and continuous, so we have proved $\mathcal{F}^* \mathcal{F} = I$ on $\mathcal{S}(\mathbb{R}^n)$; the proof that $\mathcal{F} \mathcal{F}^* = I$ on $\mathcal{S}(\mathbb{R}^n)$ is identical.

It remains to verify (3.13). We observe that $p(\varepsilon, x)$, defined by (3.11), is an entire holomorphic function of $x \in \mathbb{C}^n$, for any $\varepsilon > 0$. It is convenient to verify that

$$(3.16) \quad p(\varepsilon, ix) = (4\pi\varepsilon)^{-n/2} e^{-|x|^2/4\varepsilon}, \quad x \in \mathbb{R}^n,$$

from which (3.13) follows by analytic continuation. Now

$$\begin{aligned}
 p(\varepsilon, ix) &= (2\pi)^{-n} \int e^{-x \cdot \xi - \varepsilon |\xi|^2} d\xi \\
 (3.17) \quad &= (2\pi)^{-n} e^{|x|^2/4\varepsilon} \int e^{-|x/2\sqrt{\varepsilon} + \sqrt{\varepsilon}\xi|^2} d\xi \\
 &= (2\pi)^{-n} \varepsilon^{-n/2} e^{|x|^2/4\varepsilon} \int_{\mathbb{R}^n} e^{-|\xi|^2} d\xi.
 \end{aligned}$$

To prove (3.16), it remains to show that

$$(3.18) \quad \int_{\mathbb{R}^n} e^{-|\xi|^2} d\xi = \pi^{n/2}.$$

Indeed, if

$$(3.19) \quad A = \int_{-\infty}^{\infty} e^{-\xi^2} d\xi,$$

then the left side of (3.18) is equal to A^n . But for $n = 2$ we can use polar coordinates:

$$(3.20) \quad A^2 = \int_{\mathbb{R}^2} e^{-|\xi|^2} d\xi = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta = \pi.$$

This completes the proof of the identity (3.16) and hence of (3.13).

In light of (3.7) and the Fourier inversion formula (3.8), we see that, for $u, v \in \mathcal{S}(\mathbb{R}^n)$,

$$(3.21) \quad (\mathcal{F}u, \mathcal{F}v) = (u, v) = (\mathcal{F}^*u, \mathcal{F}^*v).$$

Thus $\mathcal{F}$ and $\mathcal{F}^*$ extend uniquely from $\mathcal{S}(\mathbb{R}^n)$ to isometries on $L^2(\mathbb{R}^n)$ and are inverses to each other. Thus we have the Plancherel theorem:

Proposition 3.2. *The Fourier transform*

$$(3.22) \quad \mathcal{F} : L^2(\mathbb{R}^n) \longrightarrow L^2(\mathbb{R}^n)$$

is unitary, with inverse $\mathcal{F}^$.*

The inversion formulas of Propositions 3.1 and 3.2 do not provide for the inversion of $\mathcal{F}$ in (3.2). We will obtain this as a byproduct of the Fourier inversion formula for tempered distributions, in the next section.

We make a remark about the computation of the Fourier integral (3.11), done above via analytic continuation. The following derivation does not make any direct use of complex analysis. It suffices to handle the case $\varepsilon = 1/2$, that is, to show

$$(3.23) \quad \hat{G}(\xi) = e^{-|\xi|^2/2} \quad \text{if } G(x) = e^{-|x|^2/2}, \text{ on } \mathbb{R}^n.$$

We have interchanged the roles of x and ξ compared to those in (3.11) and (3.13). It suffices to get (3.23) in the case $n = 1$, by the obvious multiplicativity. Now the Gaussian function $G(x) = e^{-x^2/2}$ satisfies the differential equation

$$(3.24) \quad \left(\frac{d}{dx} + x\right)G(x) = 0.$$

By the intertwining property (3.5), it follows that $(d/d\xi + \xi)\hat{G}(\xi) = 0$, and uniqueness of solutions to this ODE yields $\hat{G}(\xi) = Ce^{-\xi^2/2}$. The constant C is evaluated via the identity (3.20); $C = 1$; and we are done.

As for the necessity of computing the Fourier integral (3.11) to prove the Fourier inversion formula, let us note the following. For *any* $g \in \mathcal{S}(\mathbb{R}^n)$ with $g(0) = 1$, ($g(\xi) = e^{-|\xi|^2}$ being an example), we have (replacing ε by δ^2), just as in (3.9),

$$(3.25) \quad \begin{aligned} \mathcal{F}^* \mathcal{F} f(x) &= (2\pi)^{-n} \lim_{\delta \searrow 0} \iint f(y) g(\delta \xi) e^{i(x-y) \cdot \xi} dy d\xi \\ &= \lim_{\delta \searrow 0} \int f(y) h_\delta(x-y) dy, \end{aligned}$$

where

$$(3.26) \quad \begin{aligned} h_\delta(x) &= (2\pi)^{-n} \int g(\delta \xi) e^{ix \cdot \xi} d\xi \\ &= (2\pi)^{-n/2} \delta^{-n} \tilde{g}(\delta^{-1}x). \end{aligned}$$

By the peaked nature of h_δ as $\delta \rightarrow 0$, we see that the limit in (3.25) is equal to

$$(3.27) \quad C f(x),$$

where

$$(3.28) \quad C = \int h_1(x) dx = (2\pi)^{-n/2} \int \tilde{g}(x) dx.$$

The argument (3.25)–(3.27) shows that C is independent of the choice of $g \in \mathcal{S}(\mathbb{R}^n)$, and we need only find a single example g such that $\tilde{g}(x)$ can be evaluated explicitly and then the integral on the right in (3.28) can be evaluated explicitly. In most natural examples one picks g to be even, so $\tilde{g} = \hat{g}$.

We remark that one does not need to have $g \in \mathcal{S}(\mathbb{R}^n)$ in the argument above; it suffices to have $g \in L^1(\mathbb{R}^n)$, bounded and continuous, and such that $\hat{g} \in L^1(\mathbb{R}^n)$. An example, in the case $n = 1$, is

$$(3.29) \quad g(\xi) = e^{-|\xi|}.$$

In this case, elementary calculations give

$$(3.30) \quad \hat{g}(x) = \left(\frac{2}{\pi}\right)^{1/2} \frac{1}{x^2 + 1};$$

compare (5.21). In this case, (3.28) can be evaluated in terms of the arctangent. Another example, in the case $n = 1$, is

$$(3.31) \quad g(\xi) = \begin{cases} 1 - |\xi| & \text{if } |\xi| \leq 1, \\ 0 & \text{if } |\xi| \geq 1. \end{cases}$$

In this case,

$$(3.32) \quad \hat{g}(x) = (2\pi)^{-1/2} \left(\frac{\sin \frac{1}{2}x}{\frac{1}{2}x} \right)^2,$$

and (3.28) can be evaluated by the method of residues. The calculation of (3.32) can be achieved by evaluating

$$\int_0^1 (1 - \xi) \cos x\xi \, d\xi$$

via an integration by parts, though there is a more painless way, mentioned below.

We now make some comments on the relation between the Fourier transform and convolutions. The convolution $u * v$ of two functions on $\mathbb{R}^n$ is defined by

$$(3.33) \quad \begin{aligned} u * v(x) &= \int u(y)v(x - y) \, dy \\ &= \int u(x - y)v(y) \, dy. \end{aligned}$$

Note that $u * v = v * u$. If $u, v \in \mathcal{S}(\mathbb{R}^n)$, so is $u * v$. Also

$$\|u * v\|_{L^p(\mathbb{R}^n)} \leq \|u\|_{L^1} \|v\|_{L^p},$$

so the convolution has a unique, continuous extension to a bilinear map

$$(3.34) \quad L^1(\mathbb{R}^n) \times L^p(\mathbb{R}^n) \longrightarrow L^p(\mathbb{R}^n),$$

for $1 \leq p < \infty$; one can directly perceive that this also works for $p = \infty$. Note that the right side of (3.10), for any $\varepsilon > 0$, is an example of a convolution. Computing the Fourier transform of (3.33) leads immediately to the formula

$$(3.35) \quad \mathcal{F}(u * v)(\xi) = (2\pi)^{n/2} \hat{u}(\xi) \hat{v}(\xi).$$

We also note that if

$$(3.36) \quad P = \sum_{|\alpha| \leq k} a_\alpha D^\alpha$$

is a constant-coefficient differential operator, we have

$$(3.37) \quad P(u * v) = (Pu) * v = u * (Pv)$$

if $u, v \in \mathcal{S}(\mathbb{R}^n)$. This also generalizes; if $u \in \mathcal{S}(\mathbb{R}^n), v \in L^p(\mathbb{R}^n)$, the first identity continues to hold; as we will see in the next section, so does the second identity, once we are able to interpret what it means.

We mention the following simple application of (3.35), to a short calculation of (3.32). With g given by (3.31), we have $g = g_1 * g_1$, where

$$(3.38) \quad g_1(\xi) = 1 \text{ for } \xi \in \left[-\frac{1}{2}, \frac{1}{2}\right], \quad 0 \text{ otherwise.}$$

Thus

$$(3.39) \quad \hat{g}_1(x) = (2\pi)^{-1/2} \int_{-1/2}^{1/2} e^{-ix\xi} d\xi = (2\pi)^{-1/2} \frac{\sin \frac{1}{2}x}{\frac{1}{2}x},$$

and then (3.32) follows immediately from (3.35).

Exercises

1. Show that $\mathcal{F} : L^1(\mathbb{R}^n) \rightarrow C_0(\mathbb{R}^n)$, where $C_0(\mathbb{R}^n)$ denotes the space of functions v , continuous on $\mathbb{R}^n$, such that $v(\xi) \rightarrow 0$ as $|\xi| \rightarrow \infty$.
(Hint: Use the denseness of $\mathcal{S}(\mathbb{R}^n)$ in $L^1(\mathbb{R}^n)$.)

This result is the Riemann-Lebesgue lemma for the Fourier transform.

2. Show that the Fourier transforms (3.1) and (3.22) coincide on $L^1(\mathbb{R}^n) \cap L^2(\mathbb{R}^n)$.
3. For $f \in L^1(\mathbb{R})$, set $S_R f(x) = (2\pi)^{-1/2} \int_{-R}^R \hat{f}(\xi) e^{ix\xi} d\xi$. Show that

$$S_R f(x) = D_R * f(x) = \int_{-\infty}^{\infty} D_R(x-y) f(y) dy,$$

where

$$D_R(x) = (2\pi)^{-1} \int_{-R}^R e^{ix\xi} d\xi = \frac{\sin Rx}{\pi x}.$$

Compare Exercise 5 of §1.

4. Show that $f \in L^2(\mathbb{R}) \Rightarrow S_R f \rightarrow f$ in L^2 -norm as $R \rightarrow \infty$.
5. Show that there exist $f \in L^1(\mathbb{R})$ such that $S_R f \notin L^1(\mathbb{R})$ for any $R \in (0, \infty)$.
(Hint: Note that $D_R \notin L^1(\mathbb{R})$.)
6. For $f \in L^1(\mathbb{R})$, set

$$C_R f(x) = (2\pi)^{-1/2} \int_{-R}^R \left(1 - \frac{|\xi|}{R}\right) \hat{f}(\xi) e^{ix\xi} d\xi.$$

Show that $C_R f(x) = E_R * f(x)$, where

$$E_R(x) = (2\pi)^{-1} \int_{-R}^R \left(1 - \frac{|\xi|}{R}\right) e^{ix\xi} d\xi = \frac{2}{\pi R} \left[\frac{\sin \frac{1}{2}Rx}{x} \right]^2.$$

Note that $E_R \in L^1(\mathbb{R})$. Show that, for $1 \leq p < \infty$,

$$f \in L^p(\mathbb{R}) \implies C_R f \rightarrow f \text{ in } L^p\text{-norm, as } R \rightarrow \infty.$$

We say the Fourier transform of f is Cesaro-summable if $C_R f \rightarrow f$ as $R \rightarrow \infty$.

In Exercises 7–13, suppose $f \in \mathcal{S}(\mathbb{R})$ has the following properties: $f \geq 0$, $\int_{-\infty}^{\infty} f(x) dx = 1$, and $\int_{-\infty}^{\infty} xf(x) dx = 0$. Set $F(\xi) = (2\pi)^{1/2} \hat{f}(\xi)$. The point of the exercises is to obtain a version of the *central limit theorem*.

7. Show that $F(0) = 1$, $F'(0) = 0$, $F''(0) = -2a < 0$. Also, $\xi \neq 0 \Rightarrow |F(\xi)| < 1$.
8. Set $F_n(\xi) = F(\xi/\sqrt{n})^n$. Relate $(2\pi)^{-1/2} \tilde{F}_n(x)$ to the convolution of n copies of f .
9. Show that there exist $A > 0$ and $G \in C^\infty([-A, A])$ such that $f(\xi) = e^{-a\xi^2} G(\xi)$ for $|\xi| \leq A$, and $G(0) = 1$, $G'(0) = G''(0) = 0$. Hence

$$F_n(\xi) = e^{-a\xi^2} G(n^{-1/2}\xi)^n, \quad \text{for } |\xi| \leq A\sqrt{n}.$$

10. Show that $|G(\xi/\sqrt{n})^n - 1| \leq Cn^{-\alpha}$ if $|\xi| \leq n^{(1/2-\alpha)/3}$, for n large. Fix $\alpha \in (0, \frac{1}{2})$, and set $\gamma = (1/2 - \alpha)/3 \in (0, \frac{1}{6})$.
11. Show that, for $|\xi| \geq n^\gamma$, $|F(\xi/\sqrt{n})| \leq 1 - \frac{1}{2}an^{-(1-2\gamma)}$, for n large, so $|F_n(\xi)| \leq e^{-an^{2\gamma}/4} = \delta_n$. Deduce that

$$\int_{|\xi| \geq n^\gamma} |F_n(\xi)| d\xi \leq C \delta_n^{(n-1)/n} \sqrt{n} \rightarrow 0, \quad \text{as } n \rightarrow \infty.$$

12. From Exercises 9–11, deduce that $F_n \rightarrow e^{-a\xi^2}$ in $L^1(\mathbb{R})$ as $n \rightarrow \infty$.
13. Deduce now that $(2\pi)^{-1/2} \tilde{F}_n \rightarrow (4\pi a)^{-1/2} e^{-x^2/4a}$ in both $C_0(\mathbb{R})$ and $L^1(\mathbb{R})$, as $n \rightarrow \infty$. Relate this to the central limit theorem of probability theory. Weaken the hypotheses on f as much as you can. (*Hint:* In passing from the C_0 -result to the L^1 -result, positivity of $\tilde{F}_n$ will be useful.)

14. With $p_\varepsilon(x) = (4\pi\varepsilon)^{-1/2} e^{-x^2/4\varepsilon}$, as in (3.13) for $n = 1$, show that, for any $u(x)$, continuous and compactly supported on $\mathbb{R}$, $p_\varepsilon * u \rightarrow u$ uniformly as $\varepsilon \rightarrow 0$. Show that for each $\varepsilon > 0$, $p_\varepsilon * u(x)$ is the restriction to $\mathbb{R}$ of an entire holomorphic function of $x \in \mathbb{C}$.

15. Using Exercise 14, prove the *Weierstrass approximation theorem*:

Any $f \in C([a, b])$ is a uniform limit of polynomials.

(*Hint:* Extend f to u as above, approximate u by $p_\varepsilon * u$, and expand this in a power series.)

16. Suppose $f \in \mathcal{S}(\mathbb{R}^n)$ is supported in $B_R = \{x \in \mathbb{R}^n : |x| \leq R\}$. Show that $\hat{f}(\xi)$ is holomorphic in $\xi \in \mathbb{C}^n$ and satisfies

$$(3.40) \quad |\hat{f}(\xi + i\eta)| \leq C_N \langle \xi \rangle^{-N} e^{R|\eta|}, \quad \xi, \eta \in \mathbb{R}^n.$$

17. Conversely, suppose $g(\xi) = \hat{f}(\xi) \in \mathcal{S}(\mathbb{R}^n)$ has a holomorphic extension to $\mathbb{C}^n$ satisfying (3.40). Show that f is supported in $|x| \leq R$.

(*Hint:* With $\omega = x/|x|$, $r \geq 0$, write

$$(3.41) \quad f(x) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} \hat{f}(\xi + ir\omega) e^{ix \cdot \xi - rx \cdot \omega} d\xi,$$

with

$$|\hat{f}(\xi + ir\omega) e^{ix \cdot \xi - rx \cdot \omega}| \leq C_N \langle \xi \rangle^{-N} e^{r(R-|x|)},$$

and let $r \rightarrow +\infty$.)

This is a basic case of the *Paley-Wiener theorem*.

18. Given $f \in L^1(\mathbb{R}^n)$, show that f is supported in B_R if and only if $\hat{f}(\xi)$ is holomorphic in $\mathbb{C}^n$, satisfying

$$|\hat{f}(\xi + i\eta)| \leq C e^{R|\eta|}, \quad \xi, \eta \in \mathbb{R}^n.$$

Reconsider this problem after reading §4.

19. Show that

$$f(x) = \frac{1}{\cosh x} \implies \hat{f}(\xi) = \frac{\sqrt{\pi/2}}{\cosh \frac{\pi}{2}\xi}.$$

(Hint: See (A.13)–(A.15).)

4. Distributions and tempered distributions

L. Schwartz's theory of distributions has proved to be not only a wonderful tool in partial differential equations, but also a device that lends clarity to many aspects of Fourier analysis. We sketch the basic concepts of distribution theory here, making use of such basic concepts as Fréchet spaces and weak topologies, which are treated in Appendix A, Functional Analysis.

We begin with the concept of a tempered distribution. This is a continuous linear functional

$$(4.1) \quad w : \mathcal{S}(\mathbb{R}^n) \longrightarrow \mathbb{C},$$

where $\mathcal{S}(\mathbb{R}^n)$ is the Schwartz space defined in §3. The space $\mathcal{S}(\mathbb{R}^n)$ has a topology, determined by the seminorms

$$(4.2) \quad p_k(u) = \sum_{|\alpha| \leq k} \sup_{x \in \mathbb{R}^n} \langle x \rangle^k |D^\alpha u(x)|.$$

The distance function

$$(4.3) \quad d(u, v) = \sum_{k=0}^{\infty} 2^{-k} \frac{p_k(u - v)}{1 + p_k(u - v)}$$

makes $\mathcal{S}(\mathbb{R}^n)$ a complete metric space; with such a topology it is a Fréchet space. For a linear map w as in (4.1) to be continuous, it is necessary and sufficient that, for some k, C ,

$$(4.4) \quad |w(u)| \leq C p_k(u), \quad \text{for all } u \in \mathcal{S}(\mathbb{R}^n).$$

The action of w is often written as follows:

$$(4.5) \quad w(u) = \langle u, w \rangle.$$

The set of all continuous linear functionals on $\mathcal{S}(\mathbb{R}^n)$ is denoted

$$(4.6) \quad \mathcal{S}'(\mathbb{R}^n)$$

and is called the space of tempered distributions.

The space $\mathcal{S}'(\mathbb{R}^n)$ has a topology, called the weak* topology, or sometimes simply the weak topology, in terms of which a directed family w_γ converges

to w weakly in $\mathcal{S}'(\mathbb{R}^n)$ if and only if, for each $u \in \mathcal{S}(\mathbb{R}^n)$, $\langle u, w_\gamma \rangle \rightarrow \langle u, w \rangle$. One can also consider the strong topology on $\mathcal{S}'(\mathbb{R}^n)$, the topology of uniform convergence on bounded subsets of $\mathcal{S}(\mathbb{R}^n)$, but we will not consider this explicitly. For more on the topology of $\mathcal{S}$ and $\mathcal{S}'$, see [H], [Sch], and [Yo]. We now consider examples of tempered distributions.

There is a natural injection

$$(4.7) \quad L^p(\mathbb{R}^n) \hookrightarrow \mathcal{S}'(\mathbb{R}^n),$$

for any $p \in [1, \infty]$, given by

$$(4.8) \quad \langle u, f \rangle = \int u(x)f(x) \, dx, \quad u \in \mathcal{S}(\mathbb{R}^n), \, f \in L^p(\mathbb{R}^n).$$

Similarly any finite measure on $\mathbb{R}^n$ gives an element of $\mathcal{S}'(\mathbb{R}^n)$. The basic example is the Dirac “delta function” δ , defined by

$$(4.9) \quad \langle u, \delta \rangle = u(0).$$

Also, each differential operator $D_j = -i\partial/\partial x_j$ acts on $\mathcal{S}'(\mathbb{R}^n)$, by the definition

$$(4.10) \quad \langle u, D_j w \rangle = -\langle D_j u, w \rangle, \quad u \in \mathcal{S}, \, w \in \mathcal{S}'.$$

Iterating, we see that each $D^\alpha = D_1^{\alpha_1} \cdots D_n^{\alpha_n}$ acts on $\mathcal{S}'$:

$$(4.11) \quad D^\alpha : \mathcal{S}'(\mathbb{R}^n) \longrightarrow \mathcal{S}'(\mathbb{R}^n),$$

and we have

$$(4.12) \quad \langle u, D^\alpha w \rangle = (-1)^{|\alpha|} \langle D^\alpha u, w \rangle$$

for $u \in \mathcal{S}, w \in \mathcal{S}'$. Similarly,

$$\langle u, fw \rangle = \langle fu, w \rangle$$

defines fw for $w \in \mathcal{S}'$, provided that f and each of its derivatives is polynomially bounded.

To illustrate, consider on $\mathbb{R}$ the Heaviside function

$$(4.13) \quad H(x) = \begin{cases} 1 & \text{if } x \geq 0, \\ 0 & \text{if } x < 0. \end{cases}$$

Then $H \in L^\infty(\mathbb{R}) \subset \mathcal{S}'(\mathbb{R})$, and the definition (4.10) gives

$$(4.14) \quad \frac{d}{dx} H = \delta$$

as a consequence of the fundamental theorem of calculus. The derivative of δ is characterized by

$$(4.15) \quad \langle u, \delta' \rangle = -u'(0)$$

in this case.

The Fourier transform $\mathcal{F} : \mathcal{S}(\mathbb{R}^n) \rightarrow \mathcal{S}(\mathbb{R}^n)$, studied in §3, extends to $\mathcal{S}'$ by the formula

$$(4.16) \quad \langle u, \mathcal{F}w \rangle = \langle \mathcal{F}u, w \rangle;$$

we can also set

$$(4.17) \quad \langle u, \mathcal{F}^*w \rangle = \langle \mathcal{F}^*u, w \rangle$$

to get

$$(4.18) \quad \mathcal{F}, \mathcal{F}^* : \mathcal{S}'(\mathbb{R}^n) \longrightarrow \mathcal{S}'(\mathbb{R}^n).$$

The maps (4.18) are continuous when $\mathcal{S}'(\mathbb{R}^n)$ is given the weak* topology, as follows easily from the definitions.

The Fourier inversion formula of Proposition 3.1 yields:

Proposition 4.1. *We have*

$$(4.19) \quad \mathcal{F}^*\mathcal{F} = \mathcal{F}\mathcal{F}^* = I \quad \text{on } \mathcal{S}'(\mathbb{R}^n).$$

Proof. Using (4.16) and (4.17), if $u \in \mathcal{S}$, $w \in \mathcal{S}'$,

$$\langle u, \mathcal{F}^*\mathcal{F}w \rangle = \langle \mathcal{F}^*u, \mathcal{F}w \rangle = \langle \mathcal{F}\mathcal{F}^*u, w \rangle = \langle u, w \rangle,$$

and a similar analysis works for $\mathcal{F}\mathcal{F}^*w$.

As an example of a Fourier transform of a tempered distribution, the definition gives directly

$$(4.20) \quad \mathcal{F}\delta = (2\pi)^{-n/2};$$

the Fourier transform of the delta function is a constant function. One has the same result for $\mathcal{F}^*\delta$. By the Fourier inversion formula,

$$(4.21) \quad \mathcal{F}1 = (2\pi)^{n/2}\delta.$$

Next, let us consider on the line $\mathbb{R}$, for any $\varepsilon > 0$,

$$(4.22) \quad \begin{aligned} H_\varepsilon(x) &= e^{-\varepsilon x} & \text{for } x \geq 0, \\ &0 & \text{for } x < 0. \end{aligned}$$

We have, by elementary calculation,

$$(4.23) \quad \hat{H}_\varepsilon(\xi) = (2\pi)^{-1/2} \int_0^\infty e^{-\varepsilon x - i\xi x} dx = (2\pi)^{-1/2} (\varepsilon + i\xi)^{-1},$$

for each $\varepsilon > 0$. Now it is clear that

$$(4.24) \quad H_\varepsilon \rightarrow H \quad \text{as } \varepsilon \searrow 0, \quad \text{in } \mathcal{S}'(\mathbb{R}),$$

in the weak* topology. It follows that

$$(4.25) \quad \hat{H}(\xi) = (2\pi)^{-1/2} \lim_{\varepsilon \searrow 0} (\varepsilon + i\xi)^{-1} \quad \text{in } \mathcal{S}'(\mathbb{R}).$$

In particular, the limit on the right *exists* in $\mathcal{S}'(\mathbb{R})$. Changing the sign of x in (4.22)–(4.24) and noting that $H(-x) = 1 - H(x)$, we also have

$$(4.26) \quad (2\pi)^{1/2}\delta - \hat{H} = (2\pi)^{-1/2} \lim_{\varepsilon \searrow 0} (\varepsilon - i\xi)^{-1} \text{ in } \mathcal{S}'(\mathbb{R}).$$

Let us set

$$(4.27) \quad (\xi \pm i0)^{-1} = \lim_{\varepsilon \searrow 0} (\xi \pm i\varepsilon)^{-1}.$$

Then (4.25) and (4.26) give

$$(4.28) \quad \hat{H} = -i(2\pi)^{-1/2}(\xi - i0)^{-1}$$

and

$$(4.29) \quad (\xi + i0)^{-1} - (\xi - i0)^{-1} = -2\pi i\delta.$$

The last identity is often called the Plemelj jump relation. Also, subtracting (4.25) from (4.26) gives

$$(4.30) \quad (\xi + i0)^{-1} + (\xi - i0)^{-1} = (2\pi)^{1/2}i \widehat{\text{sgn}}(\xi),$$

where

$$(4.31) \quad \text{sgn}(x) = \begin{cases} 1 & \text{if } x \geq 0, \\ -1 & \text{if } x < 0. \end{cases}$$

It is also an easy exercise to show that

$$(4.32) \quad (\xi + i0)^{-1} + (\xi - i0)^{-1} = 2 \, PV\left(\frac{1}{\xi}\right),$$

where the “principal value” distribution

$$(4.33) \quad PV\left(\frac{1}{x}\right) \in \mathcal{S}'(\mathbb{R})$$

is defined by

$$(4.34) \quad \begin{aligned} \left\langle u, PV\left(\frac{1}{x}\right) \right\rangle &= \lim_{h \searrow 0} \int_{\mathbb{R} \setminus (-h, h)} \frac{u(x)}{x} \, dx \\ &= \lim_{h \searrow 0} \int_h^\infty \left[\frac{u(x)}{x} - \frac{u(-x)}{x} \right] \, dx \\ &= \int_0^\infty \frac{u(x) - u(-x)}{x} \, dx. \end{aligned}$$

Note that if we replace the left side of (4.29) by

$$\frac{1}{\xi + i\varepsilon} - \frac{1}{\xi - i\varepsilon} = -\frac{2i\varepsilon}{\xi^2 + \varepsilon^2} = -2i \frac{1}{\varepsilon} \cdot \frac{1}{(\xi/\varepsilon)^2 + 1},$$

the conclusion (4.29) is a special case of the following obvious result.

Proposition 4.2. *If $f \in L^1(\mathbb{R}^n)$, $\int f(x)dx = C_0$, then, as $\varepsilon \rightarrow 0$,*

$$(4.35) \quad \varepsilon^{-n} f(\varepsilon^{-1}x) \rightarrow C_0 \delta \text{ in } \mathcal{S}'(\mathbb{R}^n).$$

That δ is the limit of a sequence of elements of $\mathcal{S}(\mathbb{R}^n)$ is a special case of the fact that $\mathcal{S}(\mathbb{R}^n)$ is dense in $\mathcal{S}'(\mathbb{R}^n)$, which will be established shortly.

Given $w \in \mathcal{S}'(\mathbb{R}^n)$, if $\Omega \subset \mathbb{R}^n$ is open, one says w vanishes on Ω provided $\langle u, w \rangle = 0$ for all $u \in C_0^\infty(\Omega)$. By a partition of unity argument, it follows that if w vanishes on Ω_j , then it vanishes on their union; w is said to be *supported* on a closed set $K \subset \mathbb{R}^n$ if w vanishes on $\mathbb{R}^n \setminus K$. The smallest closed set K on which w is supported exists; it is denoted $\text{supp } w$. Note that if w vanishes on all of $\mathbb{R}^n$, then $w = 0$, since $C_0^\infty(\mathbb{R}^n)$ is dense in $\mathcal{S}(\mathbb{R}^n)$. (See the first part of the proof of Proposition 4.4 below.) If $w \in \mathcal{S}'(\mathbb{R}^n)$ is supported on a compact set $K \subset \mathbb{R}^n$, we say w has compact support. The space of compactly supported distributions on $\mathbb{R}^n$ is denoted by

$$(4.36) \quad \mathcal{E}'(\mathbb{R}^n).$$

If $w \in \mathcal{S}'(\mathbb{R}^n)$ is supported on a compact set $K \subset \mathbb{R}^n$, then w can be extended to a continuous linear functional

$$(4.37) \quad w : C^\infty(\mathbb{R}^n) \longrightarrow \mathbb{C}$$

by setting

$$(4.38) \quad \langle u, w \rangle = \langle \chi u, w \rangle, \quad u \in C^\infty(\mathbb{R}^n),$$

for any $\chi \in C_0^\infty(\mathbb{R}^n)$ such that $\chi = 1$ on a neighborhood of K . The space $C^\infty(\mathbb{R}^n)$ is also a Fréchet space, with topology defined by the seminorms

$$(4.39) \quad p_{R,k}(u) = \sup_{|x| \leq R} \sum_{|\alpha| \leq k} |D^\alpha u(x)|.$$

To say a linear map (4.37) is continuous is to say there exist R, k , and C such that

$$(4.40) \quad |w(u)| \leq C p_{R,k}(u), \quad \text{for all } u \in C^\infty(\mathbb{R}^n).$$

Such a linear functional restricts to $\mathcal{S}(\mathbb{R}^n) \subset C^\infty(\mathbb{R}^n)$, so it defines an element of $\mathcal{S}'(\mathbb{R}^n)$, and from (4.40) it follows that such an element must be supported in the compact set $\{x \in \mathbb{R}^n : |x| \leq R\}$. Thus the space (4.36) is precisely the dual space of $C^\infty(\mathbb{R}^n)$.

Fourier transforms of compactly supported distributions have some special properties.

Proposition 4.3. *If $w \in \mathcal{E}'(\mathbb{R}^n)$, then $\hat{w} \in C^\infty(\mathbb{R}^n)$ and, with $e_\xi(x) = e^{-ix \cdot \xi}$,*

$$(4.41) \quad \hat{w}(\xi) = (2\pi)^{-n/2} \langle e_\xi, w \rangle,$$

for all $\xi \in \mathbb{R}^n$. Furthermore, $\hat{w}$ extends to an entire holomorphic function of $\xi \in \mathbb{C}^n$.

Proof. For any $u \in \mathcal{S}(\mathbb{R}^n)$, $\langle u, \hat{w} \rangle = \langle \hat{u}, w \rangle$. Now we can write

$$(4.42) \quad \hat{u}(x) = (2\pi)^{-n/2} \int u(\xi) e_\xi(x) d\xi,$$

the integral converging in the Fréchet space topology of $C^\infty(\mathbb{R}^n)$, and the continuity of w acting on $C^\infty(\mathbb{R}^n)$ implies

$$\langle \hat{u}, w \rangle = (2\pi)^{-n/2} \int u(\xi) \langle e_\xi, w \rangle d\xi,$$

which gives (4.41). The right side of (4.41) is clearly holomorphic in $\xi \in \mathbb{C}^n$.

We next obtain the promised denseness of $\mathcal{S}(\mathbb{R}^n)$ in $\mathcal{S}'(\mathbb{R}^n)$.

Proposition 4.4. $C_0^\infty(\mathbb{R}^n)$ is dense in $\mathcal{S}'(\mathbb{R}^n)$, with its weak* topology.

Proof. Pick $\varphi \in C_0^\infty(\mathbb{R}^n)$, $\varphi(0) = 1$. It is easy to see that, given $w \in \mathcal{S}'(\mathbb{R}^n)$, φw is well defined, by $\langle u, \varphi w \rangle = \langle \varphi u, w \rangle$. Also, if $\varphi_j(x) = \varphi(x/j)$, then for $u \in \mathcal{S}(\mathbb{R}^n)$,

$$(4.43) \quad \varphi_j u \rightarrow u \text{ in } \mathcal{S}(\mathbb{R}^n),$$

which gives sequential denseness of $C_0^\infty(\mathbb{R}^n)$ in $\mathcal{S}(\mathbb{R}^n)$. Hence $\varphi_j w \rightarrow w$ in $\mathcal{S}'(\mathbb{R}^n)$ as $j \rightarrow \infty$. Since $\mathcal{F}$ and $\mathcal{F}^*$ are continuous on $\mathcal{S}'(\mathbb{R}^n)$, we have $\mathcal{F}^*(\varphi_j \hat{w}) \rightarrow w$ as $j \rightarrow \infty$. Now for each j , $w_{jk} = \varphi_k(\mathcal{F}^* \varphi_j \hat{w}) \rightarrow \mathcal{F}^*(\varphi_j \hat{w})$ as $k \rightarrow \infty$. But by Proposition 4.3, $\mathcal{F}^*(\varphi_j \hat{w})$ is smooth, so $w_{jk} \in C_0^\infty(\mathbb{R}^n)$, and the result follows.

One useful result is the following classification of distributions supported at a single point.

Proposition 4.5. If $w \in \mathcal{S}'(\mathbb{R}^n)$ is supported by $\{0\}$, then there exist k and complex numbers a_α such that

$$(4.44) \quad w = \sum_{|\alpha| \leq k} a_\alpha D^\alpha \delta.$$

Proof. We can suppose w satisfies the estimate (4.4). Thus w extends to $\mathcal{B}_k$, the closure of $\mathcal{S}(\mathbb{R}^n)$ in the space of C^k -functions on $\mathbb{R}^n$ for which the norm p_k is finite. By hypothesis, w annihilates the linear space $\mathcal{E}_0$ of elements of $C_0^\infty(\mathbb{R}^n)$ vanishing on a neighborhood of 0; thus w annihilates the closure of $\mathcal{E}_0$ in $\mathcal{B}_k$; call this closure $\mathcal{E}_k$. It is not hard to prove that

$$(4.45) \quad \mathcal{E}_k = \{u \in \mathcal{B}_k : D^\alpha u(0) = 0 \text{ for } |\alpha| \leq k\}.$$

See Exercise 7 below for some hints. Now, for general $u \in \mathcal{B}_k$, write

$$(4.46) \quad u(x) = \chi \left[\sum_{|\alpha| \leq k} \frac{u^{(\alpha)}(0)}{\alpha!} x^\alpha \right] + u^b(x),$$

where $\chi \in C_0^\infty(\mathbb{R}^n)$, $\chi(x) = 1$ for $|x| < 1$, and $u^b \in \mathcal{E}_k$. Applying w to both sides, we have an expression of the form (4.44), with $a_\alpha = (-1)^{|\alpha|} \langle \chi x^\alpha, w \rangle / \alpha!$.

As an application of Proposition 4.5, we establish the following result, which is an extension of the classical Liouville theorem for harmonic functions.

Proposition 4.6. *Suppose $u \in \mathcal{S}'(\mathbb{R}^n)$ satisfies*

$$(4.47) \quad \Delta u = 0 \quad \text{in } \mathbb{R}^n.$$

Then u is a polynomial in $(x_1, \dots, x_n)$.

Proof. As in §3, the identity

$$(4.48) \quad \Delta u = f \in \mathcal{S}'(\mathbb{R}^n)$$

is equivalent to

$$(4.49) \quad -|\xi|^2 \hat{u} = \hat{f} \quad \text{in } \mathcal{S}'(\mathbb{R}^n).$$

In particular, (4.47) for $u \in \mathcal{S}'(\mathbb{R}^n)$ implies

$$(4.50) \quad |\xi|^2 \hat{u} = 0 \quad \text{in } \mathcal{S}'(\mathbb{R}^n).$$

This of course implies

$$(4.51) \quad \text{supp } \hat{u} \subset \{0\}.$$

By Proposition 4.5, $\hat{u}$ must have the form (4.44), and the result follows.

It is clear that any nonconstant polynomial blows up, so we have:

Corollary 4.7. *If u is harmonic on $\mathbb{R}^n$ and bounded, then u is constant.*

This is the classical version of the Liouville theorem. We remind the reader of one of its uses. If $p(z)$ is a polynomial on $\mathbb{C}$, and if it has no zeros, then $q(z) = 1/p(z)$ is holomorphic (hence harmonic) on all of $\mathbb{C}$; clearly, $|q(z)| \rightarrow 0$ as $|z| \rightarrow \infty$ if $\deg p \geq 1$. Corollary 4.7 yields the obvious contradiction that $q(z)$ would have to be constant. This proves the fundamental theorem of algebra: Any nonconstant polynomial $p(z)$ must have a complex root.

Consider the function

$$(4.52) \quad \Psi(z) = \frac{1}{z}.$$

This is holomorphic on $\mathbb{C} \setminus \{0\}$. It is integrable near 0 and bounded outside a neighborhood of 0, and hence it defines an element of $\mathcal{S}'(\mathbb{R}^2)$. $(\partial/\partial\bar{z})\Psi \in \mathcal{S}'(\mathbb{R}^2)$ is supported at $\{0\}$. In fact, we claim:

Proposition 4.8. *We have*

$$(4.53) \quad \frac{\partial}{\partial\bar{z}} \frac{1}{z} = \pi\delta.$$

Proof. Let $u \in C_0^\infty(\mathbb{R}^2)$. We have

$$(4.54) \quad \begin{aligned} \left\langle u, \frac{\partial}{\partial\bar{z}} \frac{1}{z} \right\rangle &= - \iint_{\mathbb{R}^2} \frac{\partial u}{\partial\bar{z}} \frac{1}{z} dx dy \\ &= - \lim_{\varepsilon \rightarrow 0} \iint_{\mathbb{R}^2 \setminus B_\varepsilon} \frac{\partial(z^{-1}u)}{\partial\bar{z}} dx dy, \end{aligned}$$

where $B_\varepsilon = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < \varepsilon^2\}$. By Green's formula, in the form (2.37), the right side is equal to

$$(4.55) \quad \frac{1}{2i} \int_{\partial B_\varepsilon} \frac{u}{z} dz,$$

which is clearly equal in the limit $\varepsilon \rightarrow 0$ to $\pi u(0)$. This proves the proposition.

We say $(\pi z)^{-1}$ is a *fundamental solution* of $(\partial/\partial\bar{z})$. We will say more about the use of fundamental solutions later. Let us look at the task of producing a fundamental solution for the Laplace operator Δ on $\mathbb{R}^n$. In view of the rotational invariance of Δ , we are led to look for a function of $r = |x|$, for $x \neq 0$. The form

$$(4.56) \quad \Delta = \frac{\partial^2}{\partial r^2} + \frac{n-1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \Delta_S,$$

for Δ in polar coordinates, where Δ_S is the Laplace operator on the unit sphere S^{n-1} , shows that, for $n \geq 3$,

$$(4.57) \quad |x|^{2-n}$$

is harmonic for $x \neq 0$. As it is locally integrable near 0, it defines an element of $\mathcal{S}'(\mathbb{R}^n)$. We have the following result.

Proposition 4.9. *If $n \geq 3$,*

$$(4.58) \quad \Delta(|x|^{2-n}) = C_n \delta \text{ on } \mathbb{R}^n,$$

with $C_n = -(n-2) \cdot \text{Area}(S^{n-1})$. Also,

$$(4.59) \quad \Delta(\log|x|) = C_2 \delta \text{ on } \mathbb{R}^2,$$

with $C_2 = 2\pi$.

Proof. This will use Green's formula, in the form

$$(4.60) \quad \int_{\Omega} (\Delta u \cdot v - u \cdot \Delta v) \, dx = \int_{\partial\Omega} \left[v \frac{\partial u}{\partial \nu} - u \frac{\partial v}{\partial \nu} \right] \, dS.$$

Let $u \in C_0^\infty(\mathbb{R}^n)$ be arbitrary. Let $v = |x|^{2-n}$, and let $\Omega = \Omega_\varepsilon = \mathbb{R}^n \setminus B_\varepsilon$, where $B_\varepsilon = \{x \in \mathbb{R}^n : |x| < \varepsilon\}$. We have

$$(4.61) \quad \begin{aligned} \langle \Delta u, |x|^{2-n} \rangle &= \lim_{\varepsilon \rightarrow 0} \int_{\Omega_\varepsilon} \Delta u \cdot |x|^{2-n} \, dx \\ &= \lim_{\varepsilon \rightarrow 0} \int_{\Omega_\varepsilon} [\Delta u \cdot |x|^{2-n} - u \cdot \Delta |x|^{2-n}] \, dx \end{aligned}$$

since $\Delta |x|^{2-n} = 0$ for $x \neq 0$. Applying (4.60), we have this equal to

$$(4.62) \quad - \lim_{\varepsilon \rightarrow 0} \int_{\partial B_\varepsilon} \left[\varepsilon^{2-n} \frac{\partial u}{\partial r} - (2-n) \varepsilon^{1-n} u \right] \, dS.$$

Since the area of ∂B_ε is $\varepsilon^{n-1} \cdot \text{Area } S^{n-1}$, this limit is seen to be

$$(4.63) \quad -(n-2)u(0) \cdot \text{Area } S^{n-1},$$

which proves (4.58). The proof of (4.59) is similar.

Calculations yielding expressions for the area of S^{n-1} will be given in the appendix to this chapter.

Note that the equation

$$(4.64) \quad \Delta \Phi = \delta \quad \text{on } \mathbb{R}^n,$$

with $\Phi \in \mathcal{S}'(\mathbb{R}^n)$, is equivalent to

$$(4.65) \quad -|\xi|^2 \hat{\Phi} = (2\pi)^{-n/2}.$$

If $n \geq 3$, $|\xi|^{-2} \in L_{\text{loc}}^1(\mathbb{R}^n)$ and one solution to (4.65) is

$$(4.66) \quad \hat{\Phi}(\xi) = -(2\pi)^{-n/2} |\xi|^{-2} \in \mathcal{S}'(\mathbb{R}^n),$$

in such a case. We can relate this directly to (4.58) as follows. The orthogonal group $O(n)$ acts on $\mathcal{S}(\mathbb{R}^n)$, by

$$(4.67) \quad \pi(g)u(x) = u(g^{-1}x), \quad x \in \mathbb{R}^n, \, g \in O(n),$$

and this extends to an action on $\mathcal{S}'(\mathbb{R}^n)$, via $\langle u, \pi(g)v \rangle = \langle \pi(g^{-1})u, v \rangle$. This action commutes with $\mathcal{F}$. Thus the Fourier transform of an element like $|x|^{2-n}$ which is invariant under the $O(n)$ -action will also be $O(n)$ -invariant. There is also the dilation action on $\mathcal{S}(\mathbb{R}^n)$,

$$(4.68) \quad D(s)u(x) = u(sx), \quad s > 0, \, x \in \mathbb{R}^n,$$

which extends to $\mathcal{S}'(\mathbb{R}^n)$, via $\langle u, D(t)v \rangle = t^{-n} \langle D(1/t)u, v \rangle$. We have

$$(4.69) \quad D(s)\mathcal{F} = s^{-n}\mathcal{F}D(s^{-1}).$$

The element $|x|^{2-n} \in \mathcal{S}'(\mathbb{R}^n)$ is homogeneous of degree $2-n$, that is,

$$(4.70) \quad D(s)(|x|^{2-n}) = s^{2-n}|x|^{2-n},$$

so $\mathcal{F}(|x|^{2-n})$ will be homogeneous of degree -2 . This establishes that $\Phi(x) = C_n|x|^{2-n}$ satisfies (4.66), up to a constant factor. Note that $\delta \in \mathcal{S}'(\mathbb{R}^n)$ is homogeneous of degree $-n$. Since the Laplace operator Δ decreases the order of homogeneity of a distribution by two units, these considerations of orthogonal invariance and homogeneity directly suggest a constant times $|x|^{2-n}$ as a suitable candidate for a fundamental solution for the Laplace operator on $\mathbb{R}^n$.

We mention some extensions of the convolution

$$(4.71) \quad u * v(x) = \int u(y)v(x-y) dy,$$

which gives a bilinear map

$$(4.72) \quad \mathcal{S}(\mathbb{R}^n) \times \mathcal{S}(\mathbb{R}^n) \rightarrow \mathcal{S}(\mathbb{R}^n).$$

Note that if $u, v, w \in \mathcal{S}(\mathbb{R}^n)$,

$$(4.73) \quad \langle u * v, w \rangle = \langle u, v^\# * w \rangle,$$

where $v^\#(x) = v(-x)$, so the convolution extends in a straightforward way to

$$(4.74) \quad \mathcal{S}(\mathbb{R}^n) \times \mathcal{S}'(\mathbb{R}^n) \rightarrow \mathcal{S}'(\mathbb{R}^n),$$

with $\mathcal{S}(\mathbb{R}^n) \times \mathcal{E}'(\mathbb{R}^n) \rightarrow \mathcal{S}(\mathbb{R}^n)$, and hence

$$(4.75) \quad \mathcal{E}'(\mathbb{R}^n) \times \mathcal{S}'(\mathbb{R}^n) \rightarrow \mathcal{S}'(\mathbb{R}^n).$$

In either case, the identity

$$(4.76) \quad \mathcal{F}(u * v) = (2\pi)^{n/2} \hat{u}(\xi) \hat{v}(\xi)$$

continues to hold. For more on this, see Exercises 11–13 below. If P is any constant-coefficient differential operator, then

$$(4.77) \quad P(u * v) = (Pu) * v = u * Pv$$

in cases (4.74) and (4.75). For example, if $\Phi \in \mathcal{S}'(\mathbb{R}^n)$ and

$$(4.78) \quad P\Phi = \delta$$

(we say Φ is a fundamental solution of P), then a solution to

$$(4.79) \quad Pu = f,$$

for any given $f \in \mathcal{E}'(\mathbb{R}^n)$, is provided by

$$(4.80) \quad u = f * \Phi.$$

An object more general than a tempered distribution is a *distribution*. In general, a distribution on $\mathbb{R}^n$ is a continuous linear map

$$(4.81) \quad w : C_0^\infty(\mathbb{R}^n) \longrightarrow \mathbb{C}.$$

Here, continuity can be characterized as follows. For each $\varphi \in C_0^\infty(\mathbb{R}^n)$, the identity $\langle u, \varphi w \rangle = \langle \varphi u, w \rangle$ makes φw a linear functional on $C^\infty(\mathbb{R}^n)$. We require that each such linear functional be continuous, in the sense specified in (4.40). For further discussion, including a direct discussion of the natural “inductive limit” topology on $C_0^\infty(\mathbb{R}^n)$, see [RS] and [Sch]. The space of all distributions on $\mathbb{R}^n$ is denoted by

$$(4.82) \quad \mathcal{D}'(\mathbb{R}^n).$$

More generally, if M is any smooth, paracompact manifold, the space of continuous linear functionals on $C^\infty(M)$ is denoted by $\mathcal{E}'(M)$ and the space of continuous linear functionals on $C_0^\infty(M)$ is denoted by $\mathcal{D}'(M)$. Of course, if M is compact, $\mathcal{E}'(M) = \mathcal{D}'(M)$.

The case $M = \mathbb{T}^n$ is of interest, with respect to Fourier series. Given $w \in \mathcal{D}'(\mathbb{T}^n)$, we can define

$$(4.83) \quad \mathcal{F}w(k) = \hat{w}(k) = (2\pi)^{-n} \langle e_k, w \rangle,$$

where

$$(4.84) \quad e_k(\theta) = e^{-ik \cdot \theta} \in C^\infty(\mathbb{T}^n).$$

Since w must satisfy an estimate of the form

$$(4.85) \quad |\langle u, w \rangle| \leq C \|u\|_{C^\ell(\mathbb{T}^n)},$$

it is clear that

$$(4.86) \quad \mathcal{F} : \mathcal{D}'(\mathbb{T}^n) \longrightarrow s'(\mathbb{Z}^n),$$

where

$$(4.87) \quad s'(\mathbb{Z}^n) = \{a : \mathbb{Z}^n \rightarrow \mathbb{C} : |a(k)| \leq C \langle k \rangle^N \text{ for some } C, N\}$$

consists of polynomially bounded functions on $\mathbb{Z}^n$. Note that $s'(\mathbb{Z}^n)$ is the dual space to $s(\mathbb{Z}^n)$, defined in §1, and the map $\mathcal{F}$ in (4.86) is the adjoint of the map $\mathcal{F}^* : s(\mathbb{Z}^n) \longrightarrow C^\infty(\mathbb{T}^n)$ given by (1.11) and (1.12). Here we use the Hermitian inner product $(u, w) = \langle u, \bar{w} \rangle = \overline{\langle \bar{u}, w \rangle}$. The map $\mathcal{F} : C^\infty(\mathbb{T}^n) \rightarrow s(\mathbb{Z}^n)$ given by (1.1) and (1.6) also has an adjoint

$$(4.88) \quad \mathcal{F}^* : s'(\mathbb{Z}^n) \rightarrow \mathcal{D}'(\mathbb{T}^n),$$

extending the map (1.11)–(1.12), which, we recall, is

$$(4.89) \quad (\mathcal{F}^* a)(\theta) = \sum_{k \in \mathbb{Z}^n} a(k) e^{ik \cdot \theta}.$$

The Fourier inversion formulas

$$(4.90) \quad \mathcal{F}^* \mathcal{F} = I \text{ on } C^\infty(\mathbb{T}^n), \quad \mathcal{F} \mathcal{F}^* = I \text{ on } s(\mathbb{Z}^n),$$

extend by duality (or by continuity, and denseness of $C^\infty(\mathbb{T}^n)$ in $\mathcal{D}'(\mathbb{T}^n)$ and of $s(\mathbb{Z}^n)$ in $s'(\mathbb{Z}^n)$) to

$$(4.91) \quad \mathcal{F}^* \mathcal{F} = I \text{ on } \mathcal{D}'(\mathbb{T}^n), \quad \mathcal{F} \mathcal{F}^* = I \text{ on } s'(\mathbb{Z}^n);$$

consequently, the map (4.86) is an isomorphism.

Exercises

1. Define $M_f u$ by $\langle v, M_f u \rangle = \langle f v, u \rangle$, for $v \in \mathcal{S}(\mathbb{R}^n)$, $u \in \mathcal{S}'(\mathbb{R}^n)$. $M_f u$ is also denoted by $f u$. Show that $M_f : \mathcal{S}'(\mathbb{R}^n) \rightarrow \mathcal{S}'(\mathbb{R}^n)$ continuously, provided $f \in C^\infty(\mathbb{R}^n)$ and each derivative is polynomially bounded, that is, $|D^\alpha f(x)| \leq C_\alpha \langle x \rangle^{N(\alpha)}$.
2. Show that the identity $\xi^\alpha D_\xi^\beta \mathcal{F} f(\xi) = (-1)^{|\beta|} \mathcal{F}(D^\alpha x^\beta f)(\xi)$ from §3 continues to hold for $f \in \mathcal{S}'(\mathbb{R}^n)$.
3. Calculate $\mathcal{F}$ of x^α and of $D^\alpha \delta$.
4. Verify the identity (4.32) involving $PV(1/\xi)$.
5. Give a proof of Proposition 4.4, that $\mathcal{S}(\mathbb{R}^n)$ is dense in $\mathcal{S}'(\mathbb{R}^n)$, using convolutions in place of the Fourier transform.
6. Show the denseness of $\mathcal{S}(\mathbb{R}^n)$ in $\mathcal{S}'(\mathbb{R}^n)$ follows from the Hahn-Banach theorem. See Appendix A for a discussion of the Hahn-Banach theorem. On the other hand, sharpen the proof of Proposition 4.4, to obtain *sequential* denseness.
7. Prove the identity (4.45), used in the proof of Proposition 4.5.
(Hint: Fix $\eta \in C^\infty(\mathbb{R}^n)$ so that $\eta(x) = 0$ for $|x| \leq 1/2$, 1 for $|x| \geq 1$. Show that $\eta(Rx)u \rightarrow u$ in $\mathcal{B}_k$, as $R \rightarrow \infty$, for any $u \in \mathcal{E}_k$. This, plus a couple of further approximations, yields (4.45).)
8. Let $f(x, y) = 1/z \in \mathcal{S}'(\mathbb{R}^2)$. Using (4.53), compute $\mathcal{F} f \in \mathcal{S}'(\mathbb{R}^2)$. Using Proposition 4.9, compute the Fourier transform of $\log |x|$ on $\mathbb{R}^2$, and of $|x|^{2-n}$ on $\mathbb{R}^n$, $n \geq 3$. Reconsider this problem after reading §8.
9. Let $u \in \mathcal{E}'(\mathbb{R}^n)$, and suppose $\langle p, u \rangle = 0$ for every *polynomial* p on $\mathbb{R}^n$. Show that $u = 0$. (Hint: Show that $D^\alpha \hat{u}(0) = 0$ for all α ; but $\hat{u}$ is analytic.) Show that this result implies the Weierstrass approximation theorem, discussed in Exercise 15 of §3.
10. Let $f \in C^\infty(\mathbb{R}^n)$ be real-valued, $\Sigma = \{x : f(x) = 0\}$. Define $\delta(f(x)) \in \mathcal{D}'(\mathbb{R}^n)$ to be

$$(4.92) \quad \delta(f(x)) = \lim_{\varepsilon \searrow 0} \delta_\varepsilon(f(x)),$$

where $\delta_\varepsilon(x) = 1/\varepsilon$ for $|t| < \varepsilon/2$, 0 otherwise, provided this limit exists, with respect to the weak* topology on $\mathcal{D}'(\mathbb{R}^n)$. Show that if $\nabla f \neq 0$ on Σ , the limit does exist and that, for $u \in C_0^\infty(\mathbb{R}^n)$,

$$\langle u, \delta(f(x)) \rangle = \int_{\Sigma} u(y) |\nabla f(y)|^{-1} dS(y),$$

where dS is the $(n-1)$ -dimensional measure on Σ . Consider cases where the limit in (4.92) exists though ∇f vanishes on a variety in Σ .

11. Using an argument like that in the proof of Proposition 4.5, show that if $u \in \mathcal{S}'(\mathbb{R}^n)$ has support in a closed ball B , then, for some C, k ,

$$|\langle f, u \rangle| \leq C \sup_{x \in B, |\alpha| \leq k} |D^\alpha f(x)|.$$

(Hint: Establish the following analogue of (4.45). If $\mathcal{E}_B$ is the linear space of elements of $C_0^\infty(\mathbb{R}^n)$ vanishing on a neighborhood of B , and $\mathcal{E}_k$ is the closure of $\mathcal{E}_B$ in $\mathcal{B}_k$, then

$$\mathcal{E}_k = \{u \in \mathcal{B}_k : u = 0 \text{ on } B\}.$$

Then show that if $u : \mathcal{B}_k \rightarrow \mathbb{C}$ is continuous and $\text{supp } u \subset B$,

$$\langle f, u \rangle = \langle E\rho(f), u \rangle,$$

where $\rho(f) = f|_B$ and $E : C^k(B) \rightarrow \mathcal{B}_k$ is an extension operator. For help in constructing E , look ahead to §4 of Chapter 4.)

12. If $u \in \mathcal{E}'(\mathbb{R}^n)$, show that $\hat{u} \in C^\infty(\mathbb{R}^n)$ satisfies an estimate

$$(4.93) \quad |\hat{u}(\xi)| \leq C \langle \xi \rangle^m, \quad \xi \in \mathbb{R}^n,$$

for some $m \in \mathbb{R}$. More generally, show that if a distribution u has support in $B_R = \{x \in \mathbb{R}^n : |x| \leq R\}$, then

$$(4.94) \quad |\hat{u}(\xi + i\eta)| \leq C \langle \xi + i\eta \rangle^m e^{R|\eta|}, \quad \xi, \eta \in \mathbb{R}^n.$$

(Hint. Use (4.39)–(4.41). For (4.94), use the result of Exercise 11.)

13. Given the formula (4.76) for $\mathcal{F}(u*v)$ when $u \in \mathcal{S}(\mathbb{R}^n)$, $v \in \mathcal{S}'(\mathbb{R}^n)$, show that if $u \in \mathcal{S}(\mathbb{R}^n)$ and $v \in \mathcal{E}'(\mathbb{R}^n)$, then $\mathcal{F}(u*v) \in \mathcal{S}(\mathbb{R}^n)$, hence $u*v \in \mathcal{S}(\mathbb{R}^n)$, as asserted above (4.75). (Hint: Use (4.93).)
14. Show that the convolution product extends to

$$\mathcal{E}'(\mathbb{R}^n) \times \mathcal{D}'(\mathbb{R}^n) \longrightarrow \mathcal{D}'(\mathbb{R}^n), \quad \mathcal{E}'(\mathbb{R}^n) \times \mathcal{E}'(\mathbb{R}^n) \rightarrow \mathcal{E}'(\mathbb{R}^n).$$

15. Given $u \in \mathcal{E}'(\mathbb{R}^n)$, show that there exist $k \in \mathbb{Z}^+$, $f \in L^2(\mathbb{R}^n)$ such that $u = (1 - \Delta)^k f$.

(Hint: Obtain $\langle \xi \rangle^{-2k+m} \in L^2(\mathbb{R}^n)$.)

Show that there exist compactly supported $f_\alpha \in L^2(\mathbb{R}^n)$ such that $u = \sum_{|\alpha| \leq k} D^\alpha f_\alpha$.

16. Assume that $u \in \mathcal{S}'(\mathbb{R}^n)$ and $\hat{u}$ is holomorphic in $\mathbb{C}^n$ and satisfies (4.94). Show that u is supported in the ball B_R . This is the distributional version of the Paley-Wiener theorem.

(Hint: Pick $\varphi \in C_0^\infty(\mathbb{R}^n)$, supported in B_1 , $\int \varphi dx = 1$, let $\varphi_\varepsilon(x) = \varepsilon^{-n} \varphi(x/\varepsilon)$, and consider $u_\varepsilon = \varphi_\varepsilon * u \in \mathcal{S}(\mathbb{R}^n)$. Apply Exercises 17 and 18 of §3.)

5. The classical evolution equations

In this section we analyze solutions to the classical heat equation on $\mathbb{R}^+ \times \mathbb{R}^n$, to the Laplace equation on $\mathbb{R}_+^{n+1}$, and to the wave equation on $\mathbb{R} \times \mathbb{R}^n$. We begin with the heat equation for $u = u(t, x)$,

$$(5.1) \quad \frac{\partial u}{\partial t} - \Delta u = 0,$$

where Δ is the Laplace operator on $\mathbb{R}^n$,

$$(5.2) \quad \Delta u = \frac{\partial^2 u}{\partial x_1^2} + \cdots + \frac{\partial^2 u}{\partial x_n^2}.$$

We pose an initial condition

$$(5.3) \quad u(0, x) = f(x).$$

We suppose that $f \in \mathcal{S}'(\mathbb{R}^n)$, and we look for a solution $u \in C^\infty(\overline{\mathbb{R}}^+, \mathcal{S}'(\mathbb{R}^n))$, via Fourier analysis. Taking the Fourier transform of u with respect to x , we obtain the ODE with parameters

$$(5.4) \quad \frac{\partial \hat{u}}{\partial t} = -|\xi|^2 \hat{u}(t, \xi),$$

with initial condition

$$(5.5) \quad \hat{u}(0, \xi) = \hat{f}(\xi).$$

The unique solution to (5.4)–(5.5) is

$$(5.6) \quad \hat{u}(t, \xi) = e^{-t|\xi|^2} \hat{f}(\xi).$$

Set

$$(5.7) \quad G(t, \xi) = e^{-t|\xi|^2}.$$

Note that, for each $t > 0$, $G(t, \cdot) \in \mathcal{S}(\mathbb{R}^n)$. By (4.76), we have

$$(5.8) \quad u(t, x) = (2\pi)^{-n/2} \hat{G}(t, \cdot) * f(x).$$

The computation of the Fourier transform of such a Gaussian function was made in §3. From (3.18), we deduce that

$$(5.9) \quad u(t, x) = p(t, \cdot) * f(x),$$

where

$$(5.10) \quad p(t, x) = (4\pi t)^{-n/2} e^{-|x|^2/4t},$$

for $t > 0$. The function $p(t, x)$ is called the fundamental solution to the heat equation. It satisfies

$$(5.11) \quad \begin{aligned} (\partial/\partial t - \Delta)p &= 0, \quad \text{for } t > 0, \\ \lim_{t \searrow 0} p(t, x) &= \delta(x) \quad \text{in } \mathcal{S}'(\mathbb{R}^n). \end{aligned}$$

We record what Fourier analysis has yielded for the heat equation.

Proposition 5.1. *The heat equation (5.1)–(5.3), with $f \in \mathcal{S}'(\mathbb{R}^n)$, has a unique solution $u \in C^\infty(\overline{\mathbb{R}}^+, \mathcal{S}'(\mathbb{R}^n))$, given by (5.9)–(5.10). The solution is C^∞ on $(0, \infty) \times \mathbb{R}^n$. If $f \in \mathcal{S}(\mathbb{R}^n)$, then $u \in C^\infty(\overline{\mathbb{R}}^+, \mathcal{S}(\mathbb{R}^n))$.*

Note carefully that uniqueness of the solution is asserted only within the class $C^\infty(\overline{\mathbb{R}}^+, \mathcal{S}'(\mathbb{R}^n))$; this entails bounds on the solution considered, near

infinity. If one removes such growth restrictions, uniqueness fails. There exist nontrivial solutions to

$$(5.12) \quad \left(\frac{\partial}{\partial t} - \Delta\right)v = 0, \quad \text{for } t > 0, \quad v(0, x) = 0,$$

outlined in Exercise 2 at the end of this section. For fixed $t > 0$, such solutions blow up too fast to belong to $\mathcal{S}'(\mathbb{R}^n)$.

In view of the boundedness and continuity properties of (5.7), we have the following:

Corollary 5.2. *Suppose $f \in L^2(\mathbb{R}^n)$. Then the solution u to (5.1)–(5.3) of Proposition 5.1 belongs to $C(\overline{\mathbb{R}}^+, L^2(\mathbb{R}^n))$.*

We cannot say that $u \in C^\infty(\overline{\mathbb{R}}^+, L^2(\mathbb{R}^n))$, or even that $\partial u / \partial t$ belongs to $C(\overline{\mathbb{R}}^+, L^2(\mathbb{R}^n))$, in such a case, without further restrictions on f . The appropriate behavior of $\partial^j u / \partial t^j$ in such a case is best described in terms of *Sobolev spaces*, which will be discussed in Chapter 4.

Next we look at the following boundary problem for functions harmonic in an upper half space:

$$(5.13) \quad \left(\frac{\partial^2}{\partial y^2} + \Delta\right)u(y, x) = 0, \quad \text{for } y > 0, \quad x \in \mathbb{R}^n,$$

$$(5.14) \quad u(0, x) = f(x).$$

Here, Δ is given by (5.2). In view of such simple examples as

$$(5.15) \quad u(y, x) = y,$$

which satisfy (5.13) and (5.14) with $f = 0$, we will need to make appropriate restrictions on u in order to obtain uniqueness. As before, we suppose that $f \in \mathcal{S}'(\mathbb{R}^n)$ and look for $u \in C^\infty(\overline{\mathbb{R}}^+, \mathcal{S}'(\mathbb{R}^n))$. Fourier transforming with respect to x gives the second-order ODE, with parameters,

$$(5.16) \quad \left(\frac{d^2}{dy^2} - |\xi|^2\right)\hat{u}(y, \xi) = 0,$$

$$(5.17) \quad \hat{u}(0, \xi) = \hat{f}(\xi).$$

The general solution to (5.16)–(5.17), for any fixed $\xi \neq 0$, is

$$(5.18) \quad \hat{u}(y, \xi) = c_0(\xi)e^{y|\xi|} + c_1(\xi)e^{-y|\xi|},$$

with $c_0(\xi) + c_1(\xi) = \hat{f}(\xi)$. Let us restrict attention to f such that $\hat{f}(\xi)$ is continuous, and look for $\hat{u}(y, \xi)$ continuous in (y, ξ) ; as (5.15) illustrates, things can be more complicated if $\hat{u}(y, \cdot)$ is a singular distribution near $\xi = 0$. In view of the blow-up of $e^{y|\xi|}$ as $y \nearrow \infty$, it is natural to require that $c_0(\xi) = 0$, so

$$(5.19) \quad \hat{u}(y, \xi) = e^{-y|\xi|}\hat{f}(\xi).$$

In partial analogy with Proposition 5.1, we have obtained the following result.

Proposition 5.3. *Let $f \in \mathcal{S}'(\mathbb{R}^n)$, and suppose $\hat{f}$ is continuous. Then there is a unique solution $u(y, x)$ of (5.13) and (5.14), belonging to the space $C^\infty(\overline{\mathbb{R}^+}, \mathcal{S}'(\mathbb{R}^n))$ and satisfying the condition that $\hat{u}(y, \xi)$ is continuous on $\overline{\mathbb{R}^+} \times \mathbb{R}^n$, and furthermore that $u(y, \cdot)$ is a bounded function of y taking values in $\mathcal{S}'(\mathbb{R}^n)$. It is given by (5.19).*

Note that $\hat{f}(\xi)$ is continuous provided f is a finite measure. It is also continuous if $f \in \mathcal{E}'(\mathbb{R}^n)$.

We want to find the “fundamental solution” $P(y, x)$ for (5.13)–(5.14), corresponding to $f = \delta$. In other words,

$$(5.20) \quad \hat{P}(y, \xi) = (2\pi)^{-n/2} e^{-y|\xi|}.$$

This computation is elementary in the case $n = 1$. We have

$$(5.21) \quad \begin{aligned} P(y, x) &= (2\pi)^{-1} \int_{-\infty}^{\infty} e^{-y|\xi| + ix\xi} d\xi \\ &= (2\pi)^{-1} \left[\int_0^{\infty} e^{-y\xi + ix\xi} d\xi + \int_0^{\infty} e^{-y\xi - ix\xi} d\xi \right] \\ &= (2\pi)^{-1} [(y - ix)^{-1} + (y + ix)^{-1}] \\ &= \frac{1}{\pi} \frac{y}{y^2 + x^2}. \end{aligned}$$

For $n > 1$, a direct calculation of such a Fourier transform is not so elementary. One way to perform the computation is to use the following *subordination identity*:

$$(5.22) \quad e^{-yA} = \frac{y}{2\pi^{1/2}} \int_0^{\infty} e^{-y^2/4t} e^{-tA^2} t^{-3/2} dt, \quad A > 0, y > 0.$$

We will give a proof of (5.22) shortly. First we will show how it leads to a computation of the Fourier transform of (5.20). We let $A = |\xi|$, and we use our prior computation of the Fourier transform of $e^{-|\xi|^2}$. Thus, for any $n \geq 1$,

$$(5.23) \quad \begin{aligned} P(y, x) &= (2\pi)^{-n} \int e^{-y|\xi| + ix \cdot \xi} d\xi \\ &= (2\pi)^{-n} \frac{y}{2\pi^{1/2}} \int_0^{\infty} e^{-y^2/4t} \left[\int e^{-t|\xi|^2 + ix \cdot \xi} d\xi \right] t^{-3/2} dt, \end{aligned}$$

and substituting in the calculation (5.7)–(5.10), we have

$$(5.24) \quad \begin{aligned} P(y, x) &= (4\pi)^{-(n+1)/2} y \int_0^\infty e^{-y^2/4t} e^{-|x|^2/4t} t^{-(n+3)/2} dt \\ &= c_n \frac{y}{(y^2 + |x|^2)^{(n+1)/2}}, \end{aligned}$$

where the last integral is evaluated using the substitution $s = 1/t$. The constant c_n is given by

$$(5.25) \quad c_n = \pi^{-(n+1)/2} \Gamma\left(\frac{n+1}{2}\right),$$

where $\Gamma(z) = \int_0^\infty e^{-s} s^{z-1} ds$ is Euler's gamma function, which is discussed in the appendix to this chapter. Note that $c_1 = 1/\pi$, so the calculation (5.24) agrees with (5.21), in the case $n = 1$.

The observation that the calculations (5.21) and (5.24) coincide for $n = 1$ can be used to provide a simple proof of the subordination identity (5.22). Indeed, with $|\xi|$ substituted for A in (5.22), we know that the operation of Fourier multiplication by the left side coincides with the operation of Fourier multiplication by the right side, so the two functions of $|\xi|$ ($\xi \in \mathbb{R}$) must coincide.

There are other proofs of (5.22). In Appendix A we note the equivalence of such an identity and a classical identity involving Euler's gamma function. While the proof of (5.22) given above is complete, it leaves one with an unsatisfied feeling, since the right side of the formula (5.22) seems to have been pulled out of a hat. We want to introduce a setting where such a formula arises naturally, a setting involving the use of operator notations, as follows. For a decent function f defined on $[0, \infty)$, define $f(\sqrt{-\Delta})$ on $\mathcal{S}(\mathbb{R}^n)$, or on $L^2(\mathbb{R}^n)$, or even on $\mathcal{S}'(\mathbb{R}^n)$, when it makes sense, by

$$(5.26) \quad (f(\sqrt{-\Delta})u)^\wedge(\xi) = f(|\xi|)\hat{u}(\xi).$$

Thus, the content of (5.10) is

$$(5.27) \quad e^{t\Delta}\delta(x) = (4\pi t)^{-n/2} e^{-|x|^2/4t},$$

for $t > 0$. The formula (5.24) is a formula for $e^{-y\sqrt{-\Delta}}\delta(x)$, $x \in \mathbb{R}^n$, and the formula (5.21) is the special case of this for $n = 1$.

We will approach the subordination identity via the formula

$$(5.28) \quad (\lambda^2 - \Delta)^{-1} = \int_0^\infty e^{-(\lambda^2 - \Delta)t} dt,$$

for the *resolvent* $(\lambda^2 - \Delta)^{-1}$ of the Laplace operator Δ . This identity follows via the Fourier transform, as in (5.26), from

$$(5.29) \quad (\lambda^2 + |\xi|^2)^{-1} = \int_0^\infty e^{-(\lambda^2 + |\xi|^2)t} dt.$$

In order to derive the subordination identity, we will apply both sides of (5.28) to δ ; we will do this in the special case of $\Delta = d^2/dx^2$ acting on

$\mathcal{S}'(\mathbb{R}^1)$. For the special case $\xi \in \mathbb{R}^1$, we have the Fourier integral formula

$$(5.30) \quad \int_{-\infty}^{\infty} (\lambda^2 + \xi^2)^{-1} e^{ix\xi} d\xi = \frac{\pi}{\lambda} e^{-\lambda|x|} \quad (\lambda > 0),$$

a fact that can be established either by residue calculus or by applying the Fourier inversion formula to the computation (5.21). Thus applying both sides of (5.28) to $\delta \in \mathcal{S}'(\mathbb{R})$ and using $e^{t\Delta}\delta = (4\pi t)^{-1/2} e^{-x^2/4t}$ in this case, we have

$$(5.31) \quad \lambda^{-1} e^{-\lambda|x|} = \frac{1}{\sqrt{\pi}} \int_0^{\infty} e^{-x^2/4t} e^{-\lambda^2 t} t^{-1/2} dt.$$

Making the change of variables $y = |x|$, $A = \lambda$, and taking the y -derivative of the resulting identity give (5.22). Also note that taking the A -derivative of (5.22) gives (5.31). The identity (5.31) is also called the subordination identity. One can see that it arises very naturally in this context, from (5.28). We will return to the calculation of $(\lambda^2 - \Delta)^{-1}\delta(x)$ in case $n > 1$, later in this section.

We next consider the wave equation on $\mathbb{R} \times \mathbb{R}^n$:

$$(5.32) \quad \frac{\partial^2 u}{\partial t^2} - \Delta u = 0,$$

with initial data

$$(5.33) \quad u(0, x) = f(x), \quad u_t(0, x) = g(x).$$

As before, we suppose that f and g belong to $\mathcal{S}'(\mathbb{R}^n)$ and look for a solution u in $C^\infty(\mathbb{R}, \mathcal{S}'(\mathbb{R}^n))$. Taking the Fourier transform with respect to x again yields an ODE with parameters:

$$(5.34) \quad \frac{d^2 \hat{u}}{dt^2} + |\xi|^2 \hat{u} = 0,$$

$$(5.35) \quad \hat{u}(0, \xi) = \hat{f}(\xi), \quad \hat{u}_t(0, \xi) = \hat{g}(\xi).$$

The general solution to (5.28) (for $\xi \neq 0$) is of the form

$$\hat{u}(t, \xi) = c_2(\xi) \sin t|\xi| + c_1(\xi) \cos t|\xi|,$$

which it is convenient to write as

$$\hat{u}(t, \xi) = c_0(\xi) |\xi|^{-1} \sin t|\xi| + c_1(\xi) \cos t|\xi|,$$

since the right side is well defined for any $c_0, c_1 \in \mathcal{S}'(\mathbb{R}^n)$. The initial conditions (5.35) imply $c_1 = \hat{f}$, $c_0 = \hat{g}$, so the solution to (5.34)–(5.35) is

$$(5.36) \quad \hat{u}(t, \xi) = \hat{g}(\xi) |\xi|^{-1} \sin t|\xi| + \hat{f}(\xi) \cos t|\xi|.$$

This is clearly the unique solution in $C^\infty(\mathbb{R}, \mathcal{S}'(\mathbb{R}^n))$, if $f, g \in \mathcal{S}'(\mathbb{R}^n)$. The uniqueness in $C^\infty(\mathbb{R}, \mathcal{S}'(\mathbb{R}^n))$ for general $f, g \in \mathcal{S}'(\mathbb{R}^n)$ will be proved shortly.

If $f = 0$, $g = \delta$, the solution given by (5.36) is called the fundamental solution, or the “Riemann function.” Of course, it is actually a distribution. It is characterized by

$$(5.37) \quad \hat{R}(t, \xi) = (2\pi)^{-n/2} |\xi|^{-1} \sin t|\xi|.$$

We want a direct formula for $R(t, x)$. We will be able to deduce this formula from the formula (5.24) for the Fourier transform of $e^{-y|\xi|}$, via analytic continuation. To bring in the factor $|\xi|^{-1}$, integrate (5.24) with respect to y . Thus, if

$$(5.38) \quad \hat{F}(y, \xi) = (2\pi)^{-n/2} |\xi|^{-1} e^{-y|\xi|},$$

which belongs to $\mathcal{S}'(\mathbb{R}^n)$ for each $y \geq 0$ if $n \geq 2$, we have

$$(5.39) \quad F(y, x) = c'_n (y^2 + |x|^2)^{-(n-1)/2}$$

with

$$(5.40) \quad c'_n = \frac{c_n}{n-1} = \frac{1}{2} \pi^{-(n+1)/2} \Gamma\left(\frac{n-1}{2}\right).$$

This has been verified for real $y > 0$. But (5.38) is holomorphic in y , with values in $\mathcal{S}'(\mathbb{R}^n)$, for all y such that $\operatorname{Re} y > 0$. Also, it is continuous in the right half-plane $\{y \in \mathbb{C} : \operatorname{Re} y \geq 0\}$. In view of the continuity of the Fourier transform on $\mathcal{S}'(\mathbb{R}^n)$, we deduce that if

$$(5.41) \quad \hat{\Phi}(t, \xi) = (2\pi)^{-n/2} |\xi|^{-1} e^{it|\xi|},$$

$t \in \mathbb{R}$, then

$$(5.42) \quad \Phi(t, x) = \lim_{\varepsilon \searrow 0} c'_n (|x|^2 - (t - i\varepsilon)^2)^{-(n-1)/2},$$

the limit existing in $\mathcal{S}'(\mathbb{R}^n)$ for each $t \in \mathbb{R}$, since $\hat{\Phi}(t, \cdot) = \lim_{\varepsilon \searrow 0} \hat{\Phi}(t - i\varepsilon, \cdot)$ in $\mathcal{S}'(\mathbb{R}^n)$. Consequently, for the Riemann function, we have

$$(5.43) \quad R(t, x) = \lim_{\varepsilon \searrow 0} c'_n \operatorname{Im}(|x|^2 - (t - i\varepsilon)^2)^{-(n-1)/2}.$$

Note that if $|x| > |t|$, $\lim_{\varepsilon \searrow 0} (|x|^2 - (t - i\varepsilon)^2)^{-(n-1)/2} = (|x|^2 - t^2)^{-(n-1)/2}$ is real, so

$$(5.44) \quad R(t, x) = 0, \quad \text{for } |x| > |t|.$$

This is a reflection of the *finite propagation speed*, which was discussed in Chapter 2. Note also that if n is odd, then $(n-1)/2$ is an integer, so (5.43) vanishes also for $|x| < |t|$. In other words,

$$(5.45) \quad n \geq 3 \text{ odd} \Rightarrow \operatorname{supp} R(t, \cdot) \subset \{x \in \mathbb{R}^n : |x| = |t|\}.$$

This is the *strict Huygens principle*. Of course, it does not hold when n is even. When $n = 2$, the computation of the limit in (5.43) is elementary.

We have

$$(5.46) \quad \begin{aligned} R(t, x) &= c'_2(t^2 - |x|^2)^{-1/2} \cdot \operatorname{sgn}(t), & \text{for } |x| < |t|, \\ &0, & \text{for } |x| > |t|, \end{aligned}$$

for $n = 2$. For $n = 3$, the Plemelj jump relation (4.29) yields

$$(5.47) \quad R(t, x) = (4\pi t)^{-1} \delta(|x| - |t|).$$

The discussion above has to be modified for $n = 1$, since (5.41) is not locally integrable near $\xi = 0$ in that case. This simple case ($n = 1$) was treated in §1 of Chapter 2; see (1.24)–(1.28) in that chapter.

The solution to (5.32)–(5.33) given by (5.36) can be expressed as

$$(5.48) \quad u(t, \cdot) = R(t, \cdot) * g + \frac{\partial}{\partial t} R(t, \cdot) * f.$$

We record our result on solutions to the wave equation.

Proposition 5.4. *Given $f, g \in \mathcal{S}'(\mathbb{R}^n)$, there is a unique solution $u \in C^\infty(\mathbb{R}, \mathcal{S}'(\mathbb{R}^n))$ to the initial-value problem (5.32)–(5.33). It is given by (5.48).*

Proof. The only point remaining to be established is the uniqueness. Suppose $u \in C^\infty(\mathbb{R}, \mathcal{S}'(\mathbb{R}^n))$ solves (5.32)–(5.33) with $f = g = 0$. Then

$$v(t, \cdot) = u(t, \cdot) * \varphi$$

solves the same equation, for any $\varphi \in C_0^\infty(\mathbb{R}^n)$; $\varphi = \varphi(x)$. We have $v \in C^\infty(\mathbb{R} \times \mathbb{R}^n)$. Thus the energy estimates of Chapter 2 are applicable to v , and we have $v = 0$ everywhere. Taking a sequence $\varphi_j \in C_0^\infty(\mathbb{R}^n)$ approaching δ , we have $v_j \rightarrow u$; since each $v_j = 0$, it follows that $u = 0$, and the proof is complete.

We note that the argument above yields uniqueness for u in the class $C^\infty(\mathbb{R}, \mathcal{D}'(\mathbb{R}^n))$. We also remark that any $u \in \mathcal{D}'(\mathbb{R} \times \mathbb{R}^n)$ solving (5.32) actually belongs to $C^\infty(\mathbb{R}, \mathcal{D}'(\mathbb{R}^n))$, and that (5.48) gives the unique solution to (5.32)–(5.33) for any $f, g \in \mathcal{D}'(\mathbb{R}^n)$. Justification of these statements is left as an exercise.

Returning to the operator notation (5.26), we have

$$(5.49) \quad R(t, x) = (-\Delta)^{-1/2} \sin t\sqrt{-\Delta} \delta(x)$$

and

$$(5.50) \quad \frac{\partial}{\partial t} R(t, x) = \cos t\sqrt{-\Delta} \delta(x).$$

We also denote (5.49) by $R(t)$ and (5.50) by $R'(t)$.

Having introduced in (5.28) the notion of synthesizing some operators from other operators, we want to mention the particular desirability of synthesizing functions of the Laplace operator from the fundamental solution

of the wave equation. If $\varphi(s)$ is an even function, the Fourier inversion formula implies

$$(5.51) \quad \varphi(\sqrt{-\Delta}) = (2\pi)^{-1/2} \int_{-\infty}^{\infty} \hat{\varphi}(t) \cos t\sqrt{-\Delta} dt.$$

Note that

$$(5.52) \quad \cos t\sqrt{-\Delta} u = R'(t) * u,$$

where $R(t)$ is the Riemann function constructed above. We have the following rather general calculation of $\varphi(\sqrt{-\Delta})\delta$, using the formula (5.37) for the Riemann function on $\mathbb{R}^n$.

Proposition 5.5. *Let $\varphi \in \mathcal{S}(\mathbb{R})$ be even. Then, on $\mathbb{R}^n$, we have*

$$(5.53) \quad \varphi(\sqrt{-\Delta})\delta(x) = \frac{1}{\sqrt{2\pi}} \left[-\frac{1}{2\pi r} \frac{\partial}{\partial r} \right]^k \hat{\varphi}(r)$$

if $n = 2k + 1$ is odd, and

$$(5.54) \quad \varphi(\sqrt{-\Delta})\delta(x) = \sqrt{\frac{2}{\pi}} \int_r^{\infty} \left\{ \left[-\frac{1}{2\pi s} \frac{\partial}{\partial s} \right]^k \hat{\varphi}(s) \right\} \frac{s}{\sqrt{s^2 - r^2}} ds$$

if $n = 2k$ is even. Here, $r = |x|$.

Proof. When $n = 3$, we have

$$(5.55) \quad \begin{aligned} \varphi(\sqrt{-\Delta})\delta &= -\frac{2}{\sqrt{2\pi}} \int_0^{\infty} \hat{\varphi}'(t) (4\pi t)^{-1} \delta(r-t) dt \\ &= \frac{1}{\sqrt{2\pi}} \left[-\frac{1}{2\pi r} \frac{\partial}{\partial r} \right] \hat{\varphi}(r), \end{aligned}$$

giving (5.53) in this case. When $n = 2$, we have, from (5.46),

$$(5.56) \quad \varphi(\sqrt{-\Delta})\delta = -\frac{2c'_2}{\sqrt{2\pi}} \int_r^{\infty} \hat{\varphi}'(t) (t^2 - r^2)^{-1/2} dt,$$

giving (5.54) in this case, once one checks that $c'_2 = (1/2)\pi^{-3/2}\Gamma(1/2) = 1/2\pi$. To pass to the general case, we note that if $R_n(t, r)$ denotes the formula (5.43) for the Riemann function, in view of the evaluation of c'_n in (5.40), we have the formal relation

$$(5.57) \quad R_{n+2}(t, r) = \left[-\frac{1}{2\pi r} \frac{\partial}{\partial r} \right] R_n(t, r),$$

so (5.53) and (5.54) follow by induction.

It is clear that Proposition 5.5 holds for a more general class of even functions φ than those in $\mathcal{S}(\mathbb{R})$, by simple limiting arguments. For example, the function $\varphi(s) = (\lambda^2 + s^2)^{-1}$, $\lambda > 0$, giving the resolvent of Δ , can be

treated. We leave the formulation of general results on classes of φ which can be treated as an exercise.

In the case of using Proposition 5.5 to treat the resolvent of Δ , we have the following formula. With $\varphi(s) = (\lambda^2 + s^2)^{-1}$, from (5.30) we have

$$(5.58) \quad \hat{\varphi}(t) = \left(\frac{\pi}{2}\right)^{1/2} \lambda^{-1} e^{-\lambda|t|},$$

to plug into (5.53)–(5.54). For example, for $n = 3$, we have

$$(5.59) \quad (\lambda^2 - \Delta)^{-1} \delta = (4\pi|x|)^{-1} e^{-\lambda|x|} \quad \text{on } \mathbb{R}^3.$$

Note that computing $(\lambda^2 - \Delta)^{-1} \delta$ by evaluating the right side of (5.28) gives in this case

$$(5.60) \quad (\lambda^2 - \Delta)^{-1} \delta = \int_0^\infty e^{-|x|^2/4t} e^{-\lambda^2 t} (4\pi t)^{-3/2} dt;$$

comparing (5.59) and (5.60) again reveals the subordination identity, in the original form (5.22).

The fact that the answer comes out “in closed form” for n odd is a consequence of the strict Huygens principle. For n even, one tends not to get elementary functions. Note that, for $n = 2$, the formula (5.54) gives

$$(5.61) \quad (\lambda^2 - \Delta)^{-1} \delta = c \int_{|x|}^\infty e^{-\lambda s} (s^2 - r^2)^{-1/2} ds \quad \text{on } \mathbb{R}^2;$$

the formula (5.28) gives

$$(5.62) \quad (\lambda^2 - \Delta)^{-1} \delta = \frac{1}{4\pi} \int_0^\infty e^{-|x|^2/4t-t} t^{-1} dt \quad \text{on } \mathbb{R}^2.$$

Both of these integrals can be expressed in terms of the modified Bessel function K_0 ; we say a little more about this in the next section; see (6.46)–(6.54).

In general, the use of results on the wave equation together with (5.51) provides a tool of tremendous power and flexibility in the analysis of numerous functions of the Laplace operator. We will see more of this in Chapter 8.

To end this section, we re-derive formulas for the solution to the wave equation (5.32)–(5.33), and then re-derive the formulas (5.53)–(5.54). Recall that the solution to the wave equation on $\mathbb{R} \times \mathbb{R}^n$ is given by

$$(5.63) \quad u(t, x) = \cos t\sqrt{-\Delta} f(x) + \frac{\sin t\sqrt{-\Delta}}{\sqrt{-\Delta}} g(x).$$

We first derive formulas for these solution operators, in case

$$(5.64) \quad n = 2k + 1,$$

by comparing two formulas for $e^{t\Delta} f(x)$. This approach follows material in [PT].

The first formula for $e^{t\Delta}$ is

$$(5.65) \quad \begin{aligned} e^{t\Delta} f(x) &= (4\pi t)^{-n/2} \int_{\mathbb{R}^n} e^{-|y|^2/4t} f(x-y) dy \\ &= (4\pi t)^{-n/2} A_{n-1} \int_0^\infty \bar{f}_x(r) r^{n-1} e^{-r^2/4t} dr, \end{aligned}$$

where A_{n-1} is the area of the unit sphere S^{n-1} in $\mathbb{R}^n$, and

$$(5.66) \quad \bar{f}_x(r) = \frac{1}{A_{n-1}} \int_{S^{n-1}} f(x + r\omega) dS(\omega).$$

Note that $\bar{f}_x(r)$ is well defined for all $r \in \mathbb{R}$, and $\bar{f}_x(-r) = \bar{f}_x(r)$. The first identity in (4) follows, via Fourier analysis, from the evaluation of the Gaussian integral

$$(5.67) \quad \int_{\mathbb{R}^n} e^{-t|\xi|^2 + ix \cdot \xi} d\xi = \left(\frac{\pi}{t}\right)^{n/2} e^{-|x|^2/4t}.$$

The second identity in (5.65) follows by switching to spherical polar coordinates, $y = r\omega$, and using $dy = r^{n-1} dr dS(\omega)$.

The second formula for $e^{t\Delta}$ is

$$(5.68) \quad e^{t\Delta} f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty \hat{h}_t(s) \cos s\sqrt{-\Delta} f(x) ds,$$

with $h_t(\sigma) = e^{-t\sigma^2}$, hence, by (5), $\hat{h}_t(s) = (2t)^{-1/2} e^{-s^2/4t}$. This is a special case of (5.51).

Setting $4t = 1/\lambda$ and comparing the formulas (5.65) and (5.68), we have (with $v(s, x) = \cos s\sqrt{-\Delta} f(x)$)

$$(5.69) \quad \int_0^\infty v(s, x) e^{-\lambda s^2} ds = \frac{A_{n-1}}{2} \left(\frac{\lambda}{\pi}\right)^{(n-1)/2} \int_0^\infty \bar{f}_x(r) r^{n-1} e^{-\lambda r^2} dr,$$

for all $\lambda > 0$. The key to getting a formula for $v(s, x)$ from this is to make the factor $\lambda^{(n-1)/2}$ on the right side of (5.69) disappear.

Bringing in the hypothesis (5.64), we use the identity

$$(5.70) \quad -\frac{1}{2r} \frac{d}{dr} e^{-\lambda r^2} = \lambda e^{-\lambda r^2}$$

to write the right side of (5.69) as

$$(5.71) \quad C_n \int_0^\infty r^{2k} \bar{f}_x(r) \left(-\frac{1}{2r} \frac{d}{dr}\right)^k e^{-\lambda r^2} dr.$$

Repeated integration by parts shows that this is equal to

$$(5.72) \quad C_n \int_0^\infty r \left(\frac{1}{2r} \frac{d}{dr}\right)^k [r^{2k-1} \bar{f}_x(r)] e^{-\lambda r^2} dr.$$

Now it follows from uniqueness of Laplace transforms (see the exercises) that

$$(5.73) \quad \cos t\sqrt{-\Delta}f(x) = C_n t \left(\frac{1}{2t} \frac{d}{dt} \right)^k [t^{2k-1} \bar{f}_x(t)],$$

for well behaved functions f on $\mathbb{R}^n$, when $n = 2k + 1$. By (5.69), we have

$$(5.74) \quad C_n = \frac{1}{2} \pi^{-(n-1)/2} A_{n-1}.$$

We can also compute C_n directly in (5.73), by considering $f = 1$. Then $\bar{f}_x = 1$ and $u = 1$, so

$$(5.75) \quad 1 = C_n t \left(\frac{1}{2t} \frac{d}{dt} \right)^k t^{2k-1} = C_n \left(k - \frac{1}{2} \right) \left(k - \frac{3}{2} \right) \cdots \frac{1}{2},$$

i.e.,

$$(5.76) \quad C_n = \frac{1}{(k - \frac{1}{2})(k - \frac{3}{2}) \cdots \frac{1}{2}}, \quad n = 2k + 1.$$

This simply means

$$(5.77) \quad A_{2k} = \frac{2\pi^k}{(k - \frac{1}{2})(k - \frac{3}{2}) \cdots \frac{1}{2}},$$

a formula that is frequently derived by looking at Gaussian integrals. See formulas (A.4)–(A.9) at the end of this chapter.

To compute $(\sin t\sqrt{-\Delta})/\sqrt{-\Delta}$, we use

$$(5.78) \quad \frac{\sin t\sqrt{-\Delta}}{\sqrt{-\Delta}} g(x) = \int_0^t \cos s\sqrt{-\Delta} g(x) ds.$$

From (5.73), if $k \geq 1$,

$$(5.79) \quad \cos t\sqrt{-\Delta} g(x) = \frac{C_n}{2} \frac{d}{dt} \left(\frac{1}{2t} \frac{d}{dt} \right)^{k-1} [t^{2k-1} \bar{g}_x(t)],$$

so (5.78) becomes

$$(5.80) \quad \frac{\sin t\sqrt{-\Delta}}{\sqrt{-\Delta}} g(x) = \frac{C_n}{2} \left(\frac{1}{2t} \frac{d}{dt} \right)^{k-1} [t^{2k-1} \bar{g}_x(t)].$$

The formulas (5.73) and (5.80) are for $t > 0$. For arbitrary $t \in \mathbb{R}$, use

$$(5.81) \quad \cos t\sqrt{-\Delta} = \cos(-t)\sqrt{-\Delta}, \quad \sin t\sqrt{-\Delta} = -\sin(-t)\sqrt{-\Delta}.$$

The case $k = 0$ is exceptional. Then (5.79) does not work. Instead, we have

$$(5.82) \quad \cos t\sqrt{-\Delta} g(x) = \frac{1}{2} [g(x+t) + g(x-t)],$$

and (5.78) gives

$$(5.83) \quad \begin{aligned} \frac{\sin t\sqrt{-\Delta}}{\sqrt{-\Delta}} g(x) &= \frac{1}{2} \int_0^t [g(x+s) + g(x-s)] ds \\ &= \frac{1}{2} \int_{-t}^t g(x+s) ds, \end{aligned}$$

for $n = 1$.

Let us take another look at the case $k = 1$, when $n = 3$. From (5.76), $C_2 = 2$, and then (5.80) gives

$$(5.84) \quad \begin{aligned} \frac{\sin t\sqrt{-\Delta}}{\sqrt{-\Delta}} g(x) &= t\bar{g}_x(t) \\ &= \frac{t}{4\pi} \int_{S^2} g(x+t\omega) dS(\omega) \\ &= \frac{1}{4\pi t} \int_{|y|=|t|} g(x+y) dS(y), \end{aligned}$$

which is equivalent to (5.47).

To solve the wave equation (5.32)–(5.33) for $u = u(t, x)$, $t \in \mathbb{R}$, $x \in \mathbb{R}^n$, $n = 2k$, we can use the following device, known as the method of descent. Set

$$(5.85) \quad F(x, x_{n+1}) = f(x), \quad G(x, x_{n+1}) = g(x),$$

and solve for $U = U(t, x, x_{n+1})$ the wave equation

$$(5.86) \quad \partial_t^2 U - \Delta_{n+1} U = 0, \quad U(0) = F, \quad \partial_t U(0) = G.$$

Then U is independent of x_{n+1} and

$$(5.87) \quad u(t, x) = U(t, x, 0).$$

In particular,

$$(5.88) \quad \cos t\sqrt{-\Delta} f(x) = \cos t\sqrt{-\Delta_{n+1}} F(x, 0),$$

and

$$(5.89) \quad \frac{\sin t\sqrt{-\Delta}}{\sqrt{-\Delta}} g(x) = \frac{\sin t\sqrt{-\Delta_{n+1}}}{\sqrt{-\Delta_{n+1}}} G(x, 0).$$

Using the formula (5.73) for waves on $\mathbb{R}^{n+1} = \mathbb{R}^{2k+1}$, we have, for $v(t, x) = \cos t\sqrt{-\Delta} f(x)$,

$$(5.90) \quad v(t, x) = C_{n+1} t \left(\frac{1}{2t} \frac{d}{dt} \right)^k [t^{2k-1} f_x^\#(t)],$$

where

$$\begin{aligned}
 f_x^\#(r) &= \overline{F}_{(x,0)}(r) \\
 &= \frac{1}{A_n} \int_{S^n} F((x,0) + r\omega) dS(\omega) \\
 &= \frac{2}{A_n} \int_{S_+^n} f(x + r\omega^b) dS(\omega),
 \end{aligned}
 \tag{5.91}$$

with

$$S_+^n = \{\omega = (\omega^b, \omega_{n+1}) \in S^n : \omega_{n+1} \geq 0\}.$$

Here A_n is the n -dimensional area of S^n and, we recall,

$$C_{n+1} = \frac{1}{2} \pi^{-n/2} A_n.$$

Note that $f_x^\#(r) = f_x^\#(-r)$. To proceed, map $B = \{y \in \mathbb{R}^n : |y| \leq 1\}$ to S_+^n by

$$y \mapsto (y, \psi(y)), \quad \psi(y) = \sqrt{1 - |y|^2}.$$

Then

$$dS(\omega) = \sqrt{1 + |\nabla \psi(y)|^2} dy = \frac{dy}{\sqrt{1 - |y|^2}},$$

and we get

$$f_x^\#(r) = \frac{2}{A_n} \int_{|y| \leq 1} f(x + ry) \frac{dy}{\sqrt{1 - |y|^2}}.$$

Using the identity

$$\int_{|y| \leq 1} h(y) dy = \int_0^1 \int_{S^{n-1}} H(\rho\omega) \rho^{n-1} dS(\omega) d\rho,$$

we get (for $r > 0$)

$$\begin{aligned}
 f_x^\#(r) &= \frac{2}{A_n} \int_0^1 \int_{S^{n-1}} f(x + r\rho\omega) \frac{\rho^{n-1}}{\sqrt{1 - \rho^2}} dS(\omega) d\rho \\
 &= 2 \frac{A_{n-1}}{A_n} \int_0^1 \overline{f}_x(\rho r) \frac{\rho^{n-1}}{\sqrt{1 - \rho^2}} d\rho \\
 &= 2 \frac{A_{n-1}}{A_n} \frac{1}{r^{n-1}} \int_0^r \overline{f}_x(s) \frac{s^{n-1}}{\sqrt{r^2 - s^2}} ds.
 \end{aligned}
 \tag{5.98}$$

Plugging this in (5.90), we get, for a function f on $\mathbb{R}^n = \mathbb{R}^{2k}$, $t > 0$,

$$(5.99) \quad \cos t\sqrt{-\Delta}f(x) = 2\frac{A_{n-1}}{A_n}C_{n+1}t\left(\frac{1}{2t}\frac{d}{dt}\right)^k \int_0^t \bar{f}_x(s) \frac{s^{n-1}}{\sqrt{t^2-s^2}} ds.$$

Note that

$$(5.100) \quad 2\frac{A_{n-1}}{A_n}C_{n+1} = \pi^{-n/2}A_{n-1} = \frac{2}{\Gamma(n/2)} = \frac{2}{\Gamma(k)}.$$

Similarly, we have

$$(5.101) \quad \frac{\sin t\sqrt{-\Delta}}{\sqrt{-\Delta}}g(x) = \frac{C_{n+1}}{2}\left(\frac{1}{2t}\frac{d}{dt}\right)^{k-1} [t^{2k-1}g_x^\#(t)],$$

where $g_x^\#(|t|)$ is as in (5.86), with G in place of F , hence as in (5.98), with $\bar{g}_x(s)$ in place of $\bar{f}_x(s)$. Consequently, for $t > 0$,

$$(5.102) \quad \frac{\sin t\sqrt{-\Delta}}{\sqrt{-\Delta}}g(x) = \frac{A_{n-1}}{A_n}C_{n+1}\left(\frac{1}{2t}\frac{d}{dt}\right)^{k-1} \int_0^t \bar{g}_x(s) \frac{s^{n-1}}{\sqrt{t^2-s^2}} ds,$$

and $(A_{n-1}/A_n)C_{n+1} = 1/\Gamma(k)$.

If we specialize to $n = 2$ ($k = 1$), we get

$$(5.103) \quad \begin{aligned} \frac{\sin t\sqrt{-\Delta}}{\sqrt{-\Delta}}g(x) &= \int_0^t \bar{g}_x(s) \frac{s}{\sqrt{t^2-s^2}} ds \\ &= \frac{1}{2\pi} \int_0^t \int_{S^1} g(x-s\omega) \frac{s}{\sqrt{t^2-s^2}} dS(\omega) ds \\ &= \frac{1}{2\pi} \int_{|y|\leq t} \frac{g(x-y)}{\sqrt{t^2-|y|^2}} dy, \end{aligned}$$

which is equivalent to (5.46).

We turn to a re-derivation of the formulas in Proposition 5.5. As before, we use (5.51), plus formulas for $\cos t\sqrt{-\Delta}$. This time, we combine (5.51) with the formula (5.79), for a function f on $\mathbb{R}^n = \mathbb{R}^{2k+1}$, where C_n is given by (5.74). We get

$$(5.104) \quad \begin{aligned} \varphi(\sqrt{-\Delta})f(x) &= \frac{C_n}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{\varphi}(t) \left(\frac{d}{dt}\frac{1}{2t}\right)^k [t^{n-1}\bar{f}_x(t)] dt \\ &= \frac{C_n}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(-\frac{1}{2t}\frac{d}{dt}\right)^k \hat{\varphi}(t) \cdot t^{n-1}\bar{f}_x(t) dt. \end{aligned}$$

Now

$$\begin{aligned}
 (5.105) \quad & \int_0^\infty \Phi(r) r^{n-1} \bar{f}_x(r) dr \\
 &= \frac{1}{A_{n-1}} \int_0^\infty \int_{S^{n-1}} f(x - r\omega) \Phi(r) r^{n-1} dS(\omega) dr \\
 &= \frac{1}{A_{n-1}} \int_{\mathbb{R}^n} f(x - y) \Phi(|y|) dy.
 \end{aligned}$$

Hence, using (5.74), we obtain from (5.104) that

$$(5.106) \quad \varphi(\sqrt{-\Delta})f(x) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}^n} \Phi_n(|y|) f(x - y) dy,$$

for $n = 2k + 1$, where

$$(5.107) \quad \Phi_{2k+1}(r) = \left(-\frac{1}{2\pi r} \frac{d}{dr} \right)^k \hat{\varphi}(r).$$

Another way to write (5.106) is

$$(5.108) \quad \varphi(\sqrt{-\Delta})\delta(x) = \frac{1}{\sqrt{2\pi}} \Phi_n(|x|), \quad x \in \mathbb{R}^n,$$

which is equivalent to (5.53).

REMARK. The case $k = 0$ of (5.107) can be seen directly by the Fourier inversion theorem, without use of the calculations (5.104)–(5.105).

We seek an analogous formula for $\varphi(\sqrt{-\Delta})f(x)$ when f is a function on $\mathbb{R}^n$ with $n = 2k$. We get this by the following extension of the method of descent, which gives

$$(5.109) \quad \cos t\sqrt{-\Delta}f(x) = \cos t\sqrt{-\Delta_{n+1}}F(x, 0),$$

with

$$(5.110) \quad F(x, x_{n+1}) = f(x).$$

From this and (5.51), we get

$$\begin{aligned}
 (5.111) \quad & \varphi(\sqrt{-\Delta})f(x) = \varphi(\sqrt{-\Delta_{n+1}})F(x, 0) \\
 &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}^{2k+1}} \Phi_{2k+1}(|(y, y_{n+1})|) F(x - y, y_{n+1}) dy dy_{n+1} \\
 &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}^{2k+1}} \Phi_{2k+1}((|y|^2 + s^2)^{1/2}) f(x - y) dy ds,
 \end{aligned}$$

or

$$(5.112) \quad \varphi(\sqrt{-\Delta})f(x) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}^n} \Phi_{2k}(|y|)f(x-y) dy,$$

where

$$(5.113) \quad \Phi_{2k}(r) = \int_{-\infty}^{\infty} \Phi_{2k+1}(\sqrt{r^2 + s^2}) ds,$$

and Φ_{2k+1} is given by (5.107). The change of variable $t = \sqrt{r^2 + s^2}$ gives

$$(5.114) \quad \Phi_{2k}(r) = 2 \int_r^{\infty} \Phi_{2k+1}(t) \frac{t}{\sqrt{t^2 - r^2}} dt.$$

Another way to write (5.112) is

$$(5.115) \quad \varphi(\sqrt{-\Delta})\delta(x) = \sqrt{\frac{2}{\pi}} \Phi_{2k}(|x|),$$

which is equivalent to (5.54).

Exercises

1. A function $g \in C^\infty(\mathbb{R}^n)$ is said to be in the Gevrey class $G^\sigma(\mathbb{R}^n)$ provided, for each compact $K \subset \mathbb{R}^n$, there exist C and R , such that

$$(5.116) \quad |D^\alpha g(x)| \leq CR^k k^{\sigma k}, \quad |\alpha| = k, \quad x \in K.$$

The class $G^1(\mathbb{R}^n)$ is equal to the space of real-analytic functions on $\mathbb{R}^n$. If $\sigma > 1$, show that there exist *compactly supported* elements of $G^\sigma(\mathbb{R}^n)$, not identically zero.

Remark: This is part of the Denjoy-Carleman theorem; see [Ru].

2. Consider the following “sideways heat equation” for $u = u(t, x)$ on $\mathbb{R} \times \mathbb{R}$:

$$u_t = u_{xx}, \quad u(t, 0) = g(t), \quad u_x(t, 0) = 0.$$

Show that if $g \in G^\sigma(\mathbb{R})$ for some $\sigma \in (1, 2)$, then a solution on all of $\mathbb{R} \times \mathbb{R}$ is given by the convergent series

$$(5.117) \quad u(t, x) = \sum_{k=0}^{\infty} \frac{x^{2k}}{(2k)!} g^{(k)}(t),$$

which is the power series for the “formal” object $(\cos x \sqrt{-\partial/\partial t})g$. Using Exercise 1, find nontrivial solutions to $u_t = u_{xx}$ which are supported in a strip $a \leq t \leq b$. This construction is due to J. Rauch.

(*Hint:* To prove convergence, use Stirling’s formula:

$$(5.118) \quad n! \sim (2\pi n)^{1/2} \cdot e^{-n} \cdot n^n \quad \text{as } n \rightarrow \infty.)$$

3. Given the formula (5.37) for $\hat{R}(t, \xi)$, that is,

$$\hat{R}(t, \xi) = (2\pi)^{-n/2} |\xi|^{-1} \sin t|\xi|, \quad \xi \in \mathbb{R}^n,$$

show that the fact that $R(t, \cdot) \in \mathcal{S}'(\mathbb{R}^n)$ is supported in $B_{|t|} = \{x \in \mathbb{R}^n : |x| \leq |t|\}$ follows from the Paley-Wiener theorem for distributions, given in Exercise 16 of §4.

Exercises 4–7 provide justification for passing from (5.69)–(5.71) to (5.73).

4. Take $v \in L^1(\mathbb{R}^+)$ and assume

$$\int_0^\infty v(s) e^{-\lambda s^2} ds = 0, \quad \forall \lambda > 0.$$

Deduce that $v \equiv 0$. (*Hint.* Use the Stone-Weierstrass theorem to show that if $e_\lambda(s) = e^{-\lambda s^2}$, then the linear span of $\{e_\lambda : \lambda > 0\}$ is dense in $C_*(\overline{\mathbb{R}^+})$, the space of continuous functions on $\mathbb{R}^+ = [0, \infty)$ that vanish at ∞ . Hence the hypothesis implies $\int_0^\infty v(s) f(s) ds = 0$ for all $f \in C_*(\overline{\mathbb{R}^+})$.)

5. Show that if instead we assume $v \in L^\infty(\mathbb{R}^+)$, the result of Exercise 4 still holds. (*Hint.* Consider $v_\varepsilon(s) = e^{-\varepsilon s^2} v(s)$.)
 6. Show that if $f, g \in \mathcal{S}(\mathbb{R}^n)$, then the solution u to (5.32)–(5.33) is bounded and continuous on $\mathbb{R} \times \mathbb{R}^n$. Hence deduce the validity of (5.73) for $f \in \mathcal{S}(\mathbb{R}^n)$.
 7. Extend the range of validity of (5.73) from $f \in \mathcal{S}(\mathbb{R}^n)$ to other function spaces, including $f \in C^\infty(\mathbb{R}^n)$.

6. Radial distributions, polar coordinates, and Bessel functions

The rotational invariance of the Laplace operator on $\mathbb{R}^n$ directly suggests the use of polar coordinates; one has

$$(6.1) \quad \Delta = \frac{\partial^2}{\partial r^2} + \frac{n-1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \Delta_S,$$

where Δ_S is the Laplace operator on the unit sphere S^{n-1} . This formula has been used in (4.56) and follows from the formula given in Chapter 2 for the Laplace operator in a general coordinate system; see (4.4) in that chapter.

Related is the fact that in treating the equations of §5 via Fourier analysis, one computes the Fourier transforms of various radial functions, such Fourier transforms also being radial functions (or rotationally invariant tempered distributions). Bessel functions arise naturally in either approach, and we will develop a little of the theory of Bessel functions here. More results on Bessel functions will appear in Chapter 8, which discusses spectral theory, and in Chapter 9, which treats scattering theory. One can find a great deal more material on this subject in the treatise [Wat].

We begin by considering the Fourier transform of a radial function, $F(x) = f(r)$, $r = |x|$. We have

$$(6.2) \quad \hat{F}(\xi) = (2\pi)^{-n/2} \int_0^\infty f(r) \psi_n(r|\xi|) r^{n-1} dr,$$

where

$$(6.3) \quad \psi_n(|\xi|) = \Psi_n(\xi)$$

with

$$(6.4) \quad \Psi_n(\xi) = \int_{S^{n-1}} e^{i\xi \cdot \omega} dS(\omega).$$

In other words, with A_{n-2} the volume of S^{n-2} ,

$$(6.5) \quad \psi_n(r) = A_{n-2} \int_{-1}^1 e^{irs} (1-s^2)^{(n-3)/2} ds.$$

From (A.4) we have $A_{n-2} = 2\pi^{(n-1)/2} / \Gamma((n-1)/2)$. It is common to write

$$(6.6) \quad \psi_n(r) = (2\pi)^{n/2} r^{1-n/2} J_{n/2-1}(r),$$

where, for general ν satisfying $\operatorname{Re} \nu > -1/2$, the Bessel function $J_\nu(z)$ is defined to be

$$(6.7) \quad J_\nu(z) = \left[\Gamma\left(\frac{1}{2}\right) \Gamma\left(\nu + \frac{1}{2}\right) \right]^{-1} \left(\frac{z}{2}\right)^\nu \int_{-1}^1 (1-t^2)^{\nu-1/2} e^{izt} dt.$$

For example, $\psi_2(r) = 2\pi J_0(r)$. Since $(1-t^2)^{\nu-1/2}$ is even in t , one can replace e^{izt} by $\cos zt$ in this formula. Now, (6.2) becomes

$$(6.8) \quad \hat{F}(\xi) = |\xi|^{1-n/2} \int_0^\infty f(r) J_{n/2-1}(r|\xi|) r^{n/2} dr.$$

We want to consider the ODE, known as Bessel's equation, solved by $J_\nu(r)$. First we consider the case $\nu = n/2 - 1$. Since Ψ_n is the Fourier transform of a measure supported on the unit sphere, we have that

$$(6.9) \quad (\Delta + 1)\Psi_n = 0.$$

Using the polar coordinate expression (6.1) for Δ , we have

$$(6.10) \quad \left(\frac{d^2}{dr^2} + \frac{n-1}{r} \frac{d}{dr} + 1 \right) \psi_n(r) = 0.$$

Substituting (6.6) yields Bessel's equation

$$(6.11) \quad \left[\frac{d^2}{dr^2} + \frac{1}{r} \frac{d}{dr} + \left(1 - \frac{\nu^2}{r^2} \right) \right] J_\nu(r) = 0$$

in case $\nu = n/2 - 1$. We want to verify this for all ν , from the integral formula (6.7). This is an exercise, but we will present the details to one

approach, which yields further interesting identities for Bessel functions. For notational simplicity, let us set

$$c(\nu) = \left[\Gamma\left(\frac{1}{2}\right) \Gamma\left(\nu + \frac{1}{2}\right) \right]^{-1} \cdot 2^{-\nu}.$$

Differentiating (6.7) with respect to z yields

$$(6.12) \quad \begin{aligned} \frac{d}{dz} J_\nu(z) &= \left(\frac{\nu c(\nu)}{z} \right) z^\nu \int_{-1}^1 e^{izt} (1-t^2)^{\nu-1/2} dt \\ &\quad + ic(\nu) z^\nu \int_{-1}^1 e^{izt} t (1-t^2)^{\nu-1/2} dt. \end{aligned}$$

The first term on the right is equal to $(\nu/z) J_\nu(z)$. The second is equal to

$$\begin{aligned} &\frac{c(\nu)}{z} z^\nu \int_{-1}^1 \left(\frac{d}{dt} e^{izt} \right) t (1-t^2)^{\nu-1/2} dt \\ &= -\frac{c(\nu)}{z} z^\nu \int_{-1}^1 e^{izt} \left[(1-t^2)^{\nu-1/2} - (2\nu-1)t^2 (1-t^2)^{\nu-1-1/2} \right] dt \\ &= -\frac{c(\nu)}{z} z^\nu \int_{-1}^1 e^{izt} \left[2\nu(1-t^2)^{\nu-1/2} - (2\nu-1)(1-t^2)^{\nu-1-1/2} \right] dt \\ &= -\frac{2\nu}{z} J_\nu(z) + \frac{(2\nu-1)c(\nu)}{c(\nu-1)} J_{\nu-1}(z). \end{aligned}$$

Since $c(\nu)/c(\nu-1) = 1/(2\nu-1)$, we have the formula

$$(6.13) \quad \frac{d}{dz} J_\nu(z) = -\frac{\nu}{z} J_\nu(z) + J_{\nu-1}(z),$$

or

$$(6.14) \quad \left(\frac{d}{dz} + \frac{\nu}{z} \right) J_\nu(z) = J_{\nu-1}(z).$$

As we have stated, the formula (6.7) for $J_\nu(z)$ is convergent for $\operatorname{Re} \nu > -1/2$. The formula (6.14) provides an analytic continuation for all complex ν . In fact, one can see directly that the integral in (6.7) is meromorphic in ν , with simple poles at $\nu + 1/2 = -1, -2, \dots$. The factor $\Gamma(\nu + 1/2)^{-1}$ cancels these poles. This serves to explain the desirability of throwing in this factor in the definition (6.7) of $J_\nu(z)$. Of course, the factor $\Gamma(1/2)^{-1}$ is more arbitrary.

Next, we note that

$$\begin{aligned} J_{\nu+1}(z) &= c(\nu+1) z^{\nu+1} \int_{-1}^1 e^{izt} (1-t^2)^{\nu+1-1/2} dt \\ &= -ic(\nu+1) z^\nu \int_{-1}^1 \left(\frac{d}{dt} e^{izt} \right) (1-t^2)^{\nu+1-1/2} dt \\ &= -ic(\nu+1)(2\nu+1) z^\nu \int_{-1}^1 e^{izt} t (1-t^2)^{\nu-1/2} dt, \end{aligned}$$

and since $c(\nu + 1) = c(\nu)/(2\nu + 1)$, this is equal to the negative of the second term on the right in (6.12). Hence we have

$$(6.15) \quad \left(\frac{d}{dz} - \frac{\nu}{z}\right)J_\nu(z) = -J_{\nu+1}(z),$$

complementing (6.14). Putting together (6.14) and (6.15), we have

$$(6.16) \quad \left(\frac{d}{dz} - \frac{\nu-1}{z}\right)\left(\frac{d}{dz} + \frac{\nu}{z}\right)J_\nu(z) = -J_\nu(z),$$

which is equivalent to Bessel's equation, (6.11). Note that adding and subtracting (6.14) and (6.15) produce the identities

$$(6.17) \quad \begin{aligned} 2J'_\nu(z) &= J_{\nu-1}(z) - J_{\nu+1}(z), \\ \frac{2\nu}{z}J_\nu(z) &= J_{\nu-1}(z) + J_{\nu+1}(z). \end{aligned}$$

Note that, by analytic continuation, $J_{-\nu}(z)$ is also a solution to Bessel's equation. This equation, for each ν , has a two-dimensional solution space. We will examine when $J_\nu(z)$ and $J_{-\nu}(z)$ are linearly independent. First, we will obtain a power-series expansion for $J_\nu(z)$. This is done by replacing e^{itz} by its power-series expansion in (6.7). To simplify the expression for the coefficients, one uses identities for the beta function and the gamma function established in Appendix A. From (6.7), we have

$$(6.18) \quad J_\nu(z) = \left[\Gamma\left(\frac{1}{2}\right)\Gamma\left(\nu + \frac{1}{2}\right)\right]^{-1} \left(\frac{z}{2}\right)^\nu \sum_{k=0}^{\infty} \frac{1}{(2k)!} \int_{-1}^1 (izt)^{2k} (1-t^2)^{\nu-1/2} dt.$$

The identity (A.24) implies

$$\int_{-1}^1 t^{2k} (1-t^2)^{\nu-1/2} dt = \frac{\Gamma(k + \frac{1}{2})\Gamma(\nu + \frac{1}{2})}{\Gamma(k + \nu + 1)},$$

so

$$J_\nu(z) = \frac{(z/2)^\nu}{\Gamma(\frac{1}{2})\Gamma(\nu + \frac{1}{2})} \sum_{k=0}^{\infty} \frac{1}{(2k)!} (iz)^{2k} \frac{\Gamma(k + \frac{1}{2})\Gamma(\nu + \frac{1}{2})}{\Gamma(k + \nu + 1)}.$$

Setting $(2k)! = \Gamma(2k + 1)$, and using the duplication formula (A.22), which implies

$$\frac{\Gamma(k + \frac{1}{2})}{\Gamma(\frac{1}{2})\Gamma(2k + 1)} = \frac{2^{-2k}}{\Gamma(k + 1)},$$

we obtain the formula

$$(6.19) \quad J_\nu(z) = \left(\frac{z}{2}\right)^\nu \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\Gamma(k + \nu + 1)} \left(\frac{z}{2}\right)^{2k}.$$

This follows from (6.18) if $\operatorname{Re} \nu > -1/2$, and then for general ν by analytic continuation. In particular, we note the leading behavior as $z \rightarrow 0$,

$$(6.20) \quad J_\nu(z) = \frac{z^\nu}{2^\nu \Gamma(\nu+1)} + O(z^{\nu+1})$$

and

$$(6.21) \quad J'_\nu(z) = \frac{z^{\nu-1}}{2^\nu \Gamma(\nu)} + O(z^\nu).$$

The leading coefficients are nonzero as long as ν is not a negative integer (or 0, for (6.21)).

From the expression (6.19) it is clear that $J_\nu(z)$ and $J_{-\nu}(z)$ are linearly independent provided ν is not an integer. On the other hand, comparison of power series shows

$$(6.22) \quad J_{-n}(z) = (-1)^n J_n(z), \quad n = 0, 1, 2, \dots$$

We want to construct a basis of solutions to Bessel's equation, uniformly good for all ν . This construction can be motivated by a calculation of the *Wronskian*.

Generally, for a pair of solutions u_1 and u_2 to a second-order ODE

$$a(z)u'' + b(z)u' + c(z)u = 0,$$

u_1 and u_2 are linearly independent if and only if their Wronskian

$$(6.23) \quad W(z) = W(u_1, u_2)(z) = u_1 u'_2 - u_2 u'_1$$

is nonvanishing. Note that the Wronskian satisfies the first-order ODE

$$(6.24) \quad W'(z) = -\frac{b}{a} W(z).$$

In the case of Bessel's equation, (6.11), this becomes

$$(6.25) \quad W'(z) = -\frac{W(z)}{z},$$

so

$$(6.26) \quad W(z) = \frac{K}{z},$$

for some K (independent of z , but perhaps depending on ν). If $u_1 = J_\nu$, $u_2 = J_{-\nu}$, we can compute K by considering the limiting behavior of $W(z)$ as $z \rightarrow 0$. From (6.20) and (6.21), we get

$$(6.27) \quad W(J_\nu, J_{-\nu})(z) = -\left[\frac{1}{\Gamma(\nu)\Gamma(1-\nu)} - \frac{1}{\Gamma(\nu+1)\Gamma(-\nu)} \right] \frac{1}{z} = -2 \frac{\sin \pi \nu}{\pi z},$$

making use of the identity (A.10). This recaptures the observation that J_ν and $J_{-\nu}$ are linearly independent, and consequently a basis of solutions to (6.11), if and only if ν is not an integer.

To construct a basis of solutions uniformly good for all ν , it is natural to set

$$(6.28) \quad Y_\nu(z) = \frac{J_\nu(z) \cos \pi\nu - J_{-\nu}(z)}{\sin \pi\nu}$$

when ν is not an integer, and define

$$(6.29) \quad Y_n(z) = \lim_{\nu \rightarrow n} Y_\nu(z) = \frac{1}{\pi} \left[\frac{\partial J_\nu(z)}{\partial \nu} - (-1)^n \frac{\partial J_{-\nu}(z)}{\partial \nu} \right] \Big|_{\nu=n}.$$

We have

$$(6.30) \quad W(J_\nu, Y_\nu)(z) = \frac{2}{\pi z},$$

for all ν . Another important pair of solutions to Bessel's equation is the pair of Hankel functions

$$(6.31) \quad H_\nu^{(1)}(z) = J_\nu(z) + iY_\nu(z), \quad H_\nu^{(2)}(z) = J_\nu(z) - iY_\nu(z).$$

For $H_\nu^{(1)}$, there is the integral formula

$$(6.32) \quad H_\nu^{(1)}(z) = \frac{2e^{-\pi i \nu}}{i\sqrt{\pi}\Gamma(\nu + \frac{1}{2})} \left(\frac{z}{2}\right)^\nu \int_1^\infty e^{izt} (t^2 - 1)^{\nu-1/2} dt,$$

for $\text{Re } \nu > -1/2$, $\text{Im } z > 0$. Another formula, valid for $\text{Re } \nu > \frac{1}{2}$ and $\text{Re } z > 0$, is

$$(6.33) \quad H_\nu^{(1)}(z) = \left(\frac{2}{\pi z}\right)^{1/2} \frac{e^{i(z-\pi\nu/2-\pi/4)}}{\Gamma(\nu + \frac{1}{2})} \int_0^\infty e^{-s} s^{\nu-1/2} \left(1 - \frac{s}{2iz}\right)^{\nu-1/2} ds.$$

To prove these identities, one can show as above that each of the right sides of (6.32) and (6.33) satisfies the same recursion formulas as $J_\nu(z)$ and hence solves the Bessel equation; thus it is a linear combination of $J_\nu(z)$ and $Y_\nu(z)$. The coefficients can be found by examining the limiting behavior as $z \rightarrow 0$, to establish the asserted identity. Hankel functions are important in scattering theory; see Chapter 9.

It is worth pointing out that the Bessel functions $J_{k+1/2}(z)$, etc., for k an integer, are elementary functions, particularly since they arise in analysis on odd-dimensional Euclidean space. For $\nu = k + 1/2$, the integrand in (6.7) involves $(1-t^2)^k$, so the integral can be evaluated explicitly. We have, in particular,

$$(6.34) \quad J_{1/2}(z) = \left(\frac{2}{\pi z}\right)^{1/2} \sin z.$$

Then (6.14) gives

$$(6.35) \quad J_{-1/2}(z) = \left(\frac{2}{\pi z}\right)^{1/2} \cos z,$$

which by (6.28) is equal to $-Y_{1/2}(z)$. Applying (6.16) and (6.14) repeatedly gives

$$(6.36) \quad J_{k+1/2}(z) = (-1)^k \left\{ \prod_{j=1}^k \left(\frac{d}{dz} - \frac{j - \frac{1}{2}}{z} \right) \right\} \frac{\sin z}{\sqrt{2\pi z}}$$

and the same sort of formula for $J_{-k-1/2}(z)$, with the $(-1)^k$ removed, and $\sin z$ replaced by $\cos z$. Similarly,

$$(6.37) \quad H_{1/2}^{(1)}(z) = -i \left(\frac{2}{\pi z} \right)^{1/2} e^{iz},$$

with a formula for $H_{k+1/2}^{(1)}(z)$ similar to (6.36).

We now make contact between the formulas (6.2)–(6.6) and some of the formulas of §5, particularly from Proposition 5.5. Note that if $F(x) = f(|x|)$, then

$$(6.38) \quad \hat{F}(x) = (2\pi)^{n/2} f(\sqrt{-\Delta}) \delta.$$

Thus, as in (5.51), we have

$$(6.39) \quad \hat{F}(x) = (2\pi)^{n/2-1} \iint f(r) e^{itr} R'_n(t, x) dt dr,$$

where $R'_n(t, x) = (\partial/\partial t)R_n(t, x)$, and $R_n(t, x)$ is the Riemann function given by (5.43). Comparison with (6.2) gives

$$\psi_n(r|x|) = (2\pi)^{n-1} r^{1-n} \int_{-\infty}^{\infty} e^{itr} R'_n(t, x) dt$$

or, equivalently,

$$(6.40) \quad J_{n/2-1}(r|x|) = 2(2\pi)^{n/2-1} \left(\frac{|x|}{r} \right)^{n/2-1} \int_{-\infty}^{\infty} (\sin tr) R_n(t, x) dt.$$

Note that using $R_3(t, x) = (4\pi t)^{-1} \delta(|t| - |x|)$ gives again the formula (6.34) for $J_{1/2}(r)$. Note also that the recursive formula

$$R_{n+2}(t, s) = -\frac{1}{2\pi s} \frac{\partial}{\partial s} R_n(t, s)$$

used in the proof of Proposition 5.5, when substituted into (6.40), gives rise to the formula (6.15), in the case $\nu = n/2 - 1$.

Instead of synthesizing functions of Δ via the formula (5.51), we could use

$$(6.41) \quad g(-\Delta) = (2\pi)^{-1/2} \int \hat{g}(t) e^{-it\Delta} dt,$$

where the operator $e^{-it\Delta}$ is obtained from the solution operator $e^{t\Delta}$ to the heat equation by analytic continuation:

$$(6.42) \quad e^{-it\Delta} \delta(x) = (-4\pi it)^{-n/2} e^{|x|^2/4it}, \quad t \neq 0.$$

If $f(r) = g(r^2)$, with g real-valued and even, we get

$$(6.43) \quad \hat{g}(t) = \frac{4}{\sqrt{2\pi}} \int_0^\infty f(r) (\cos r^2 t) r \, dr$$

and hence

$$(6.44) \quad g(-\Delta)\delta = \frac{2}{\pi} \int_{-\infty}^\infty \int_0^\infty (-4\pi it)^{-n/2} (\cos r^2 t) e^{|x|^2/4it} f(r) r \, dr \, dt.$$

Comparison of this with (6.2) gives

$$(6.45) \quad \psi_n(r|x|) = c_n r^{2-n} \int_{-\infty}^\infty (\cos r^2 t) e^{|x|^2/4it} t^{-n/2} \, dt,$$

where, for n odd, we take $t^{-n/2} = \lim_{\varepsilon \searrow 0} (t - i\varepsilon)^{-n/2}$. Note that (6.45) is an improper integral near $t = 0$.

We will not look in detail at implications of (6.41)–(6.44), which are generally not as incisive as those of Proposition 5.5, but we will briefly make a connection with the idea, used in §5, of synthesizing operators from the heat semigroup. Recall particularly the formula for the resolvent:

$$(6.46) \quad (\lambda^2 - \Delta)^{-1} = \int_0^\infty e^{-(\lambda^2 - \Delta)t} \, dt.$$

Generalizing (5.28)–(5.31), we have, for $\lambda > 0$,

$$(6.47) \quad (\lambda^2 - \Delta)^{-1}\delta = \int_0^\infty e^{-|x|^2/4t - \lambda^2 t} (4\pi t)^{-n/2} \, dt.$$

A superficial resemblance with (6.45) suggests that this function is related to Bessel functions. This is consistent with the fact that the resolvent kernel $R_\lambda(x) = (\lambda^2 - \Delta)^{-1}\delta = R_{n,\lambda}^b(|x|)$ satisfies the ODE

$$(6.48) \quad \left[\frac{d^2}{dr^2} + \frac{n-1}{r} - \lambda^2 \right] R_{n,\lambda}^b(r) = 0, \quad r > 0,$$

as a consequence of the formula (6.1) for the Laplace operator in polar coordinates; this is similar to (6.10), with 1 replaced by $-\lambda^2$. In fact, there is the following result. From (6.47),

$$(6.49) \quad (\lambda^2 - \Delta)^{-1}\delta = (2\pi)^{-n/2} \left(\frac{|x|}{\lambda} \right)^{1-n/2} K_{n/2-1}(\lambda|x|), \quad x \in \mathbb{R}^n \setminus 0,$$

where $K_\nu(r)$ is defined by

$$(6.50) \quad K_\nu(r) = \frac{1}{2} \left(\frac{r}{2} \right)^\nu \int_0^\infty e^{-r^2/4t - t} t^{-1-\nu} \, dt.$$

Simple manipulations of (6.50) produce the following analogues of (6.14)–(6.16):

$$(6.51) \quad \begin{aligned} \left(\frac{d}{dr} - \frac{\nu}{r}\right) K_\nu(r) &= -K_{\nu+1}(r), \\ K_{\nu+1}(r) - K_{\nu-1}(r) &= \frac{2\nu}{r} K_\nu(r), \end{aligned}$$

so we have the ODE

$$(6.52) \quad \left[\frac{d^2}{dr^2} + \frac{1}{r} \frac{d}{dr} - \left(1 + \frac{\nu^2}{r^2}\right) \right] K_\nu(r) = 0, \quad r > 0,$$

which differs from Bessel's equation (6.11), only in one sign. The ODE (6.52) is solved by $J_\nu(ir)$ and by $Y_\nu(ir)$, so $K_\nu(r)$ must be a linear combination of these functions. In fact,

$$(6.53) \quad K_\nu(r) = \frac{1}{2} \pi i e^{\pi i \nu / 2} H_\nu^{(1)}(ir), \quad r > 0.$$

A proof of (6.53) can be found in [Leb], Chapter 5; see also Exercise 4 below. When $\nu = 1/2$, (6.53) follows from (6.33) and (6.34) together with the identity

$$(6.54) \quad K_{1/2}(r) = \left(\frac{\pi}{2r}\right)^{1/2} e^{-r},$$

which in turn, given (6.50), follows from the subordination identity. Then the recursion relation (6.51) and analogues for the Bessel functions imply (6.53) for ν of the form $\nu = k + 1/2$, when k is a positive integer.

We mention that it is customary to take a second, linearly independent, solution to (6.52) to be

$$(6.55) \quad I_\nu(r) = e^{-\pi i \nu / 2} J_\nu(ir), \quad r > 0.$$

The functions $K_\nu(r)$ and $I_\nu(r)$ are called modified Bessel functions; also $K_\nu(r)$ is sometimes called MacDonald's function.

Exercises

1. Using the integral formula (6.7), show that, for fixed $\nu > -1/2$, as $z \rightarrow +\infty$,

$$(6.56) \quad J_\nu(z) = \left(\frac{2}{\pi z}\right)^{1/2} \cos\left(z - \frac{\nu\pi}{2} - \frac{\pi}{4}\right) + O(z^{-3/2}).$$

(*Hint:* The endpoint contributions from the integral give exponentials times Fourier transforms of functions with simple singularities at the origin.)
Reconsider this problem after reading §§7 and 8.

Similarly, using (6.33), show that, for fixed $\nu > -1/2$, as $z \rightarrow +\infty$,

$$(6.57) \quad H_\nu^{(1)}(z) = \left(\frac{2}{\pi z}\right)^{1/2} e^{i(z - \nu\pi/2 - \pi/4)} + O(z^{-3/2}).$$

2. Using the integral formula (6.50), show that, for fixed ν , as $r \rightarrow +\infty$,

$$K_\nu(r) = \left(\frac{\pi}{2r}\right)^{1/2} e^{-r} [1 + O(r^{-1})].$$

(Hint: Use the Laplace asymptotic method, such as applied to the gamma function in the appendix to this chapter; compare (A.34)–(A.39). To implement this, rewrite (6.50) as

$$K_\nu(r) = 2^{-1-\nu} r^{-1} \int_0^\infty e^{-r(s+1/4s)} s^{-1-\nu} ds.$$

Note that $\varphi(s) = s + 1/4s$ has its minimum at $s = 1/2$.)

3. Using the definition (6.55) for $I_\nu(r)$, and plugging $z = ir$ into the integral formula (6.7) for $J_\nu(z)$, show that, for fixed $\nu > -1/2$, as $r \rightarrow +\infty$,

$$I_\nu(r) = \left(\frac{1}{2\pi r}\right)^{1/2} e^r [1 + O(r^{-1})].$$

4. Show that, for $r > 0$,

$$K_\nu(r) = \frac{1}{\Gamma(\frac{1}{2})\Gamma(\nu + \frac{1}{2})} \left(\frac{r}{2}\right)^\nu \int_1^\infty e^{-rt} (t^2 - 1)^{\nu-1/2} dt,$$

by showing that the function on the right solves the modified Bessel equation (6.52) and has the same asymptotic behavior as $r \rightarrow +\infty$ as $K_\nu(r)$ does, according to Exercise 2. Hence establish the identity (6.53).

5. For $y, a > 0$, $\xi \in \mathbb{R}^n$, consider

$$F(\xi) = e^{-y(|\xi|^2 + a^2)^{1/2}}.$$

Applying the subordination identity (5.22) to $A^2 = |\xi|^2 + a^2$, and taking Fourier transforms of both sides of the resulting identity, show that

$$\begin{aligned} \hat{F}(x) &= c_n y \int_0^\infty e^{-(y^2 + |x|^2)/4t} e^{-ta^2} t^{-(n+3)/2} dt \\ &= c'_n y a^\nu r^{-\nu} K_\nu(ar), \end{aligned}$$

with $\nu = (n+1)/2$, $r^2 = |x|^2 + y^2$.

6. Using analytic continuation involving both y and a , find an expression for the fundamental solution to

$$u_{tt} + 2au_t - \Delta u = 0,$$

for $u = u(t, x)$, $t \in \mathbb{R}$, $x \in \mathbb{R}^n$, where a is a real number. Be explicit in the case $n = 2$, using the elementary character of $K_{3/2}(z)$.

7. Show that, under the change of variable $u(r) = r^\alpha f(cr)$, Bessel's equation (6.11), $u''(r) + (1/r)u'(r) + (1 - \nu^2/r^2)u(r) = 0$, is transformed to

$$(6.58) \quad f''(r) + \frac{A}{r} f'(r) + \left(\mu^2 - \frac{\lambda^2}{r^2}\right) f(r) = 0,$$

with

$$A = 2\alpha + 1, \quad \mu = c^{-1}, \quad \lambda^2 = \nu^2 - \alpha^2.$$

8. Suppose in particular that $v(x) = f(r)w(\theta)$, $\theta \in \mathcal{O} \subset S^{n-1}$, and $\Delta_S w = -\lambda^2 w$. Show that the equation $\Delta v = -\mu^2 v$ is equivalent to (6.58), with

$A = n - 1$, so $\alpha = n/2 - 1$. Thus f is a linear combination of $r^{1-n/2}J_\nu(\mu r)$ and $r^{1-n/2}H_\nu^{(1)}(\mu r)$, with $\nu = [\lambda^2 + (n-2)^2/4]^{1/2}$.

9. Show that, complementary to (6.20), we have, for $\nu > 0$,

$$H_\nu^{(1)}(z) \sim -i \frac{\Gamma(\nu)}{\pi} \left(\frac{2}{z}\right)^\nu, \quad z \searrow 0.$$

7. The method of images and Poisson's summation formula

We discuss here techniques for solving such problems as the Dirichlet problem for the heat equation on a rectangular solid in $\mathbb{R}^n$, defined by

$$(7.1) \quad \Omega = \{x \in \mathbb{R}^n : 0 \leq x_j \leq a_j, \ 1 \leq j \leq n\}.$$

That is, we want to solve

$$(7.2) \quad \frac{\partial u}{\partial t} - \Delta u = 0,$$

for $u = u(t, x)$, $t \geq 0$, $x \in \Omega$, subject to the boundary condition

$$(7.2) \quad u|_{\mathbb{R}^+ \times \partial\Omega} = 0$$

and the initial condition

$$(7.4) \quad u(0, x) = f(x), \quad x \in \Omega.$$

There are two ways of doing this. One involves using Fourier series on the torus $\mathbb{R}^n/2\Gamma$, where Γ is the lattice in $\mathbb{R}^n$ generated by $a_j e_j$ (e_j being the standard basis of $\mathbb{R}^n$). The other is to use the solution on $\mathbb{R}^+ \times \mathbb{R}^n$ constructed in §5 together with the method of images, described below. Comparing these methods provides interesting analytical identities.

The method of images works as follows. Let $u^\#$ solve the heat equation

$$(7.5) \quad \frac{\partial u^\#}{\partial t} - \Delta u^\# = 0 \quad \text{on } \mathbb{R}^+ \times \mathbb{R}^n,$$

with initial data

$$(7.6) \quad u^\#(0, x) = f^\#(x),$$

where $f^\# = f$ on Ω and $f^\#$ is *odd* with respect to reflections across the walls of all the translates of Ω by elements of Γ . The set of such translates is a set of rectangles tiling $\mathbb{R}^n$, and $f^\#$ is uniquely determined by this prescription. Since reflections are isometries, it follows that, for each $t > 0$, $u^\#(t, \cdot)$ is odd with respect to such reflections; since $u^\#$ is smooth for $t > 0$, it must therefore vanish on all these walls. The restriction of $u^\#(t, x)$ to $\mathbb{R}^+ \times \Omega$ is hence the desired solution to (7.2)–(7.4).

The same sort of technique works for the wave equation on $\mathbb{R} \times \Omega$,

$$(7.7) \quad \frac{\partial^2 u}{\partial t^2} - \Delta u = 0 \quad \text{on } \mathbb{R} \times \Omega,$$

with Dirichlet boundary condition

$$(7.8) \quad u(t, x) = 0, \quad \text{for } x \in \partial\Omega,$$

and initial condition

$$(7.9) \quad u(0, x) = f(x), \quad u_t(0, x) = g(x) \quad \text{on } \Omega.$$

One takes odd extensions $f^\#, g^\#$, as above.

One can apply the method of images to regions other than rectangular solids. It applies when Ω is a half-space, for example; in that case, only one reflection, across the hyperplane $\partial\Omega$, is involved. Similarly, one can treat slabs, bounded by parallel hyperplanes, quadrants, and so on. One can also treat different boundary conditions. If one extends f above to be *even* with respect to these reflections, one obtains solutions with *Neumann* boundary condition satisfied on $\partial\Omega$.

Another type of boundary condition to impose is a periodic boundary condition:

$$(7.10) \quad u(t, x) = u(t, x + \gamma) \quad \text{if } x, x + \gamma \in \partial\Omega, \quad \gamma \in \Gamma.$$

The solution to (7.1), (7.4), (7.10) is obtained as follows. Let $f^0(x) = f(x)$ for $x \in \Omega$, let $f^0(x) = 0$ for $x \notin \Omega$, set

$$(7.11) \quad f^b(x) = \sum_{\gamma \in \Gamma} f^0(x + \gamma),$$

and let $u^b(t, x)$ be the solution to

$$(7.12) \quad \frac{\partial u^b}{\partial t} - \Delta u^b = 0 \quad \text{on } \mathbb{R}^+ \times \mathbb{R}^n, \quad u^b(0, x) = f^b(x).$$

Note that if $u^0(t, x)$ is defined by

$$(7.13) \quad \frac{\partial u^0}{\partial t} - \Delta u^0 = 0 \quad \text{on } \mathbb{R}^+ \times \mathbb{R}^n, \quad u^0(0, x) = f^0(x),$$

then

$$(7.14) \quad u^b(t, x) = \sum_{\gamma \in \Gamma} u^0(t, x + \gamma).$$

In this case, $u(t, x)$ is the restriction of $u^b(t, x)$ to $\mathbb{R}^+ \times \Omega$.

Let us specialize to the case $\Gamma = (2\pi\mathbb{Z})^n$, $f = \delta$. We have the fundamental solution, satisfying periodic boundary conditions, given by

$$(7.15) \quad H(t, x) = (4\pi t)^{-n/2} \sum_{k \in \mathbb{Z}^n} e^{-|x + 2\pi k|^2 / 4t}.$$

On the other hand, identifying $\mathbb{R}^n/(2\pi\mathbb{Z})^n$ with $\mathbb{T}^n$, we obtain via Fourier series

$$(7.16) \quad H(t, x) = (2\pi)^{-n} \sum_{\ell \in \mathbb{Z}^n} e^{-t|\ell|^2 + i\ell \cdot x}.$$

Comparing these formulas gives the following important case of Poisson's summation formula:

$$(7.17) \quad \sum_{k \in \mathbb{Z}^n} e^{-|x+2\pi k|^2/4t} = \left(\frac{t}{\pi}\right)^{n/2} \sum_{\ell \in \mathbb{Z}^n} e^{i\ell \cdot x - |\ell|^2 t}.$$

We now show how the special case of this for $n = 1$ implies the famous functional equation for the Riemann zeta function. With $n = 1$, $x = 0$, and $t = \pi/\tau$, (7.17) yields the identity

$$(7.18) \quad \sum_{n=-\infty}^{\infty} e^{-n^2 \pi \tau} = \left(\frac{1}{\tau}\right)^{1/2} \sum_{n=-\infty}^{\infty} e^{-\pi n^2 / \tau}.$$

In other words, with $g_1(\tau)$ denoting the left side of (7.18), we have $g_1(\tau) = \tau^{-1/2} g_1(1/\tau)$. This is a transformation formula of Jacobi. It follows that if

$$(7.19) \quad g(t) = \sum_{n=1}^{\infty} e^{-n^2 \pi t},$$

then

$$(7.20) \quad g(t) = -\frac{1}{2} + \frac{1}{2} t^{-1/2} + t^{-1/2} g(t^{-1}).$$

Now (7.19) is related to the *Riemann zeta function*

$$(7.21) \quad \zeta(s) = \sum_{n=1}^{\infty} n^{-s} \quad (\text{Re } s > 1)$$

via the *Mellin transform*, discussed briefly in Appendix A, at the end of this chapter. Indeed, we have

$$(7.22) \quad \begin{aligned} \int_0^{\infty} g(t) t^{s-1} dt &= \int_0^{\infty} \sum_{n=1}^{\infty} e^{-n^2 \pi t} t^{s-1} dt \\ &= \sum_{n=1}^{\infty} n^{-2s} \pi^{-s} \int_0^{\infty} e^{-t} t^{s-1} dt \\ &= \zeta(2s) \pi^{-s} \Gamma(s). \end{aligned}$$

Consequently, for $\text{Re } s > 1$,

$$(7.23) \quad \begin{aligned} \Gamma\left(\frac{s}{2}\right) \pi^{-s/2} \zeta(s) &= \int_0^{\infty} g(t) t^{s/2-1} dt \\ &= \int_0^1 g(t) t^{s/2-1} dt + \int_1^{\infty} g(t) t^{s/2-1} dt. \end{aligned}$$

Now, into the integral over $[0, 1]$, substitute the right side of (7.20) for $g(t)$, to obtain

$$(7.24) \quad \Gamma\left(\frac{s}{2}\right)\pi^{-s/2}\zeta(s) = \int_0^1 \left(-\frac{1}{2} + \frac{1}{2}t^{-1/2}\right)t^{s/2-1} dt \\ + \int_0^1 g(t^{-1})t^{s/2-3/2} dt + \int_1^\infty g(t)t^{s/2-1} dt.$$

We evaluate the first integral on the right, and replace t by $1/t$ in the second integral, to obtain, for $\operatorname{Re} s > 1$,

$$(7.25) \quad \Gamma\left(\frac{s}{2}\right)\pi^{-s/2}\zeta(s) = \frac{1}{s-1} - \frac{1}{s} + \int_1^\infty [t^{s/2} + t^{(1-s)/2}]g(t)t^{-1} dt.$$

Note that $g(t) \leq Ce^{-\pi t}$ for $t \in [1, \infty)$, so the integral on the right is an entire analytic function of s . Since $1/\Gamma(s/2)$ is entire, with simple zeros at $s = 0, -2, -4, \dots$, as shown in Appendix A at the end of this chapter, this implies that $\zeta(s)$ is continued as a meromorphic function on $\mathbb{C}$, with one simple pole, at $s = 1$. The punch line is this: The right side of (7.25) is *invariant* under replacing s by $1-s$. Thus we have Riemann's functional equation

$$(7.26) \quad \Gamma\left(\frac{s}{2}\right)\pi^{-s/2}\zeta(s) = \Gamma\left(\frac{1-s}{2}\right)\pi^{-(1-s)/2}\zeta(1-s).$$

The functional equation is often written in an alternative form, obtained by multiplying both sides by $\Gamma((1+s)/2)$, and using the identities

$$(7.27) \quad \Gamma\left(\frac{1-s}{2}\right)\Gamma\left(\frac{1+s}{2}\right) = \frac{\pi}{\sin \frac{1}{2}\pi(1-s)}, \\ \Gamma\left(\frac{s}{2}\right)\Gamma\left(\frac{1+s}{2}\right) = 2^{-s+1}\pi^{1/2}\Gamma(s),$$

which follow from (A.10) and (A.22). We obtain

$$(7.28) \quad \zeta(1-s) = 2^{1-s}\pi^{-s} \left(\cos \frac{\pi s}{2}\right) \Gamma(s)\zeta(s).$$

Exercises

1. Apply the method of images to find the solution to the heat equation on a half-line:

$$\frac{\partial u}{\partial t} = u_{xx}, \quad t \geq 0, \quad x \geq 0, \\ u(0, x) = f(x), \quad u(t, 0) = 0.$$

2. Similarly, treat the wave equation on a half-space:

$$u_{tt} - \Delta u = 0, \quad t \in \mathbb{R}, \quad x_1 > 0, \\ u(0, x) = f(x), \quad u_t(0, x) = g(x), \quad u(t, 0, x') = 0,$$

where $x = (x_1, x') \in \mathbb{R}^n$.

3. Given $u \in \mathcal{S}(\mathbb{R}^n)$, define $f \in C^\infty(\mathbb{T}^n)$ by $f(x) = \sum_{\nu \in \mathbb{Z}^n} u(x + 2\pi\nu)$. Show that, for $\ell \in \mathbb{Z}^n$, we have $\hat{f}(\ell) = (2\pi)^{-n/2} \hat{u}(\ell)$ and hence

$$(7.29) \quad \sum_k u(x + 2\pi k) = (2\pi)^{-n/2} \sum_\ell \hat{u}(\ell) e^{i\ell \cdot x}.$$

Show that this generalizes the identity (7.17).

4. Let (a_ℓ) be polynomially bounded, and consider $v = \sum_{\ell \in \mathbb{Z}^n} a_\ell e^{i\ell \cdot x}$, pictured as a $2\pi\mathbb{Z}^n$ -periodic (tempered) distribution on $\mathbb{R}^n$ rather than as a distribution on $\mathbb{T}^n$; $v \in \mathcal{S}'(\mathbb{R}^n)$. Show that

$$(7.30) \quad \hat{v} = (2\pi)^{n/2} \sum_{\ell \in \mathbb{Z}^n} a_\ell \delta_\ell \in \mathcal{S}'(\mathbb{R}^n).$$

Relate this to the result in Exercise 3.

5. Show that $\zeta(s)$ satisfies the identity

$$\zeta(s) = \prod_p (1 - p^{-s})^{-1}, \quad \operatorname{Re} s > 1,$$

the product taken over all the primes. This is known as the Euler product formula.

8. Homogeneous distributions and principal value distributions

Recall from §4 that the fundamental solution of the Laplace operator Δ on $\mathbb{R}^n$ is $c_n |x|^{2-n}$ (if $n \geq 3$), which is homogeneous. It is useful to consider homogeneous distributions in general. The notion of homogeneity is determined by the action of the group of dilations,

$$(8.1) \quad D(t)f(x) = f(tx), \quad t > 0.$$

Note that $D(t) : \mathcal{S}(\mathbb{R}^n) \rightarrow \mathcal{S}(\mathbb{R}^n)$. Also, if $f, g \in \mathcal{S}(\mathbb{R}^n)$, a change of variable gives

$$(8.2) \quad \int g(x) D(t)f(x) dx = t^{-n} \int f(x) D(t^{-1})g(x) dx.$$

Thus we can define

$$(8.3) \quad D(t) : \mathcal{S}'(\mathbb{R}^n) \longrightarrow \mathcal{S}'(\mathbb{R}^n)$$

by

$$(8.4) \quad \langle f, D(t)u \rangle = t^{-n} \langle D(t^{-1})f, u \rangle,$$

for $f \in \mathcal{S}(\mathbb{R}^n)$, $u \in \mathcal{S}'(\mathbb{R}^n)$. We say that $u \in \mathcal{S}'(\mathbb{R}^n)$ is homogeneous of degree m if

$$(8.5) \quad D(t)u = t^m u, \quad \text{for all } t > 0.$$

Here, m can be any complex number. Let us denote the space of elements of $\mathcal{S}'(\mathbb{R}^n)$ which are homogeneous of degree m by $\mathcal{H}_m(\mathbb{R}^n)$. It is easy to see that if $\mathcal{F}$ is the Fourier transform, then

$$(8.6) \quad \mathcal{F}D(t) = t^{-n}D(t^{-1})\mathcal{F},$$

so

$$(8.7) \quad \mathcal{F} : \mathcal{H}_m(\mathbb{R}^n) \longrightarrow \mathcal{H}_{-m-n}(\mathbb{R}^n).$$

Before we delve any further into $\mathcal{H}_m(\mathbb{R}^n)$, we should aver that one's real interest is in elements of $\mathcal{H}_m(\mathbb{R}^n)$ which are *smooth* outside the origin, so we consider

$$(8.8) \quad \mathcal{H}_m^\#(\mathbb{R}^n) = \{u \in \mathcal{H}_m(\mathbb{R}^n) : u \in C^\infty(\mathbb{R}^n \setminus 0)\}.$$

It is easy to see that

$$(8.9) \quad u \in \mathcal{H}_m^\#(\mathbb{R}^n) \Rightarrow D^\alpha u \in \mathcal{H}_{m-|\alpha|}^\#(\mathbb{R}^n) \quad \text{and} \quad x^\alpha u \in \mathcal{H}_{m+|\alpha|}^\#(\mathbb{R}^n).$$

We claim (8.7) can be strengthened as follows.

Proposition 8.1. *We have*

$$(8.10) \quad \mathcal{F} : \mathcal{H}_m^\# \longrightarrow \mathcal{H}_{-m-n}^\#(\mathbb{R}^n).$$

The only point left to prove is that if $u \in \mathcal{H}_m^\#(\mathbb{R}^n)$, then $\hat{u}$ is smooth on $\mathbb{R}^n \setminus 0$. Taking $\varphi \in C_0^\infty(\mathbb{R}^n)$, $\varphi(x) = 1$ for $|x| \leq 1$, we can write $u = \varphi u + (1 - \varphi)u = u_1 + u_2$ with $u_1 \in \mathcal{E}'(\mathbb{R}^n)$ and $u_2 \in C^\infty(\mathbb{R}^n)$, homogeneous for $|x|$ large. We know that $\hat{u}_1 \in C^\infty(\mathbb{R}^n)$, so it suffices to show that $\hat{u}_2 \in C^\infty(\mathbb{R}^n \setminus 0)$. This is a special case of the following important result.

For $m \in \mathbb{R}$, we define the class $S_1^m(\mathbb{R}^n)$ of C^∞ -functions by

$$(8.11) \quad p \in S_1^m(\mathbb{R}^n) \iff |D_x^\alpha p(x)| \leq C_\alpha \langle x \rangle^{m-|\alpha|}, \quad \text{for all } \alpha \geq 0.$$

Clearly, $S_1^m(\mathbb{R}^n) \subset \mathcal{S}'(\mathbb{R}^n)$. It is also clear that $u_2 \in S_1^{\text{Re } m}(\mathbb{R}^n)$, so the proof of Proposition 8.1 is finished once we establish the following.

Proposition 8.2. *If $p \in S_1^m(\mathbb{R}^n)$, then $\hat{p} \in C^\infty(\mathbb{R}^n \setminus 0)$. Also, if $\varphi \in C_0^\infty(\mathbb{R}^n)$, and $\varphi(x) = 1$ for $|x| < a$ ($a > 0$), then $(1 - \varphi)\hat{p} \in \mathcal{S}(\mathbb{R}^n)$.*

Proof. We will show that if β is large, then $x^\beta \hat{p}$ is bounded and continuous, and so are lots of derivatives, which will suffice. Clearly,

$$\mathcal{F} : S_1^\mu(\mathbb{R}^n) \longrightarrow L^\infty(\mathbb{R}^n) \cap C(\mathbb{R}^n), \quad \text{for } \mu < -n.$$

Now, given $p \in S_1^m(\mathbb{R}^n)$, then $D^\beta p \in S_1^{m-|\beta|}(\mathbb{R}^n)$, so

$$x^\beta \hat{p} = \mathcal{F}(D^\beta p) \in L^\infty \cap C, \quad \text{for } |\beta| > m + n,$$

and more generally $x^\alpha D^\beta p \in S_1^{m-|\beta|+|\alpha|}(\mathbb{R}^n)$, so

$$(8.12) \quad D^\alpha(x^\beta \hat{p}) = \mathcal{F}(x^\alpha D^\beta p) \in L^\infty \cap C, \quad \text{for } |\beta| > m + n + |\alpha|.$$

This proves Proposition 8.2.

Generally, there is going to be a singularity at the origin for an element of $\mathcal{H}_m^\#(\mathbb{R}^n)$. In fact, there is the following result, whose proof we leave as an exercise.

Proposition 8.3. *If there is a nonzero $u \in \mathcal{H}_m^\#(\mathbb{R}^n) \cap C^\infty(\mathbb{R}^n)$, then m is a nonnegative integer and u is a homogeneous polynomial.*

Let us consider other examples of homogeneous distributions. It is easy to see from the definition (8.4) of the action of $D(t)$ that

$$(8.13) \quad \delta \in \mathcal{H}_{-n}^\#(\mathbb{R}^n).$$

Of course, δ is zero on $\mathbb{R}^n \setminus 0$! Since $\mathcal{F}\delta = (2\pi)^{-n/2} \in \mathcal{H}_0^\#(\mathbb{R}^n)$, this result is consistent with Proposition 8.1. For more examples, choose any

$$(8.14) \quad w \in C^\infty(S^{n-1}),$$

and consider, for any $m \in \mathbb{C}$,

$$(8.15) \quad u_m(x) = |x|^m w(|x|^{-1}x), \quad x \in \mathbb{R}^n \setminus 0.$$

If $\operatorname{Re} m > -n$, then $u_m \in L_{\text{loc}}^1(\mathbb{R}^n)$, so it defines in a natural manner an element of $\mathcal{S}'(\mathbb{R}^n)$, which belongs to $\mathcal{H}_m^\#(\mathbb{R}^n)$. Thus

$$(8.16) \quad D^\alpha u_m \in \mathcal{H}_{m-|\alpha|}^\#(\mathbb{R}^n) \quad (\operatorname{Re} m > -n).$$

If $\operatorname{Re} m \leq -n$, then $u_m \notin L_{\text{loc}}^1(\mathbb{R}^n)$. In the borderline case $m = -n$, it is significant that there is a natural identification of u_m with an element of $\mathcal{S}'(\mathbb{R}^n)$, under the further condition that

$$(8.17) \quad \int_{S^{n-1}} w(x) dS = 0.$$

The element of $\mathcal{S}'(\mathbb{R}^n)$ is called a *principal value* distribution and is denoted *PV* u_m . We establish this as follows. Pick any radial $\varphi \in \mathcal{S}(\mathbb{R}^n)$ such that $\varphi(0) = 1$, such as $\varphi(x) = e^{-|x|^2}$. Then, for any $v \in \mathcal{S}(\mathbb{R}^n)$, with u_{-n} as in (8.15), $u_{-n}(x)[v(x) - v(0)\varphi(x)]$ belongs to $L^1(\mathbb{R}^n)$, so we can define

$$(8.18) \quad \langle v, \text{PV } u_{-n} \rangle = \int_{\mathbb{R}^n} u_{-n}(x)[v(x) - v(0)\varphi(x)] dx.$$

Note that (8.17) is precisely what is required to guarantee that the right side of (8.18) is independent of the choice of φ (satisfying the conditions

given above). Thus we can write, for any $t > 0$,

$$\begin{aligned}
 \langle D(t)v, PV \ u_{-n} \rangle &= \int_{\mathbb{R}^n} u_{-n}(x) [v(tx) - v(0)\varphi(tx)] \, dx \\
 (8.19) \qquad &= t^{-n} \int_{\mathbb{R}^n} u_{-n}(x/t) [v(x) - v(0)\varphi(x)] \, dx \\
 &= \langle v, PV \ u_{-n} \rangle.
 \end{aligned}$$

In light of (8.4), this implies

$$(8.20) \qquad PV \ u_{-n} \in \mathcal{H}_{-n}^{\#}(\mathbb{R}^n),$$

provided (8.17) holds. By Proposition 8.1, we have

$$(8.21) \qquad \mathcal{F}(PV \ u_{-n}) \in \mathcal{H}_0^{\#}(\mathbb{R}^n).$$

In particular, this Fourier transform is bounded. Consequently, the convolution operator

$$(8.22) \qquad Tv = (PV \ u_{-n}) * v,$$

a priori taking $\mathcal{S}(\mathbb{R}^n)$ to $\mathcal{S}'(\mathbb{R}^n)$, has the property that

$$(8.23) \qquad T : L^2(\mathbb{R}^n) \longrightarrow L^2(\mathbb{R}^n).$$

Continuity properties of such a convolution operator on $L^p(\mathbb{R}^n)$, for $1 < p < \infty$, will be demonstrated in Chapter 13.

The special one-dimensional case of a principal value distribution has been discussed in §4. In analogy with (4.34), we have the following.

Proposition 8.4. *Under the hypothesis (8.17), we have, for $v \in \mathcal{S}(\mathbb{R}^n)$,*

$$(8.24) \qquad \langle v, PV \ u_{-n} \rangle = \lim_{\varepsilon \rightarrow 0} \int_{\mathbb{R}^n \setminus B_{\varepsilon}} v(x) \ u_{-n}(x) \, dx,$$

where $B_{\varepsilon} = \{x \in \mathbb{R}^n : |x| < \varepsilon\}$.

Proof. Since $u_{-n}(x)[v(x) - v(0)\varphi(x)] \in L^1(\mathbb{R}^n)$, via (8.18), we have

$$\langle v, PV \ u_{-n} \rangle = \lim_{\varepsilon \rightarrow 0} \int_{\mathbb{R}^n \setminus B_{\varepsilon}} u_{-n}(x) [v(x) - v(0)\varphi(x)] \, dx,$$

so (8.24) follows from the observation that if (8.17) holds, then

$$\int_{\mathbb{R}^n \setminus B_{\varepsilon}} u_{-n}(x) \varphi(x) \, dx = 0, \quad \text{for all } \varepsilon > 0,$$

for any radial $\varphi \in \mathcal{S}(\mathbb{R}^n)$.

In general, if $u(x)$ has the form (8.15) with $m = -n$, then u is a sum of a term to which (8.17) applies and a constant times r^{-n} . Now one can still define a distribution in $\mathcal{S}'(\mathbb{R}^n)$, equal to r^{-n} on $\mathbb{R}^n \setminus 0$, by the prescription

$$(8.25) \quad \langle v, E_\varphi r^{-n} \rangle = \int_{\mathbb{R}^n} r^{-n} [v(x) - v(0)\varphi(x)] dx,$$

for any given radial $\varphi \in \mathcal{S}(\mathbb{R}^n)$ satisfying $\varphi(0) = 1$. This time, $E_\varphi r^{-n} \in \mathcal{S}'(\mathbb{R}^n)$ depends on the choice of φ . One has

$$(8.26) \quad E_\varphi r^{-n} - E_\psi r^{-n} = \left(\int [\psi(x) - \varphi(x)] r^{-n} dx \right) \delta.$$

Also, $E_\varphi r^{-n}$ is not homogeneous. Instead, one has

$$(8.27) \quad D(t)(E_\varphi r^{-n}) = t^{-n} E_{D(t)\varphi} r^{-n},$$

and by (8.26) this yields, after a brief calculation,

$$(8.28) \quad D(t)(E_\varphi r^{-n}) = t^{-n} E_\varphi r^{-n} + A_{n-1} t^{-n} (\log t) \delta,$$

where $A_{n-1} = \text{vol}(S^{n-1})$. This implies for the Fourier transform of $E_\varphi r^{-n}$ that

$$(8.29) \quad D(t)\mathcal{F}(E_\varphi r^{-n}) = \mathcal{F}(E_\varphi r^{-n}) + (2\pi)^{-n/2} A_{n-1} \log t,$$

which, in view of rotational invariance, implies that

$$(8.30) \quad \mathcal{F}(E_\varphi r^{-n})(\xi) = (2\pi)^{-n/2} A_{n-1} \log |\xi| + B,$$

where B is a constant, depending in an affine manner on φ . A “canonical” choice of $E_\varphi r^{-n}$ would be one for which $B = 0$; such a distribution $E_\varphi r^{-n} \in \mathcal{S}'(\mathbb{R}^n)$ is denoted $PF r^{-n}$ (for “finite part”); we have

$$(8.31) \quad \mathcal{F}(PF r^{-n})(\xi) = (2\pi)^{-n/2} A_{n-1} \log |\xi|.$$

Note that this is consistent with (4.59) when $n = 2$.

It turns out that r^{-m} , which is holomorphic in $\{m \in \mathbb{C} : \text{Re } m > -n\}$, with values in $\mathcal{S}'(\mathbb{R}^n)$, has a meromorphic continuation. This can be perceived as follows. First note that if $-n < \text{Re } m < 0$, then both r^m and r^{-n-m} belong to $L^1_{\text{loc}}(\mathbb{R}^n)$, so from Proposition 8.1 and rotational invariance we deduce that

$$(8.32) \quad \mathcal{F}(r^m) = c(m) r^{-m-n},$$

for a certain factor $c(m)$, which we want to work out. We claim that

$$(8.33) \quad \mathcal{F}(r^m) = 2^{m+n/2} \frac{\Gamma(\frac{1}{2}(m+n))}{\Gamma(-\frac{1}{2}m)} r^{-m-n},$$

for $-n < \text{Re } m < 0$. This can be deduced from (8.32) and Parseval’s identity, which gives

$$(8.34) \quad \langle u, r^m \rangle = c(m) \langle \hat{u}, r^{-m-n} \rangle.$$

If we plug in $u(x) = e^{-|x|^2/2} = \hat{u}(x)$, both sides of (8.34) can be evaluated by integrating in polar coordinates. The left side is

$$(8.35) \quad \begin{aligned} A_{n-1} \int_0^\infty r^{m+n-1} e^{-r^2/2} dr &= 2^{(m+n-1)/2} A_{n-1} \int_0^\infty s^{(m+n)/2-1} e^{-s} ds \\ &= 2^{(m+n-1)/2} \Gamma\left(\frac{1}{2}(m+n)\right) A_{n-1}, \end{aligned}$$

and the right side of (8.34) is similarly evaluated, giving (8.33).

Now the left side of (8.33) extends to be holomorphic in $\operatorname{Re} m > -n$, with values in $\mathcal{S}'(\mathbb{R}^n)$, while the right side extends to be meromorphic in $\operatorname{Re} m < 0$, with poles at $m = -n, -n-2, -n-4, \dots$, due to the factor $\Gamma((m+n)/2)$. Thus we have the desired meromorphic continuation. With r^m so defined,

$$(8.36) \quad r^m \in \mathcal{H}_m^\#(\mathbb{R}^n), \quad m \neq -n, -n-2, -n-4, \dots;$$

indeed, $D(t)r^m - t^m r^m$ is a meromorphic function of m which vanishes on a nonempty open set. As we have seen, $PF r^{-n}$ can be defined by a “renormalization,” though it does not belong to $\mathcal{H}_{-n}^\#(\mathbb{R}^n)$.

Let us now consider the possibility of extending u_m , of the form (8.15), to an element of $\mathcal{S}'(\mathbb{R}^n)$, in case

$$(8.37) \quad m = -n - j, \quad j = 1, 2, 3, \dots$$

In analogy with (8.18) and (8.25), we can define $E_{j,\varphi}u_m$ in this case by

$$(8.38) \quad \langle v, E_{j,\varphi}u_m \rangle = \int u_m(x) \left[v(x) - \sum_{|\alpha| \leq j} \frac{v^{(\alpha)}(0)}{\alpha!} x^\alpha \varphi(x) \right] dx,$$

provided $\varphi \in \mathcal{S}(\mathbb{R}^n)$ is a radial function such that $\varphi(0) = 1$ and $1 - \varphi$ vanishes to order at least j at 0; for example, we could require $\varphi(x) = 1$ for $|x| \leq c$. The dependence on φ is given by

$$(8.39) \quad E_{j,\varphi}u_m - E_{j,\psi}u_m = \sum_{|\alpha| \leq j} \beta_\alpha(\varphi - \psi) \delta^{(\alpha)},$$

where

$$(8.40) \quad \beta_\alpha(\varphi - \psi) = -\frac{1}{\alpha!} \int x^\alpha [\varphi(x) - \psi(x)] u_m(x) dx.$$

In analogy with (8.27), we have

$$(8.41) \quad D(t)E_{j,\varphi}u_m = t^m E_{j,D(t)\varphi}u_m,$$

and hence, given (8.37), by a calculation similar to that establishing (8.28),

$$(8.42) \quad \begin{aligned} D(t)(E_{j,\varphi}u_m) &= t^m E_{j,\varphi}u_m + t^m \sum_{|\alpha| < j} \gamma_\alpha (t^{|\alpha|-j} - 1) \delta^{(\alpha)} \\ &\quad + t^m \log t \sum_{|\alpha|=j} \gamma_\alpha \delta^{(\alpha)}, \end{aligned}$$

for certain constants γ_α , which depend in an affine fashion on φ . Consequently, if we set

$$(8.43) \quad \tilde{E}u_m = E_{j,\varphi}u_m - \sum_{|\alpha| < j} \gamma_\alpha \delta^{(\alpha)},$$

we have another element of $\mathcal{S}'(\mathbb{R}^n)$ which agrees with u_m on $\mathbb{R}^n \setminus 0$, and

$$(8.44) \quad D(t)(\tilde{E}u_m) = t^m \tilde{E}u_m + t^m (\log t) \sum_{|\alpha|=j} \gamma_\alpha \delta^{(\alpha)}.$$

It follows for the Fourier transform $\mathcal{F}(\tilde{E}u_m)$ that

$$D(t)\mathcal{F}(\tilde{E}u_m) = t^j \mathcal{F}(\tilde{E}u_m) + t^j (\log t) \sum_{|\alpha|=j} \gamma'_\alpha \xi^\alpha.$$

Consequently, if $\mathcal{F}(\tilde{E}u_m) = \omega(\xi)$ for $|\xi| = 1$, we have

$$\mathcal{F}(\tilde{E}u_m)(t\xi) = t^j \omega(\xi) + (\log t) \sum_{|\alpha|=j} \gamma'_\alpha (t\xi)^\alpha, \quad \text{for } |\xi| = 1,$$

and hence

$$(8.45) \quad \mathcal{F}(\tilde{E}u_m)(\xi) = w_j(\xi) + p_j(\xi) \log |\xi|,$$

where

$$(8.46) \quad w_j \in \mathcal{H}_j^\#(\mathbb{R}^n) \quad \text{and } p_j \text{ is a homogeneous polynomial, of degree } j.$$

We leave it as an exercise to the reader to construct a similar extension of u_m to an element of $\mathcal{S}'(\mathbb{R}^n)$, when $\operatorname{Re} m \leq -n$ and m is not an integer. In such a case one can produce an element of $\mathcal{H}_m^\#(\mathbb{R}^n)$; log terms do not arise.

Exercises

1. More generally than $S_1^m(\mathbb{R}^n)$, for $0 < \rho \leq 1$, define $S_\rho^m(\mathbb{R}^n)$ by

$$p \in S_\rho^m(\mathbb{R}^n) \iff |D_x^\alpha p(x)| \leq C_\alpha \langle x \rangle^{m-\rho|\alpha|}, \quad \text{for all } \alpha \geq 0.$$

Show that $\hat{p} \in C^\infty(\mathbb{R}^n \setminus 0)$ in this case, as in Proposition 8.2.

2. Define $p \in C^\infty(\mathbb{R}^n \setminus 0)$ by $p(\xi) = (i\xi_1 + |\xi'|^2)^{-1}$, $\xi = (\xi_1, \xi_2, \dots, \xi_n) = (\xi_1, \xi')$. Show that $p(\xi)$ agrees outside any neighborhood of the origin with a member of $S_{1/2}^{-1}(\mathbb{R}^n)$.

3. Prove Proposition 8.3.

4. If $-n < \operatorname{Re} m < 0$ and u_m is of the form (8.15), then u_m and $\hat{u}_m$ belong to $L_{\text{loc}}^1(\mathbb{R}^n)$, with $u_m \in \mathcal{H}_m^\#$, $\hat{u}_m \in \mathcal{H}_{-n-m}^\#$. Hence

$$\hat{u}(x) = |x|^{-n-m} W_m(|x|^{-1}x), \quad W_m \in C^\infty(S^{n-1}).$$

Study the transformation $w \mapsto W_m$. Use this to produce a meromorphic continuation of u_m .

5. Study the residue of the meromorphic distribution-valued function r^z at $z = -n$, and relate this to the failure of $PF\ r^{-n}$ to be homogeneous.
6. In case $n = 1$ and $m = -s$, the formula (8.33) says

$$\mathcal{F}(r^{-s}) = 2^{1/2-s} \frac{\Gamma(\frac{1-s}{2})}{\Gamma(\frac{s}{2})} r^{s-1}, \quad \text{for } 0 < \operatorname{Re} s < 1,$$

while Riemann's functional equation (7.26) can be written

$$\frac{\zeta(s)}{\zeta(1-s)} = \pi^{s-1/2} \frac{\Gamma(\frac{1-s}{2})}{\Gamma(\frac{s}{2})}.$$

Is this a coincidence? (See [Pat], Chapter 2.) Note that these formulas yield

$$\frac{(2\pi)^{s/2}}{\zeta(s)} \mathcal{F}(r^{-s}) = \frac{(2\pi)^{(1-s)/2}}{\zeta(1-s)} r^{s-1}, \quad 0 < \operatorname{Re} s < 1.$$

9. Elliptic operators

A partial differential operator $P(D)$ of order m ,

$$(9.1) \quad P(D) = \sum_{|\alpha| \leq m} a_\alpha D^\alpha,$$

is said to be *elliptic* provided

$$(9.2) \quad |P(\xi)| \geq C|\xi|^m, \quad \text{for } |\xi| \text{ large.}$$

Here $P(\xi) = \sum a_\alpha \xi^\alpha$. The paradigm example is the Laplace operator $\Delta = P(D)$, with $P(\xi) = -|\xi|^2$, which is elliptic of order 2. In this section we consider some important properties of solutions to

$$(9.3) \quad P(D)u = f$$

when $P(D)$ is elliptic.

The hypothesis (9.2) implies the following. If (9.2) holds for $|\xi| \geq C_1$, and if $\varphi \in C_0^\infty(\mathbb{R}^n)$ is equal to 1 for $|\xi| \leq C_1$, then

$$(9.4) \quad q(\xi) = (1 - \varphi(\xi)) P(\xi)^{-1} \in S_1^{-m}(\mathbb{R}^n),$$

where $S_1^m(\mathbb{R}^n)$ is the space defined by (8.11); we call it a space of “symbols.” Now consider

$$(9.5) \quad E = (2\pi)^{-n/2} \tilde{q} \in \mathcal{S}'(\mathbb{R}^n).$$

By Proposition 8.2, we know that E is smooth on $\mathbb{R}^n \setminus 0$ and rapidly decreasing as $|x| \rightarrow \infty$. If we set

$$(9.6) \quad v = P(D)E,$$

then

$$(9.7) \quad \hat{v}(\xi) = (2\pi)^{-n/2} (1 - \varphi(\xi)).$$

In other words,

$$(9.8) \quad P(D)E = \delta + w,$$

with

$$(9.9) \quad w = -(2\pi)^{-n/2} \tilde{\varphi} \in \mathcal{S}(\mathbb{R}^n).$$

We say E is a *parametrix* for $P(D)$. It is almost as useful as a fundamental solution, for some qualitative purposes. For example, it enables us to say a great deal about the *singular support* of a solution u to (9.3), given $f \in \mathcal{D}'(\mathbb{R}^n)$. The singular support of a general distribution $u \in \mathcal{D}'(\mathbb{R}^n)$ is defined as follows. Let $\Omega \subset \mathbb{R}^n$ be open. We say u is smooth on Ω if there exists $v \in C^\infty(\Omega)$ such that $u = v$ on Ω . The smallest set K for which u is smooth on $\mathbb{R}^n \setminus K$ is the singular support of u , denoted

$$\text{sing supp } u.$$

For example, $\text{sing supp } \delta = \{0\}$; also $\text{sing supp } |x|^{2-n} = \{0\}$, if $n \neq 2$. Now, suppose that $u \in \mathcal{E}'(\mathbb{R}^n)$ and (9.3) holds. Then

$$(9.10) \quad E * f = E * P(D)u = (P(D)E) * u = u + w * u$$

and, of course,

$$w * u \in C^\infty(\mathbb{R}^n).$$

On the other hand, it is easy to see that, for any $f \in \mathcal{E}'(\mathbb{R}^n)$,

$$(9.11) \quad \text{sing supp } E * f \subset \text{sing supp } f,$$

provided $\text{sing supp } E \subset \{0\}$. More generally, for any $f_1, f_2 \in \mathcal{E}'(\mathbb{R}^n)$, if $\text{sing supp } f_j \subset K_j$, then

$$(9.12) \quad \text{sing supp } f_1 * f_2 \subset K_1 + K_2,$$

a result we leave as an exercise.

Noting that we can multiply distributions by cut-offs $\chi \in C_0^\infty(\mathbb{R}^n)$, equal to 1 on an arbitrarily large set, we deduce the following result, known as *elliptic regularity*.

Proposition 9.1. *For any $u \in \mathcal{D}'(\mathbb{R}^n)$, if (9.3) holds with $P(D)$ elliptic, then*

$$(9.13) \quad \text{sing supp } u = \text{sing supp } f.$$

Finally, we want to make a detailed analysis of the behavior of the singularity at the origin of the parametrix E for an elliptic operator $P(D)$. Since E is given by (9.4) and (9.5), with $P(\xi) = \sum_{|\alpha| \leq m} a_\alpha \xi^\alpha$ a polynomial, it follows that, for $|\xi|$ large,

$$(9.14) \quad q(\xi) \sim \sum_{j \geq 0} q_j(\xi),$$

where each $q_j \in C^\infty(\mathbb{R}^n)$, and, for $|\xi| \geq C$, $q_j(\xi)$ is homogeneous in ξ of degree $-m-j$. The meaning of (9.14) is that, for any N ,

$$(9.15) \quad q(\xi) - \sum_{j=0}^{N-1} q_j(\xi) = r_N(\xi) \in S_1^{-m-N}(\mathbb{R}^n).$$

Consequently,

$$(9.16) \quad E \sim (2\pi)^{-n/2} \sum_{j \geq 0} \tilde{q}_j$$

in the sense that, for any K , one can take N large enough that

$$(9.17) \quad E - (2\pi)^{-n/2} \sum_{j=0}^{N-1} \tilde{q}_j = (2\pi)^{-n/2} \tilde{r}_N \in C^K(\mathbb{R}^n).$$

Now, we can replace each q_j by $q_j^b \in C^\infty(\mathbb{R}^n \setminus 0)$, equal to q_j for $|\xi|$ large, and homogeneous of degree $-m-j$ on $\mathbb{R}^n \setminus 0$, and replace each q_j^b by $q_j^\# \in \mathcal{S}'(\mathbb{R}^n)$, equal to q_j^b on $\mathbb{R}^n \setminus 0$, such that $q_j^\# \in \mathcal{H}_{-m-j}^\#$ if $m+j < n$, or in any event satisfying the counterpart of (8.44)–(8.46). Note that, for each j , $q_j - q_j^\# \in \mathcal{E}'(\mathbb{R}^n)$, so the Fourier transform of the difference belongs to $C^\infty(\mathbb{R}^n)$. We have established the following.

Proposition 9.2. *A parametrix E for an elliptic operator $P(D)$ of order m satisfies the condition that $E \in C^\infty(\mathbb{R}^n \setminus 0)$, and the singularity is given by*

$$(9.18) \quad E \sim \sum_{\ell \geq 0} (E_\ell + p_\ell(x) \log |x|),$$

where

$$(9.19) \quad E_\ell \in \mathcal{H}_{m-n+\ell}^\#(\mathbb{R}^n)$$

and $p_\ell(x)$ is a polynomial homogeneous of degree $m-n+\ell$; these log coefficients appear only for $\ell \geq n-m$.

More generally, this result holds for $E = (2\pi)^{-n/2} \tilde{q}$ whenever $q \in S_1^m$ has an expansion of the form (9.14), for any $m \in \mathbb{R}$, and log terms do not arise if m is not an integer.

Exercises

1. Using Exercises 1 and 2 of §8, establish an analogue of the regularity result in Proposition 9.1 when $P(D)$ is the (nonelliptic) “heat operator”:

$$P(D) = \frac{\partial}{\partial x_1} - \left(\frac{\partial^2}{\partial x_2^2} + \cdots + \frac{\partial^2}{\partial x_n^2} \right).$$

2. Give a detailed proof of (9.12), in order to deduce (9.11).

(Hint: Use

$$f \in \mathcal{E}'(\mathbb{R}^n), g \in C^\infty(\mathbb{R}^n) \implies f * g \in C^\infty(\mathbb{R}^n).$$

Break up f_1 and f_2 into pieces. For nonsmooth pieces, establish and use

$$\text{supp } \varphi_j \subset \tilde{K}_j \implies \text{supp } \varphi_1 * \varphi_2 \subset \tilde{K}_1 + \tilde{K}_2.$$

10. Local solvability of constant-coefficient PDE

In the previous sections we have mainly used Fourier analysis as a tool to provide explicit solutions to the classical linear PDEs. Here we use Fourier series to prove an existence theorem for solutions to a general constant-coefficient linear PDE

$$(10.1) \quad P(D)u = f.$$

We show that, given any $f \in \mathcal{D}'(\mathbb{R}^n)$, and any $R < \infty$, there exists $u \in \mathcal{D}'(\mathbb{R}^n)$ solving (10.1) on the ball $|x| < R$. This result was originally established by Malgrange and Ehrenpreis. If $f \in C^\infty(\mathbb{R}^n)$, we produce $u \in C^\infty(\mathbb{R}^n)$. We do not produce a global solution, and other references, particularly [H] and [Tre], contain much more information on solutions to (10.1) than is presented here. Our method, due to Dadok and Taylor [DT], does have the advantage of being fairly straightforward and short.

For any $\alpha \in \mathbb{R}^n$, solving (10.1) on $B_R = \{x \in \mathbb{R}^n : |x| < R\}$ is equivalent to solving

$$(10.2) \quad P(D + \alpha)v = g,$$

where $v = e^{-i\alpha \cdot x}u$ and $g = e^{-i\alpha \cdot x}f$. To solve (10.2) on B_R , we can cut off g to be supported on $B_{3R/2}$ and work on $\mathbb{R}^n/2R\mathbb{Z}^n$. Without loss of generality (altering $P(D)$), we can rescale and suppose $R = \pi$, so $\mathbb{R}^n/2R\mathbb{Z}^n = \mathbb{T}^n$. The following result then implies solvability on B_R .

Proposition 10.1. *For almost every $\alpha \in A = \{(\alpha_1, \dots, \alpha_n) : 0 \leq \alpha_\nu \leq 1\}$,*

$$(10.3) \quad P(D + \alpha) : \mathcal{D}'(\mathbb{T}^n) \longrightarrow \mathcal{D}'(\mathbb{T}^n)$$

is an isomorphism, as is $P(D + \alpha) : C^\infty(\mathbb{T}^n) \rightarrow C^\infty(\mathbb{T}^n)$.

In view of the characterizations of Fourier series of elements of $\mathcal{D}'(\mathbb{T}^n)$ and of $C^\infty(\mathbb{T}^n)$, it suffices to establish the following.

Proposition 10.2. *Let $P(\xi)$ be a polynomial of order m on $\mathbb{R}^n$. For almost all $\alpha \in A$, there are constants C, N such that*

$$(10.4) \quad |P(k + \alpha)^{-1}| \leq C\langle k \rangle^N, \quad \text{for all } k \in \mathbb{Z}^n.$$

We will prove this using the following elementary fact about the behavior of a polynomial near its zero set.

Lemma 10.3. *Let $P(\xi)$ be a polynomial of order m on $\mathbb{R}^n$, not identically zero. Then there exists $\delta > 0$ such that*

$$(10.5) \quad |P(\xi)|^{-\delta} \in L_{\text{loc}}^1(\mathbb{R}^n).$$

Before proving Lemma 10.3, we show how it yields (10.4). First, we claim that, for any polynomial of order m on $\mathbb{R}^n$, not identically zero, there exist $\delta > 0$ and M such that

$$(10.6) \quad \int |P(\xi)|^{-\delta} \langle \xi \rangle^{-M} d\xi < \infty.$$

Indeed, Lemma 10.3 guarantees

$$\int_{|\xi| \leq 1} |P(\xi)|^{-\delta} d\xi < \infty,$$

while, for M sufficiently large,

$$\int_{|\xi| \geq 1} |P(\xi)|^{-\delta} |\xi|^{-M} d\xi \leq C \int_{|\xi| \leq 1} |P(|\xi|^{-2}\xi)|^{-\delta} d\xi,$$

and Lemma 10.3 also implies that, for $\delta > 0$ small enough,

$$|P(|\xi|^{-2}\xi)|^{-\delta} \in L_{\text{loc}}^1.$$

Now, using (10.6), note that

$$(10.7) \quad \int_A \sum_{k \in \mathbb{Z}^n} |P(k + \alpha)|^{-\delta} \langle k \rangle^{-M} d\alpha \leq C \int_{\mathbb{R}^n} |P(\xi)|^{-\delta} \langle \xi \rangle^{-M} d\xi < \infty.$$

Thus, for almost all $\alpha \in A$,

$$(10.8) \quad \sum_{k \in \mathbb{Z}^n} |P(k + \alpha)|^{-\delta} \langle k \rangle^{-M} < \infty,$$

which immediately gives (10.4).

We now prove the lemma for any $\delta < 1/m$. We must prove that $|P(\xi)|^{-\delta}$ is integrable on any bounded subset of $\mathbb{R}^n$. Rotating coordinates, we can suppose that $P(\xi_1, 0)$ is a polynomial of order exactly m :

$$(10.9) \quad P(\xi_1, 0) = a_m \xi_1^m + \cdots + a_0, \quad a_m \neq 0.$$

It follows that, with $\xi' = (\xi_2, \dots, \xi_n)$,

$$(10.10) \quad P(\xi) = a_m \xi_1^m + \sum_{\ell=0}^{m-1} a_\ell(\xi') \xi_1^\ell,$$

where $a_\ell(\xi')$ is a polynomial on $\mathbb{R}^{n-1}$ of order $\leq m - \ell$. Consequently, we have

$$(10.11) \quad P(\xi) = a_m \prod_{j=1}^m (\xi_1 - \lambda_j(\xi')).$$

Hence it is clear that, for any $C_1 < \infty$, there is a $C_2 < \infty$ such that if $\delta < 1/m$,

$$(10.12) \quad \int_{-C_1}^{C_1} |P(\xi)|^{-\delta} d\xi_1 \leq C_2, \quad \text{for } |\xi'| \leq C_1.$$

This completes the proof.

Exercises

1. Consider the following boundary problem on $[0, A] \times \mathbb{T}^n$:

$$\begin{aligned} u_{tt} - \Delta u &= 0, \\ u(0, x) &= f_1(x), \quad u(A, x) = f_2(x), \end{aligned}$$

where $f_j \in C^\infty(\mathbb{T}^n)$. Show that, for almost all $A \in \mathbb{R}^+$, this has a unique solution $u \in C^\infty([0, A] \times \mathbb{T}^n)$, for all $f_j \in C^\infty(\mathbb{T}^n)$. Show that, for a dense set of A , this solvability fails.

11. The discrete Fourier transform

When doing numerical work involving Fourier series, it is convenient to discretize, and replace S^1 , pictured as the group of complex numbers of modulus 1, by the group Γ_n generated by $\omega = e^{2\pi i/n}$. One can also approximate $\mathbb{T}^d$ by $(\Gamma_n)^d$, a product of d copies of Γ_n . We will restrict attention to the case $d = 1$ here; results for general d are obtained similarly.

The cyclic group Γ_n is isomorphic to the group $\mathbb{Z}_n = \mathbb{Z}/(n)$, but we will observe a distinction between these two groups; an element of Γ_n is a certain complex number of modulus 1, and an element of $\mathbb{Z}_n$ is an equivalence class of integers. For n large, we think of Γ_n as an approximation to S^1 and $\mathbb{Z}_n$ as an approximation to $\mathbb{Z}$. We note the natural dual pairing $\Gamma_n \times \mathbb{Z}_n \rightarrow \mathbb{C}$ given by $(\omega^j, \ell) \mapsto \omega^{j\ell}$, which is well defined since $\omega^{jn} = 1$.

Now, given a function $f : \Gamma_n \rightarrow \mathbb{C}$, its discrete Fourier transform $f^\# = \Phi_n f$, mapping $\mathbb{Z}_n$ to $\mathbb{C}$, is defined by

$$(11.1) \quad f^\#(\ell) = \frac{1}{n} \sum_{\omega^j \in \Gamma_n} f(\omega^j) \omega^{-j\ell}.$$

Similarly, given a function $g : \mathbb{Z}_n \rightarrow \mathbb{C}$, its “inverse Fourier transform” $g^b : \Gamma_n \rightarrow \mathbb{C}$ is defined by

$$(11.2) \quad g^b(\omega^j) = \sum_{\ell \in \mathbb{Z}_n} g(\ell) \omega^{j\ell}.$$

The following is the Fourier inversion formula in this context.

Proposition 11.1. *The map*

$$(11.3) \quad \Phi_n : L^2(\Gamma_n) \longrightarrow L^2(\mathbb{Z}_n)$$

is a unitary isomorphism, with inverse defined by (11.2), so

$$(11.4) \quad f(\omega^j) = \sum_{\ell \in \mathbb{Z}_n} f^\#(\ell) \omega^{j\ell}.$$

Here the space $L^2(\mathbb{Z}_n)$ is defined by counting measure and $L^2(\Gamma_n)$ by $1/n$ times counting measure, that is,

$$(11.5) \quad (u, v)_{L^2(\Gamma_n)} = \frac{1}{n} \sum_{\omega^j \in \Gamma_n} u(\omega^j) \overline{v(\omega^j)}.$$

Note that if we define functions e_j on Γ_n by

$$(11.6) \quad e_j(\omega^k) = \omega^{jk},$$

then Proposition 11.1 is equivalent to:

Proposition 11.2. *The functions e_j , $1 \leq j \leq n$, form an orthonormal basis of $L^2(\Gamma_n)$.*

Proof. Since $L^2(\Gamma_n)$ has dimension n , we need only check that the e_j s are mutually orthogonal. Note that

$$(e_k, e_\ell) = \frac{1}{n} \sum_{\omega^j \in \Gamma_n} \omega^{mj}, \quad m = k - \ell.$$

Denote the sum by S_m . If we multiply by ω^m , we have a sum of the same set of powers of ω , so $S_m = \omega^m S_m$. Thus $S_m = 0$ whenever $\omega^m \neq 1$, which completes the proof. Alternatively, the series is easily summed as a finite geometrical series.

Note that the functions e_j in (11.6) are the restrictions to Γ_n of $e^{ij\theta}$ (i.e., values at $\theta = 2\pi k/n$). These restrictions depend only on the residue class of $j \bmod n$, which leads to the following simple but fundamental connection between Fourier series on S^1 and on Γ_n .

Proposition 11.3. *If $f \in C(S^1)$ has absolutely convergent Fourier series, then*

$$(11.7) \quad f^\#(\ell) = \sum_{j=-\infty}^{\infty} \hat{f}(\ell + jn).$$

We will use (11.7) as a tool to see how well a function on S^1 is approximated by discretization, involving restriction to Γ_n . Precisely, we consider the operators

$$(11.8) \quad R_n : C(S^1) \longrightarrow L^2(\Gamma_n), \quad E_n : L^2(\Gamma_n) \longrightarrow C^\infty(S^1)$$

given by

$$(11.9) \quad (R_n f)(\omega^j) = f\left(\frac{2\pi j}{n}\right),$$

for $f = f(\theta)$, $0 \leq \theta \leq 2\pi$, and

$$(11.10) \quad E_n \left(\sum_{\ell \in \mathbb{Z}_n} g(\ell) \omega^{j\ell} \right) = \sum_{\ell=-\nu}^{\nu-1} g(\ell) e^{i\ell\theta}, \quad n = 2\nu.$$

We assume $n = 2\nu$ is even; one can also treat $n = 2\nu - 1$, changing the upper limit in the last sum from $\nu - 1$ to ν . Clearly, $R_n E_n$ is the identity operator on $L^2(\Gamma_n)$. The question of interest to us is: How close is $E_n R_n f$ to f , a function on S^1 ? The answer depends on smoothness properties of f and is expressed in terms involving (typically) negative powers of n .

We compare $E_n R_n$ and the partial summing operator

$$(11.11) \quad P_n f = \sum_{\ell=-\nu}^{\nu-1} \hat{f}(\ell) e^{i\ell\theta}$$

for Fourier series. Note that

$$(11.12) \quad E_n R_n f(\theta) = \sum_{\ell=-\nu}^{\nu-1} f^\#(\ell) e^{i\ell\theta}.$$

Consequently,

$$(11.13) \quad E_n R_n f = P_n f + Q_n f,$$

with

$$(11.14) \quad Q_n f(\theta) = \sum_{\ell=-\nu}^{\nu-1} [f^\#(\ell) - \hat{f}(\ell)] e^{i\ell\theta}.$$

By (11.7), we have, for $-\nu \leq \ell \leq \nu - 1$,

$$(11.15) \quad f^\#(\ell) - \hat{f}(\ell) = \sum_{j \in \mathbb{Z} \setminus 0} \hat{f}(\ell + jn).$$

Consequently, the sup norm of $Q_n f$ is bounded by

$$(11.16) \quad \sum_{\ell=-\nu}^{\nu-1} |f^\#(\ell) - \hat{f}(\ell)| \leq \sum_{|k| \geq \nu} |\hat{f}(k)|.$$

The right side also dominates the sup norm of $f - P_n f$, proving:

Proposition 11.4. *If $f \in C(S^1)$ has absolutely convergent Fourier series, then*

$$(11.17) \quad \|f - E_n R_n f\|_{L^\infty} \leq 2 \sum_{|k| \geq \nu} |\hat{f}(k)|.$$

The estimates of various norms of $f - P_n f$ is an exercise in Fourier analysis on S^1 . There are many estimates involving Sobolev spaces; see Chapter 4. Here we note a simple estimate, for $m \geq 1$:

$$(11.18) \quad \begin{aligned} \|f - P_n f\|_{C^\ell(S^1)} &\leq \sum_{|k| \geq \nu} |k|^\ell |\hat{f}(k)| \\ &\leq C_{m\ell} \|f\|_{C^{\ell+m+1}(S^1)} \cdot n^{-m}, \end{aligned}$$

the last inequality following from (1.49). As the reader can verify, use of the proof of Proposition 1.3 can lead to a sharper estimate. As for an estimate of the contribution of Q_n to the discretization error, from (11.14)–(11.16) we easily obtain

$$(11.19) \quad \begin{aligned} \|Q_n f\|_{C^\ell(S^1)} &\leq \left(\frac{n}{2}\right)^\ell \sum_{|k| \geq \nu} |\hat{f}(k)| \\ &\leq C_{\ell m} \|f\|_{C^{\ell+m+1}(S^1)} \cdot n^{-m}. \end{aligned}$$

We reiterate that sharper estimates are possible.

Recall that solutions to a number of evolution equations are given by Fourier multipliers on $L^2(S^1)$, of the form

$$(11.20) \quad F(D)u(\theta) = \sum_{\ell=-\infty}^{\infty} F(\ell) \hat{u}(\ell) e^{i\ell\theta}.$$

We want to compare such an operator with its discretized version on $L^2(\Gamma_n)$:

$$(11.21) \quad F(D_n) \left[\sum_{\ell \in \mathbb{Z}_n} g(\ell) \omega^{j\ell} \right] = \sum_{\ell=-\nu}^{\nu-1} F(\ell) g(\ell) \omega^{j\ell}.$$

In fact, a simple calculation yields

$$(11.22) \quad E_n F(D_n) R_n u(\theta) = \sum_{\ell=-\nu}^{\nu-1} F(\ell) u^\#(\ell) e^{i\ell\theta}$$

and hence

$$(11.23) \quad E_n F(D_n) R_n u = P_n F(D) u + \Psi_n u,$$

where

$$(11.24) \quad \Psi_n u(\theta) = \sum_{\ell=-\nu}^{\nu-1} F(\ell) \left[\sum_{j \in \mathbb{Z} \setminus 0} \hat{u}(\ell + jn) \right] e^{i\ell\theta}.$$

This implies the estimate

$$(11.25) \quad \|\Psi_n u\|_{L^\infty} \leq \left[\sup_{|\ell| \leq \nu} |F(\ell)| \right] \sum_{|k| \geq \nu} |\hat{u}(k)|.$$

Also, as in (11.18), we have, for $m \geq 1$,

$$(11.26) \quad \|\Psi_n u\|_{C^\ell(S^1)} \leq C_{\ell m} \left[\sup_{|\ell| \leq \nu} |F(\ell)| \right] \|u\|_{C^{\ell+m+1}(S^1)} \cdot n^{-m}.$$

The significance of these statements is that, for u smooth and n large, the discretized $F(D_n)$ provides a very accurate approximation to $F(D)$. This is of practical importance for a number of numerical problems.

Note the distinction between D_n and the centered difference operator Δ_n , defined by

$$(\Delta_n f)(\omega^j) = \frac{n}{4\pi i} [f(\omega^{j+1}) - f(\omega^{j-1})].$$

We have, in place of (11.21),

$$(11.27) \quad F(\Delta_n) \left[\sum_{\ell \in \mathbb{Z}_n} g(\ell) \omega^{j\ell} \right] = \sum_{\ell=-\nu}^{\nu-1} F\left(\frac{n}{2\pi} \sin\left(\frac{2\pi\ell}{n}\right)\right) g(\ell) \omega^{j\ell},$$

so, for $g^b \in L^2(\Gamma_n)$ given by (11.2),

$$(11.28) \quad F(\Delta_n) g^b(\omega^j) - F(D_n) g^b(\omega^j) = \sum_{\ell=-\nu}^{\nu-1} \left[F\left(\frac{n}{2\pi} \sin\left(\frac{2\pi\ell}{n}\right)\right) - F(\ell) \right] g(\ell) \omega^{j\ell}.$$

This identity leads to a variety of estimates, of which the following is a simple example. If $|F'(\lambda)| \leq K$ for $-\nu \leq \lambda \leq \nu$, then

$$(11.29) \quad \|F(\Delta_n) u - F(D_n) u\|_{L^\infty} \leq \frac{2}{3} \cdot \pi^2 K \left[\sum_{\ell=-\nu}^{\nu-1} |\ell|^3 |u^\#(\ell)| \right] \cdot n^{-2},$$

since, for $-\pi \leq x \leq \pi$, $|\sin x - x| \leq (1/6)|x|^3$. The basic content of this is that $F(\Delta_n)$ furnishes a *second-order*-accurate approximation to $F(D)$ (as $n \rightarrow \infty$). This is an improvement over the first-order accuracy one would get by using a one-sided difference operator, such as

$$(\Delta_n^+ f)(\omega^j) = \frac{n}{2\pi i} [f(\omega^{j+1}) - f(\omega^j)],$$

but not as good as the “infinite-order accuracy” one gets for $F(D_n)$ as a consequence of (11.23)–(11.26).

Similar to the case of functions on S^1 , we have, for $u \in L^2(\Gamma_n)$,

$$(11.30) \quad F(D_n)u(\omega^j) = (k_F * u)(\omega^j) = \frac{1}{n} \sum_{\ell \in \mathbb{Z}_n} k_F(\omega^{j-\ell})u(\omega^\ell),$$

where

$$(11.31) \quad k_F(\omega^j) = \sum_{\ell=-\nu}^{\nu-1} F(\ell)\omega^{j\ell}.$$

For example, with $F(\lambda) = e^{-y|\lambda|}$, we get the discrete version of the Poisson kernel:

$$(11.32) \quad k_F(\omega^j) = p_y(\omega^j) = \sum_{\ell=-\nu}^{\nu-1} e^{-y|\ell|}\omega^{j\ell},$$

which we can write as a sum of two finite geometrical series to get

$$(11.33) \quad p_y(\omega^j) = \frac{1 - r^2 - 2r^{\nu+1}(-1)^j \cos(2\pi j/n)}{1 + r^2 - 2r \cos(2\pi j/n)} + r^\nu \omega^{-j\nu},$$

with $r = e^{-y}$ and, as usual, $\omega = e^{2\pi i/n}$, $n = 2\nu$. Compare with (1.30). The reader can produce a similar formula for n odd.

As in the case of S^1 , the sum (11.31) for the (discretized) heat kernel, with $F(\ell) = e^{-t\ell^2}$, cannot generally be simplified to an expression whose size is independent of n . However, when t is an imaginary integer, such an evaluation can be performed. Such expressions are called *Gauss sums*, and their evaluation is regarded as one of the pearls of early nineteenth-century mathematics. We present one such result here.

Proposition 11.5. *For any $n \geq 1$, even or odd,*

$$(11.34) \quad \sum_{k=0}^{n-1} e^{2\pi i k^2/n} e^{2\pi i k\ell/n} = \frac{1}{2}(1+i)e^{-\pi i \ell^2/2n} [1 + (-1)^\ell i^{-n}] n^{1/2}.$$

Proof. The sum on the left is $n \cdot f^\#(-\ell)$, where $f \in C(S^1)$ is given by

$$f(\theta) = e^{in\theta^2/2\pi}, \quad 0 \leq \theta \leq 2\pi.$$

Note that f is Lipschitz on S^1 , with a simple jump in its derivative, so $\hat{f}(k) = O(|k|^{-2})$. Hence Proposition 11.3 applies, and (11.7) yields

$$(11.35) \quad f^\#(-\ell) = \sum_{j=-\infty}^{\infty} \int_0^1 e^{2\pi i n[y^2 + (j+\ell/n)y]} dy.$$

To evaluate this, we use the “Gaussian integral” (convergent though not absolutely convergent):

$$(11.36) \quad \int_{-\infty}^{\infty} e^{2\pi i n y^2} dy = n^{-1/2} \gamma, \quad \gamma = \frac{1}{2}(1+i),$$

obtained from (3.20) by a change of variable and analytic continuation, as in (6.42). We will break up the real line as a countable union of intervals $\bigcup_k [k+a, k+a+1]$, in two different ways, and then evaluate (11.35). Note that

$$(11.37) \quad \int_{k+a}^{k+a+1} e^{2\pi i n y^2} dy = \int_0^1 e^{2\pi i n [y^2 + 2(k+a)y]} dy \cdot e^{2\pi i n (k+a)^2}.$$

If we pick $a = \ell/2n$, then $2(k+a) = 2k + \ell/n$, and as k runs over $\mathbb{Z}$, we get those integrands in (11.35) for which j is even. If we pick $a = -1/2 + \ell/2n$, then $2(k+a) = 2k - 1 + \ell/n$. Furthermore, we have

$$(11.38) \quad e^{2\pi i n (k+a)^2} = e^{\pi i \ell^2 / 2n} \quad \text{and} \quad e^{\pi i (\ell-n)^2 / 2n},$$

respectively, for these two choices of a . Thus the sum in (11.35) is equal to $n^{-1/2} \gamma$ times $e^{-\pi i \ell^2 / 2n} + e^{-\pi i (\ell-n)^2 / 2n}$, which gives the desired formula, (11.34).

The basic case of this sum is the $\ell = 0$ case:

$$(11.39) \quad \sum_{k=0}^{n-1} e^{2\pi i k^2 / n} = \frac{1}{2}(1+i)(1+i^{-n})n^{1/2} = \sigma_n \cdot n^{1/2},$$

where σ_n is periodic of period 4 in n , with

$$(11.40) \quad \sigma_0 = 1+i, \quad \sigma_1 = 1, \quad \sigma_2 = 0, \quad \sigma_3 = i.$$

This result, particularly when $n = p$ is a prime, is used as a tool to obtain fascinating number-theoretical results. For more on this, see the exercises and the references [Hua], [Land], and [Rad].

Exercises

1. Generalize the Gauss sum identity (11.34) to

$$(11.41) \quad \sum_{k=0}^{n-1} e^{2\pi i k^2 m / n} e^{2\pi i \ell k / n} = \frac{1+i}{2} \left(\frac{n}{m} \right)^{1/2} e^{-\pi i \ell^2 / 2mn} \sum_{\nu=0}^{2m-1} e^{-\pi i n \nu^2 / 2m} e^{-\pi i \nu \ell / m}.$$

(Hint: The left side is $n \cdot f^\#(-\ell)$, with

$$f(\theta) = e^{i n m \theta^2 / 2\pi}, \quad 0 \leq \theta \leq 2\pi.$$

For this, one has a formula like (11.35):

$$f^\#(-\ell) = \sum_{j=-\infty}^{\infty} \int_0^1 e^{2\pi i n m [y^2 + (1/m)(j+\ell/n)y]} dy.$$

Write $j = 2m\mu + \nu$, so

$$\sum_{j=-\infty}^{\infty} a_j = \sum_{\nu=0}^{2m-1} \sum_{j=\nu \bmod 2m} a_j.$$

For fixed ν , the sum becomes a multiple of the Gaussian integral (11.36), with n replaced by nm , and the formula (11.41) arises.)

Note the $\ell = 0$ case of this:

$$\sum_{k=0}^{n-1} e^{2\pi i k^2 m/n} = \frac{1+i}{2} \left(\frac{n}{m}\right)^{1/2} \sum_{\nu=0}^{2m-1} e^{-\pi i n \nu^2 / 2m}.$$

2. Let Δ be d^2/dx^2 on $S^1 = \mathbb{R}/(2\pi\mathbb{Z})$. Using Fourier series, show that, for $t = 2\pi m/n$, where m and n are positive integers, $e^{-it\Delta}\delta(x) = H(t, x)$ has the form

$$(11.42) \quad H\left(2\pi \frac{m}{n}, x\right) = \frac{1}{n} \sum_{\ell=0}^{n-1} G(m, n, \ell) \delta_{2\pi\ell/n}(x),$$

where $G(m, n, \ell)$ is given by the left side of (11.41). On the other hand, applying $e^{-it\Delta}$, acting on $S'(\mathbb{R})$, to $\sum_{\nu} \delta(x - 2\pi\nu)$, show that (11.42) holds, with $G(m, n, \ell)$ given by the right side of (11.41). Hence deduce another proof of this Gauss sum identity.

3. Let $\#(\ell, n)$ denote the number of solutions $k \in \mathbb{Z}_n$ to

$$\ell = k^2 \pmod{n}.$$

Show that, with $\omega = e^{2\pi i/n}$,

$$\sum_{k=0}^{n-1} \omega^{jk^2} = \sum_{\ell=0}^{n-1} \#(\ell, n) \omega^{j\ell}.$$

4. Show that, more generally,

$$\left(\sum_{k=0}^{n-1} \omega^{jk^2}\right)^\nu = \sum_{\ell=0}^{n-1} \#(\ell, n; \nu) \omega^{j\ell},$$

where $\#(\ell, n; \nu)$ denotes the number of solutions $(k_1, \dots, k_\nu) \in (\mathbb{Z}_n)^\nu$ to

$$\ell = k_1^2 + \dots + k_\nu^2 \pmod{n}.$$

5. Let p be a prime. The *Legendre symbol* $(\ell|p)$ is defined to be +1 if $\ell = k^2 \bmod p$ for some k and $\ell \neq 0$, 0 if $\ell = 0$, and -1 otherwise. If p is an odd prime, $\#(\ell, p) = (\ell|p) + 1$. The Legendre symbol has the useful *multiplicative property*: $(\ell_1 \ell_2|p) = (\ell_1|p)(\ell_2|p)$. Check this. Show that, with $\omega = e^{2\pi i/p}$, if p is an odd prime,

$$\sum_{k=0}^{p-1} \omega^{k^2} = \sum_{\ell=0}^{p-1} (\ell|p) \omega^\ell,$$

and, more generally,

$$\sum_{k=0}^{p-1} \omega^{jk^2} = \sum_{\ell=0}^{p-1} (\ell|p) \omega^{j\ell} + p\delta_{j0},$$

where $\delta_{j0} = 1$ if $j = 0 \pmod{p}$, 0 otherwise. (*Hint:* Use Exercise 3.)

6. Denoting $\sum_{k=0}^{p-1} \omega^{k^2}$ by G_p , p an odd prime, show that

$$\sum_{k=0}^{p-1} \omega^{jk^2} = (j|p) \cdot G_p + p \cdot \delta_{j0}.$$

(*Hint:* If $1 \leq j \leq p-1$, use $\sum_{\ell=0}^{p-1} (\ell|p) \omega^\ell = \sum_{\ell=0}^{p-1} (j\ell|p) \omega^{j\ell}$ and $(j\ell|p) = (j|p)(\ell|p)$.)

Denote by $S(m, n)$ the Gauss sum

$$S(m, n) = \sum_{k=0}^{n-1} e^{2\pi i k^2 m/n}.$$

Then the content of Exercise 6 is that $S(j, p) = (j|p)S(1, p)$, for $1 \leq j \leq p-1$, when p is an odd prime.

7. Assume p and q are distinct odd primes. Show that

$$S(1, pq) = S(q, p)S(p, q).$$

(*Hint:* To resum $\sum_{k=0}^{pq-1} e^{2\pi i k^2/pq}$, use the fact that, as μ runs over $\{0, 1, \dots, p-1\}$ and ν runs over $\{0, 1, \dots, q-1\}$, then $k = \mu q + \nu p$ runs once over $\mathbb{Z} \bmod pq$.)

8. From Exercises 6 and 7, it follows that when p and q are distinct odd primes,

$$(p|q)(q|p) = \frac{S(1, pq)}{S(1, p)S(1, q)}.$$

Use the evaluation (11.39) of $S(1, n)$ to deduce the *quadratic reciprocity law*:

$$(p|q)(q|p) = (-1)^{(p-1)(q-1)/4}.$$

This law, together with the complementary results

$$(-1|p) = (-1)^{(p-1)/2}, \quad (2|p) = (-1)^{(p^2-1)/8},$$

allows for an effective computation of $(\ell|p)$, as one application, but the significance of quadratic reciprocity goes beyond this. It and other implications of Gauss sums are absolutely fundamental in number theory. For material on this, see [Hua], [Land], and [Rad].

12. The fast Fourier transform

In the last section we discussed some properties of the discrete Fourier transform

$$(12.1) \quad f^\#(\ell) = \frac{1}{n} \sum_{\omega^j \in \Gamma_n} f(\omega^j) \omega^{-j\ell},$$

where $\ell \in \mathbb{Z}_n = \mathbb{Z}/(n)$ and Γ_n is the multiplicative group of unit complex numbers generated by $\omega = e^{2\pi i/n}$. We now turn to a discussion of the efficient numerical computation of the discrete Fourier transform. Note that, for any fixed ℓ , computing the right side of (12.1) involves $n - 1$ additions and n multiplications of complex numbers, plus n integer products $j\ell = m$ and looking up ω^m and $f(\omega^j)$. If the computations for varying ℓ are done independently, the total effort to compute $f^\#$ involves n^2 multiplications and $n(n - 1)$ additions of complex numbers, plus some further chores. The *fast Fourier transform* (denoted FFT) is a method for computing $f^\#$ in $Cn(\log n)$ steps, in case n is a power of 2.

The possibility of doing this arises from observing redundancies in the calculation of the Fourier coefficients $f^\#(\ell)$. Let us illustrate this in the case of Γ_4 . We can write

$$(12.2) \quad \begin{aligned} 4f^\#(0) &= [f(1) + f(i^2)] + [f(i) + f(i^3)], \\ 4f^\#(2) &= [f(1) + f(i^2)] - [f(i) + f(i^3)], \end{aligned}$$

and

$$(12.3) \quad \begin{aligned} 4f^\#(1) &= [f(1) - f(i^2)] - i[f(i) - f(i^3)], \\ 4f^\#(3) &= [f(1) - f(i^2)] + i[f(i) - f(i^3)]. \end{aligned}$$

Note that each term in square brackets appears twice. Note also that (12.2) gives the Fourier coefficients of a function on Γ_2 ; namely, if

$$(12.4) \quad {}^0f(1) = f(1) + f(-1), \quad {}^0f(-1) = f(i) + f(i^3),$$

then

$$(12.5) \quad 2f^\#(2\ell) = {}^0f^\#(\ell), \quad \text{for } \ell = 0 \text{ or } 1.$$

Similarly, if we set

$$(12.6) \quad {}^1f(1) = f(1) - f(-1), \quad {}^1f(-1) = -i[f(i) - f(i^3)],$$

then

$$(12.7) \quad 2f^\#(2\ell + 1) = {}^1f^\#(\ell), \quad \text{for } \ell = 0 \text{ or } 1.$$

This phenomenon is a special case of a more general result that leads to a fast inductive procedure for evaluating the Fourier transform $f^\#$.

Suppose $n = 2^k$; let us use the notation $G_k = \Gamma_n$. Note that G_{k-1} is a subgroup of G_k . Furthermore, there is a homomorphism of G_k onto G_{k-1} , given by $\omega^j \mapsto \omega^{2j}$. Given $f : G_k \rightarrow \mathbb{C}$, define the following functions 0f and 1f on G_{k-1} , with $\omega_1 = \omega^2$, generating G_{k-1} :

$$(12.8) \quad {}^0f(\omega_1^j) = f(\omega^j) + f(\omega^{j+n/2}),$$

$$(12.9) \quad {}^1f(\omega_1^j) = \bar{\omega}^j [f(\omega^j) - f(\omega^{j+n/2})].$$

Note that the factor $\bar{\omega}^j$ in (12.9) makes ${}^1f(\omega_1^j)$ well defined for $j \in \mathbb{Z}_{n/2}$, that is, the right side of (12.9) is unchanged if j is replaced by $j + n/2$.

Then ${}^0f^\#$ and ${}^1f^\#$, the discrete Fourier transforms of the functions 0f and 1f , respectively, are functions on $\mathbb{Z}_{n/2} = \mathbb{Z}/(2^{k-1})$.

Proposition 12.1. *We have the following identities relating the Fourier transforms of 0f , 1f , and f :*

$$(12.10) \quad 2f^\#(2\ell) = {}^0f^\#(\ell)$$

and

$$(12.11) \quad 2f^\#(2\ell + 1) = {}^1f^\#(\ell),$$

for $\ell \in \{0, 1, \dots, n/2 - 1\}$.

Proof. Recall that we set $\omega_1 = \omega^2$. Since $\omega^n = 1$ and $\omega_1^{n/2} = 1$, we have

$$(12.12) \quad \begin{aligned} n f^\#(2\ell) &= \sum_{\omega^j \in G_k} f(\omega^j) \bar{\omega}^{2j\ell} \\ &= \sum_{\omega_1^j = \omega^{2j} \in G_{k-1}} [f(\omega^j) + f(\omega^{j+n/2})] \bar{\omega}_1^{j\ell}, \end{aligned}$$

proving (12.10), and, since $\omega^{n/2} = -1$,

$$(12.13) \quad \begin{aligned} n f^\#(2\ell + 1) &= \sum_{\omega^j \in G_k} f(\omega^j) \bar{\omega}^j \bar{\omega}^{2j\ell} \\ &= \sum_{\omega_1^j = \omega^{2j} \in G_{k-1}} \bar{\omega}^j [f(\omega^j) - f(\omega^{j+n/2})] \bar{\omega}_1^{j\ell}, \end{aligned}$$

proving (12.11).

Thus the problem of computing $f^\#$, given $f \in L^2(G_k)$, is transformed, after $n/2$ multiplications and n additions of complex numbers in (12.8) and (12.9), to the problem of computing the Fourier transforms of *two* functions on G_{k-1} . After $n/4$ new multiplications and $n/2$ new additions for each of these functions 0f and 1f , that is, after an additional total of $n/2$ new multiplications and n additions, this is reduced to the problem of computing *four* Fourier transforms of functions on G_{k-2} . After k iterations, we obtain 2^k functions on $G_0 = \{1\}$, which precisely give the Fourier coefficients of f . Doing this hence takes $kn = (\log_2 n)n$ additions and $kn/2 = (\log_2 n)n/2$ multiplications of complex numbers, plus a comparable number of integer operations and fetching from memory values of given or previously computed functions.

To describe an explicit implementation of Proposition 12.1 for a computation of $f^\#$, let us identify an element $\ell \in \mathbb{Z}_n$ ($n = 2^k$) with a k -tuple $L = (L_{k-1}, \dots, L_1, L_0)$ of elements of $\{0, 1\}$ giving the binary expansion of the integer in $\{0, \dots, n-1\}$ representing ℓ (i.e., $L_0 + L_1 \cdot 2 + \dots + L_{k-1} \cdot 2^{k-1} = \ell$).

mod n). To be a little fussy, we use the notation

$$(12.14) \quad f^\#(\ell) = f^{\#\#}(L).$$

Then the formulas (12.10) and (12.11) state that

$$(12.15) \quad 2f^{\#\#}(L_{k-1}, \dots, L_1, 0) = {}^0f^{\#\#}(L_{k-1}, \dots, L_1)$$

and

$$(12.16) \quad 2f^{\#\#}(L_{k-1}, \dots, L_1, 1) = {}^1f^{\#\#}(L_{k-1}, \dots, L_1).$$

The inductive procedure described above gives, from 0f and 1f defined on G_{k-1} , the functions

$$(12.17) \quad {}^{00}f = {}^0({}^0f), \quad {}^{10}f = {}^1({}^0f), \quad {}^{01}f = {}^0({}^1f), \quad {}^{11}f = {}^1({}^1f)$$

defined on G_{k-2} , and so forth, and we see from (12.15) and (12.16) that

$$(12.18) \quad f^\#(\ell) = \frac{1}{n} {}^L f,$$

where ${}^L f = {}^L f(1)$ is defined on $G_0 = \{1\}$. From (12.8) and (12.9) we have the following inductive formula for ${}^{L_{m+1}L_m \cdots L_1} f$ on G_{k-m-1} :

$$(12.19) \quad \begin{aligned} {}^{0L_m \cdots L_1} f(\omega_{m+1}^j) &= {}^{L_m \cdots L_1} f(\omega_m^j) + {}^{L_m \cdots L_1} f(\omega_m^{j+2^{k-m-1}}), \\ {}^{1L_m \cdots L_1} f(\omega_{m+1}^j) &= \bar{\omega}_m^j [{}^{L_m \cdots L_1} f(\omega_m^j) - {}^{L_m \cdots L_1} f(\omega_m^{j+2^{k-m-1}})], \end{aligned}$$

where ω_m is the generator of G_{k-m} , defined by $\omega_0 = \omega = e^{2\pi i/n}$ ($n = 2^k$), $\omega_{m+1} = \omega_m^2$, that is, $\omega_m = \omega^{2^m}$.

When doing computations, particularly in a higher-level language, it may be easier to work with integers ℓ than with m -tuples $(L_1, \dots, L_1)$. Therefore, let us set

$$(12.20) \quad {}^{L_m \cdots L_1} f(\omega_m^j) = F_m(2^m \cdot j + \ell),$$

where

$$\ell = L_1 + L_2 \cdot 2 + \cdots + L_m \cdot 2^{m-1} \in \{0, 1, \dots, 2^m - 1\}$$

and

$$j \in \{0, 1, \dots, 2^{k-m} - 1\}.$$

Note that this precisely defines F_m on $\{0, 1, \dots, 2^k - 1\}$. For $m = 0$, we have

$$(12.21) \quad F_0(j) = f(\omega^j), \quad 0 \leq j \leq 2^k - 1.$$

The iterative formulas (12.19) give

$$(12.22) \quad \begin{aligned} F_{m+1}(2^{m+1}j + \ell) &= F_m(2^m j + \ell) + F_m(2^m j + 2^{k-1} + \ell), \\ F_{m+1}(2^m + 2^{m+1}j + \ell) &= \bar{\omega}_m^j [F_m(2^m j + \ell) - F_m(2^m j + 2^{k-1} + \ell)], \end{aligned}$$

for $0 \leq j \leq 2^{k-m-1} - 1$, $0 \leq \ell \leq 2^m - 1$. It is easy to write a computer program to implement such an iteration. The formula (12.18) for the Fourier transform of f becomes

$$(12.23) \quad f^\#(\ell) = n^{-1} F_k(\ell), \quad 0 \leq \ell \leq 2^k - 1.$$

While (12.21)–(12.23) provide an easily implementable FFT algorithm, it is not necessarily the best. One drawback is the following. In passing from F_m to F_{m+1} via (12.22), you need two different arrays of n complex numbers. A variant of (12.19), where ${}^{L_{m+1}L_m \cdots L_1} f$ is replaced by $f^{L_1 \cdots L_m L_{m+1}}$, leads to an iterative procedure where a transformation of the type (12.19) is performed “in place,” and only one such array needs to be used. If memory is expensive and one needs to make the best use of it, this savings can be important. At the end of such an iteration, one needs to perform a “bit reversal” to produce $f^\#$. Details, including sample programs, can be found in [PFTV].

On any given computer, a number of factors would influence the choice of the best FFT algorithm. These include such things as relative speed of memory access and floating-point performance, efficiency of computing trigonometric functions (e.g., whether this is implemented in hardware), degree of accuracy required, and other factors. Also, special features, such as computing the Fourier transform of a real-valued function or of a function whose Fourier transform is known to be real-valued, would affect specific computer programs designed for maximum efficiency. Working out how best to implement FFTs on various computers presents many interesting problems.

Exercises

1. Write a computer program to implement the FFT via (12.21)–(12.23). Try to make it run as fast as possible.
2. Using the FFT, write a computer program to solve numerically the initial-value problem for the heat equation $\partial u / \partial t - u_{xx} = 0$ on $\mathbb{R}^+ \times S^1$.
3. Consider multidimensional generalizations of the discrete Fourier transform, and in particular the FFT. What size three-dimensional FFT could be handled by a computer with 4 megabytes of RAM? With 256 MB?
4. Generalize the FFT algorithm to a cyclic group Γ_n with $n = 3^k$. Also, generalize to the case $n = p_1 \cdots p_k$ where p_j are “small” primes.

A. The mighty Gaussian and the sublime gamma function

The Gaussian function $e^{-|x|^2}$ on $\mathbb{R}^n$ is an object whose study yields many wonderful identities. We will use the identity

$$(A.1) \quad \int_{\mathbb{R}^n} e^{-|x|^2} dx = \pi^{n/2},$$

which was established in (3.18), to compute the area A_{n-1} of the unit sphere S^{n-1} in $\mathbb{R}^n$. This computation will bring in Euler's gamma function, and other results will flow from this. Switching to polar coordinates for the right side of (A.1), we have

$$(A.2) \quad \begin{aligned} \pi^{n/2} &= A_{n-1} \int_0^\infty e^{-r^2} r^{n-1} dr \\ &= \frac{1}{2} A_{n-1} \int_0^\infty e^{-t} t^{n/2-1} dt \\ &= \frac{1}{2} A_{n-1} \Gamma\left(\frac{n}{2}\right), \end{aligned}$$

where the gamma function is defined by

$$(A.3) \quad \Gamma(z) = \int_0^\infty e^{-t} t^{z-1} dt,$$

for $\operatorname{Re} z > 0$. Thus we have the formula

$$(A.4) \quad A_{n-1} = \frac{2\pi^{n/2}}{\Gamma(\frac{1}{2}n)}.$$

To be satisfied with this, we need an explicit evaluation of $\Gamma(n/2)$. This can be obtained from $\Gamma(1/2)$ and $\Gamma(1)$ via the following identity:

$$(A.5) \quad \begin{aligned} \Gamma(z+1) &= \int_0^\infty e^{-t} t^z dt \\ &= - \int_0^\infty \frac{d}{dt}(e^{-t}) t^z dt \\ &= z\Gamma(z), \end{aligned}$$

for $\operatorname{Re} z > 0$, where we used integration by parts. The definition (A.3) clearly gives

$$(A.6) \quad \Gamma(1) = 1.$$

Thus, for any integer $k \geq 1$,

$$(A.7) \quad \Gamma(k) = (k-1)\Gamma(k-1) = \cdots = (k-1)!.$$

Note that, for $n = 2$, we have $A_1 = 2\pi/\Gamma(1)$, so (A.6) agrees with the fact that the circumference of the unit circle is 2π (which, of course, figured

into the proof of (3.18), via (3.20)). In case $n = 1$, we have $A_0 = 2$, which by (A.4) is equal to $2\pi^{1/2}/\Gamma(1/2)$, so

$$(A.8) \quad \Gamma\left(\frac{1}{2}\right) = \pi^{1/2}.$$

Again using (A.5), we see that, when $k \geq 1$ is an integer,

$$(A.9) \quad \begin{aligned} \Gamma\left(k + \frac{1}{2}\right) &= \left(k - \frac{1}{2}\right)\Gamma\left(k - \frac{1}{2}\right) = \cdots \\ &= \left(k - \frac{1}{2}\right)\left(k - \frac{3}{2}\right) \cdots \left(\frac{1}{2}\right)\Gamma\left(\frac{1}{2}\right) \\ &= \pi^{1/2}\left(k - \frac{1}{2}\right)\left(k - \frac{3}{2}\right) \cdots \left(\frac{1}{2}\right). \end{aligned}$$

In particular, $\Gamma(3/2) = (1/2)\Gamma(1/2) = \pi^{1/2}/2$, so $A_2 = 2\pi^{3/2}/(\pi^{1/2}/2) = 4\pi$, which agrees with the well known formula for the area of the unit sphere in $\mathbb{R}^3$.

Note that while $\Gamma(z)$ defined by (A.3) is a priori holomorphic for $\operatorname{Re} z$ positive, the equation (A.5) shows that $\Gamma(z)$ has a meromorphic extension to the entire complex plane, with simple poles at $z = 0, -1, -2, \dots$. It turns out that $\Gamma(z)$ has no zeros, so $1/\Gamma(z)$ is an entire analytic function. This is a consequence of the identity

$$(A.10) \quad \Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin \pi z},$$

which we now establish. From (A.4) we have (for $0 < \operatorname{Re} z < 1$)

$$(A.11) \quad \begin{aligned} \Gamma(z)\Gamma(1-z) &= \int_0^\infty \int_0^\infty e^{-(s+t)} s^{-z} t^{z-1} ds dt \\ &= \int_0^\infty \int_0^\infty e^{-u} v^{z-1} (1+v)^{-1} du dv \\ &= \int_0^\infty (1+v)^{-1} v^{z-1} dv, \end{aligned}$$

where we have used the change of variables $u = s + t$, $v = t/s$. With $v = e^x$, the last integral is

$$(A.12) \quad \int_{-\infty}^\infty (1 + e^x)^{-1} e^{xz} dx,$$

which is holomorphic for $0 < \operatorname{Re} z < 1$, and we want to show that it is equal to the right side of (A.10) on this strip. It suffices to prove identity on the line $z = 1/2 + i\xi$, $\xi \in \mathbb{R}$; then (A.12) is equal to the Fourier integral

$$(A.13) \quad \int_{-\infty}^\infty \left(2 \cosh \frac{x}{2}\right)^{-1} e^{ix\xi} dx.$$

To evaluate this, shift the contour of integration from the real line to the line $\operatorname{Im} x = -2\pi$. There is a pole of the integrand at $x = -\pi i$, and we have

(A.13) equal to

$$(A.14) \quad - \int_{-\infty}^{\infty} \left(2 \cosh \frac{x}{2}\right)^{-1} e^{2\pi\xi} e^{ix\xi} dx - \text{Residue} \cdot (2\pi i).$$

Consequently, (A.13) is equal to

$$(A.15) \quad -2\pi i \frac{\text{Residue}}{1 + e^{2\pi\xi}} = \frac{\pi}{\cosh \pi\xi},$$

and since $\pi / \sin \pi(1/2 + i\xi) = \pi / \cosh \pi\xi$, the demonstration of (A.10) is complete.

The integral (A.3) and also the last integral in (A.11) are special cases of the *Mellin transform*:

$$(A.16) \quad \mathcal{M}f(z) = \int_0^{\infty} f(t)t^{z-1} dt.$$

If we evaluate this on the imaginary axis:

$$(A.17) \quad \mathcal{M}^{\#}f(s) = \int_0^{\infty} f(t)t^{is-1} dt,$$

given appropriate growth restrictions on f , this is related to the Fourier transform by a change of variable:

$$(A.18) \quad \mathcal{M}^{\#}f(s) = \int_{-\infty}^{\infty} f(e^x)e^{isx} dx.$$

The Fourier inversion formula and Plancherel formula imply

$$(A.19) \quad f(r) = (2\pi)^{-1} \int_{-\infty}^{\infty} (\mathcal{M}^{\#}f)(s)r^{-is} ds$$

and

$$(A.20) \quad \int_{-\infty}^{\infty} |\mathcal{M}^{\#}f(s)|^2 ds = (2\pi) \int_0^{\infty} |f(r)|^2 r^{-1} dr.$$

In some cases, as seen above, one evaluates $\mathcal{M}f(z)$ on a vertical line other than the imaginary axis, which introduces only a slight wrinkle.

An important identity for the gamma function follows from taking the Mellin transform with respect to y of both sides of the subordination identity

$$(A.21) \quad e^{-yA} = \frac{1}{2}y\pi^{-1/2} \int_0^{\infty} e^{-y^2/4t} e^{-tA^2} t^{-3/2} dt$$

(if $y > 0$, $A > 0$), established in §5; see (5.22). The Mellin transform of the left side is clearly $\Gamma(z)A^{-z}$. The Mellin transform of the right side is a double integral, which is readily converted to a product of two integrals, each defining gamma functions. After a few changes of variables, there

results the identity

$$(A.22) \quad \pi^{1/2} \Gamma(2z) = 2^{2z-1} \Gamma(z) \Gamma\left(z + \frac{1}{2}\right),$$

known as the *duplication formula* for the gamma function. In view of the uniqueness of Mellin transforms, following from (A.18) and (A.19), the identity (A.22) conversely implies (A.21). In fact, (A.22) was obtained first (by Legendre) and this argument produces one of the standard proofs of the subordination identity (A.21).

There is one further identity, which, together with (A.5), (A.10), and (A.22), completes the list of the basic elementary identities for the gamma function. Namely, if the beta function is defined by

$$(A.23) \quad B(x, y) = \int_0^1 s^{x-1} (1-s)^{y-1} ds = \int_0^1 (1+u)^{-x-y} u^{x-1} du$$

(with $u = s/(1-s)$), then

$$(A.24) \quad B(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}.$$

To prove this, note that since

$$(A.25) \quad \Gamma(z) p^{-z} = \int_0^\infty e^{-pt} t^{z-1} dt,$$

we have

$$(1+u)^{-x-y} = \frac{1}{\Gamma(x+y)} \int_0^\infty e^{-(1+u)t} t^{x+y-1} dt,$$

so

$$\begin{aligned} B(x, y) &= \frac{1}{\Gamma(x+y)} \int_0^\infty e^{-t} t^{x+y-1} \int_0^\infty e^{-ut} u^{x-1} du dt \\ &= \frac{\Gamma(x)}{\Gamma(x+y)} \int_0^\infty e^{-t} t^{y-1} dt \\ &= \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}, \end{aligned}$$

as asserted.

The four basic identities proved above are the workhorses for most applications involving gamma functions, but fundamental insight is provided by the identities (A.27) and (A.31) below. First, since $0 \leq e^{-t} - (1-t/n)^n \leq$

$e^{-t} \cdot t^2/n$ for $0 \leq t \leq n$, we have, for $\text{Re } z > 0$,

$$\begin{aligned}
 \Gamma(z) &= \int_0^\infty e^{-t} t^{z-1} dt \\
 (A.26) \quad &= \lim_{n \rightarrow \infty} \int_0^n \left(1 - \frac{t}{n}\right)^n t^{z-1} dt \\
 &= \lim_{n \rightarrow \infty} n^z \int_0^1 (1-s)^n s^{z-1} ds.
 \end{aligned}$$

Repeatedly integrating by parts gives

$$\Gamma(z) = \lim_{n \rightarrow \infty} n^z \frac{n(n-1) \cdots 1}{z(z+1) \cdots (z+n-1)} \int_0^1 s^{z+n-1} ds,$$

which yields the following result of Euler:

$$(A.27) \quad \Gamma(z) = \lim_{n \rightarrow \infty} n^z \frac{1 \cdot 2 \cdots n}{z(z+1) \cdots (z+n)}.$$

Using the identity (A.5), analytically continuing $\Gamma(z)$, we have (A.27) for all z , other than $0, -1, -2, \dots$. We can rewrite (A.27) as

$$(A.28) \quad \Gamma(z) = \lim_{n \rightarrow \infty} n^z z^{-1} (1+z)^{-1} \left(1 + \frac{z}{2}\right)^{-1} \cdots \left(1 + \frac{z}{n}\right)^{-1}.$$

If we denote by γ Euler's constant:

$$(A.29) \quad \gamma = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{2} + \cdots + \frac{1}{n} - \log n\right),$$

then (A.28) is equivalent to

$$(A.30) \quad \Gamma(z) = \lim_{n \rightarrow \infty} e^{-\gamma z} e^{z(1+1/2+\cdots+1/n)} z^{-1} (1+z)^{-1} \left(1 + \frac{z}{2}\right)^{-1} \cdots \left(1 + \frac{z}{n}\right)^{-1},$$

that is, to the Euler product expansion

$$(A.31) \quad \frac{1}{\Gamma(z)} = z e^{\gamma z} \prod_{n=1}^{\infty} \left(1 + \frac{z}{n}\right) e^{-z/n}.$$

It follows that the entire analytic function $1/\Gamma(z)\Gamma(-z)$ has the product expansion

$$(A.32) \quad \frac{1}{\Gamma(z)\Gamma(-z)} = -z^2 \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right).$$

Since $\Gamma(1-z) = -z\Gamma(-z)$, by virtue of (A.10) this last identity is equivalent to the Euler product expansion

$$(A.33) \quad \sin \pi z = \pi z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right).$$

It is quite easy to deduce the formula (A.5) from the Euler product expansion (A.31). Also, to deduce the duplication formula (A.22) from the Euler product formula is a fairly straightforward exercise.

Finally, we derive Stirling's formula, for the asymptotic behavior of $\Gamma(z)$ as $z \rightarrow +\infty$. The approach uses the Laplace asymptotic method, which has many other applications. We begin by setting $t = sz$ and then $s = e^y$ in the integral formula (A.3), obtaining

$$\begin{aligned} \Gamma(z) &= z^z \int_0^\infty e^{-z(s-\log s)} s^{-1} ds \\ (A.34) \quad &= z^z \int_{-\infty}^\infty e^{-z(e^y-y)} dy. \end{aligned}$$

The last integral is of the form

$$(A.35) \quad \int_{-\infty}^\infty e^{-z\varphi(y)} dy,$$

where $\varphi(y) = e^y - y$ has a nondegenerate minimum at $y = 0$; $\varphi(0) = 1$, $\varphi'(0) = 0$, $\varphi''(0) = 1$. If we write $1 = A(y) + B(y)$, $A \in C_0^\infty((-2, 2))$, $A(y) = 1$ for $|y| \leq 1$, then the integral (A.35) is readily seen to be

$$(A.36) \quad \int_{-\infty}^\infty A(y) e^{-z\varphi(y)} dy + O(e^{-(1+1/\epsilon)z}).$$

We can make a smooth change of variable $x = \xi(y)$ such that $\xi(y) = y + O(y^2)$, $\varphi(y) = 1 + x^2/2$, and the integral in (A.36) becomes

$$(A.37) \quad e^{-z} \int_{-\infty}^\infty A_1(x) e^{-zx^2/2} dx,$$

where $A_1 \in C_0^\infty(\mathbb{R})$, $A_1(0) = 1$, and it is easy to see that, as $z \rightarrow +\infty$,

$$(A.38) \quad \int_{-\infty}^\infty A_1(x) e^{-zx^2/2} dx \sim \left(\frac{2\pi}{z}\right)^{1/2} \left[1 + \frac{a_1}{z} + \cdots\right].$$

In fact, if $z = 1/2t$, then (A.38) is equal to $(4\pi t)^{1/2} u(t, 0)$, where $u(t, x)$ solves the heat equation, $u_t - u_{xx} = 0$, $u(0, x) = A_1(x)$. Returning to (A.34), we have Stirling's formula:

$$(A.39) \quad \Gamma(z) = \left(\frac{2\pi}{z}\right)^{1/2} z^z e^{-z} [1 + O(z^{-1})].$$

Since $n! = \Gamma(n+1)$, we have in particular that

$$(A.40) \quad n! = (2\pi n)^{1/2} n^n e^{-n} [1 + O(n^{-1})]$$

as $n \rightarrow \infty$.

Regarding this approach to the Laplace asymptotic method, compare the derivation of the stationary phase method in Appendix B of Chapter 6.

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